

UVA CS 4774: Machine Learning

Part 2: Course Session QA

Related:

- Part 1 [[HERE](#)]
- Course Announcement
Page [[HERE](#)]

Dr. Yanjun Qi

University of Virginia
Department Of Computer Science



09/30

09/30/2025 Assignments

- HW2 is due this coming Sunday midnight!
 - Using HW1 code pieces as components;
 - If you struggle with HW1, please contact TA @Haochen ASAP
- HW1 grading is work-in-progress,
 - Grades will be released by next Tuesday class time
 - We posted the guide from TA in Canvas
- Course vote:
 - **New Survey that needs your vote on**
 - 1. back to lecture in-person twice a week?
 - 2. If not, best way to use the in-person session:
 - Quiz to continue
 - + Project discussions
 - Interested in Shark Tank alike setup? idea screening, pitch talk, demo ...

Project Process

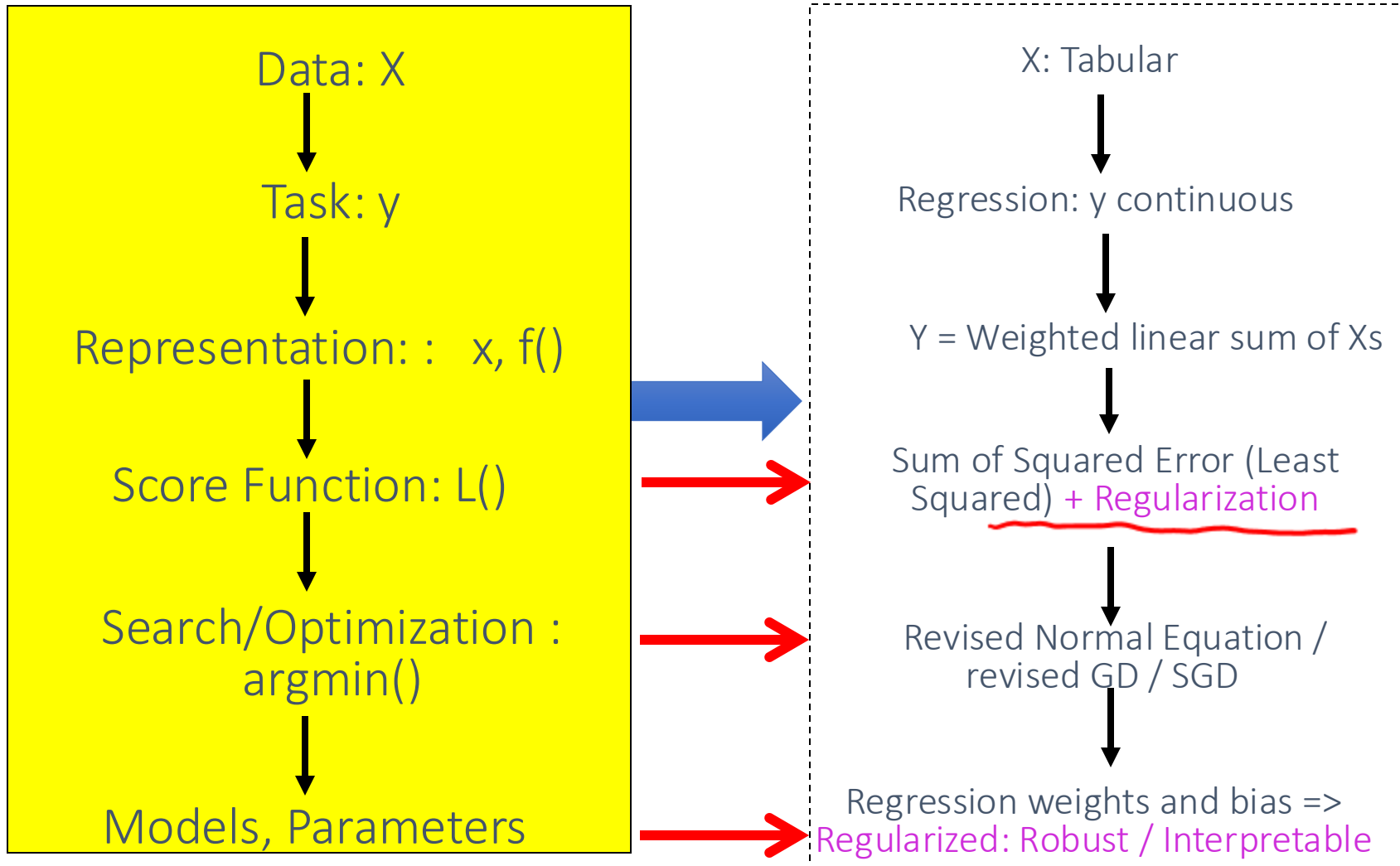


- Format:
 - Team, individual ...
 - Shark Tank alike Screening ? – https://en.wikipedia.org/wiki/Shark_Tank
 - Next week – Idea collection
- Final deliverables:
 - (1) Code (Github PR to course project repo)
 - (2) Poster presentation class wide (Date: TBD)
 - (3) Video Demo (TBD)

09/30/2025 Roadmap

- TA to go over HW1
- One UVA ML club to introduce their setups and projects
- Q5
- Review Q4
- Review QA for L5-L7

L7: Regularized multivariate linear regression

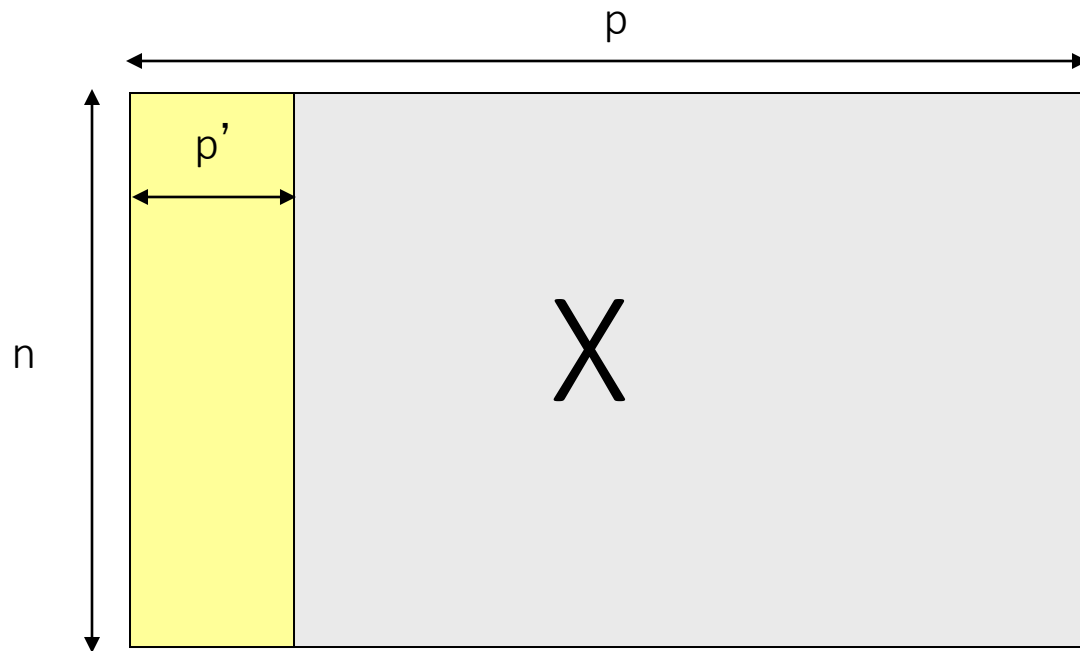


L7: Regularized multivariate linear regression

We aim to make our trained model

- 1. Generalize Well
- 2. Computational Scalable and Efficient
- 3. Trustworthy: Robust / Interpretable
 - Especially for some domains, this is about trust!

Large p , small n : How? $p \rightarrow p' \Rightarrow$ easy to understand



Regularized multivariate linear regression

• Model: $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_p x_p$

- LR estimation:

$$\arg \min_{\beta} \sum \left(Y - \hat{Y} \right)^2$$

- LASSO estimation:

$$\arg \min \sum_{i=1}^n \left(Y - \hat{Y} \right)^2 + \lambda \sum_{j=1}^p |\beta_j|$$

- Ridge regression estimation:

$$\arg \min_{\beta} \sum_{i=1}^n \left(Y - \hat{Y} \right)^2 + \lambda \sum_{j=1}^p \beta_j^2$$

Error on data

+

Regularization

9/54

Ridge Regression / L2 Regularized Regression

$$\beta^* = (X^T X)^{-1} X^T \bar{y}$$

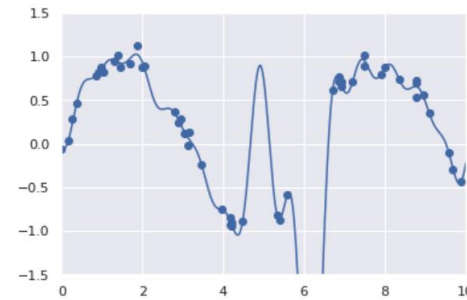


- If not **invertible**, a classical solution is to add a small positive element to diagonal

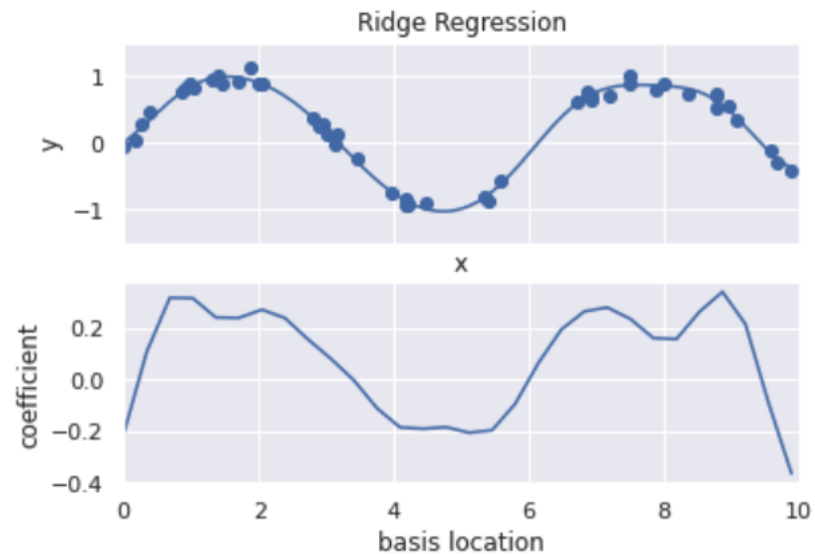
$$\beta^* = (X^T X + \lambda I)^{-1} X^T \bar{y}$$

Overfitting: Can be Handled by Regularization

A **regularizer** is an additional criteria to the loss function to make sure that we don't overfit. It's called a **regularizer** since it tries to keep the parameters more normal/regular



```
from sklearn.linear_model import Ridge
model = make_pipeline(GaussianFeatures(30), Ridge(alpha=0.1))
basis_plot(model, title='Ridge Regression')
```



WHY and How to Select λ ?

- 1. Generalization ability
 ➔ k-folds CV to decide
- 2. Control the bias and Variance of the model (details in future lectures)

L2: Squared weights penalizes large values more

L1: Sum of weights will penalize small values more

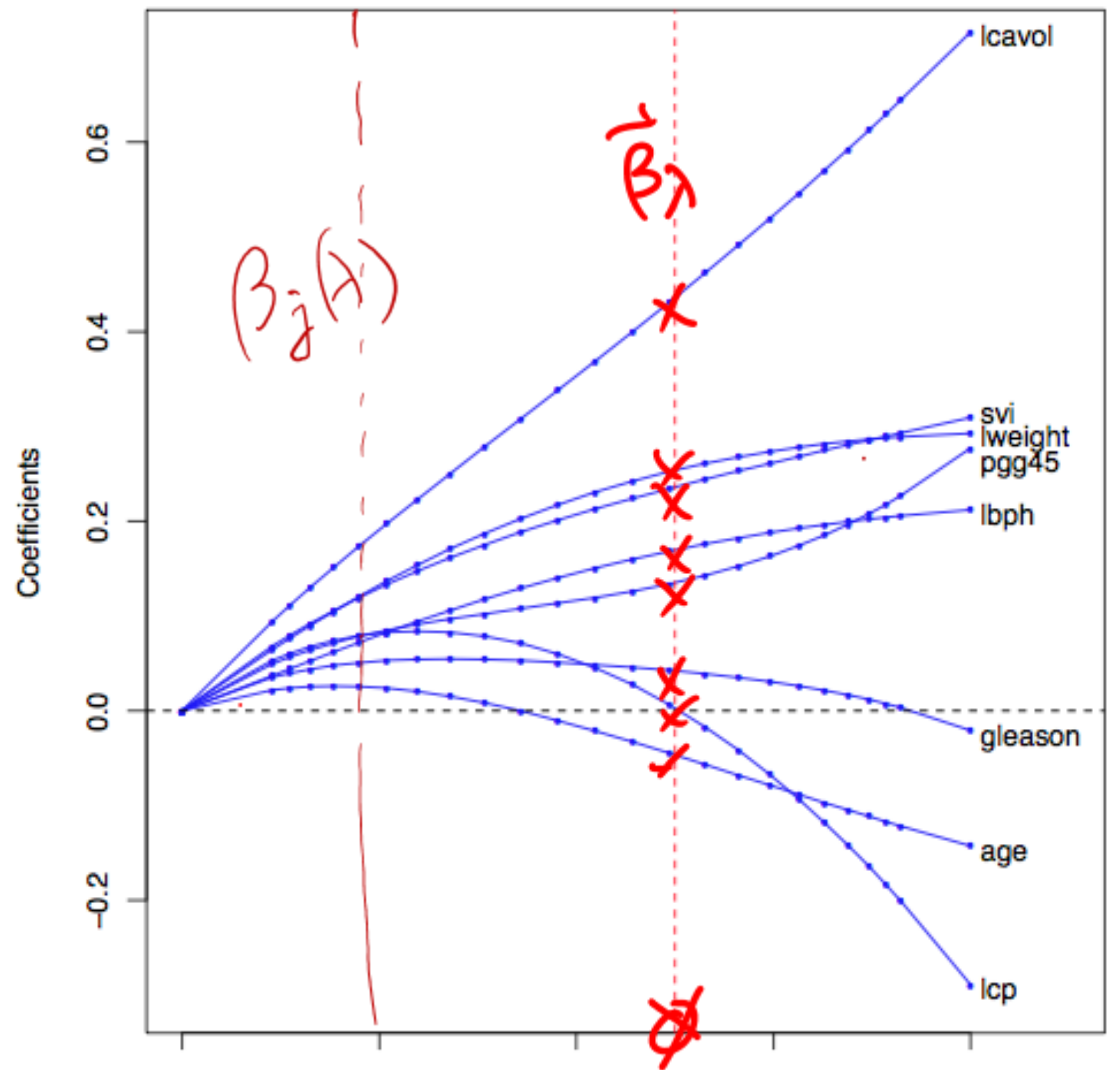
$$\sum_j \beta_j^2$$

$$\sum_j |\beta_j|$$

Regularization path of a Ridge Regression

When $X^T X = I \Rightarrow \frac{1}{1+\lambda} \beta_{OLS}$

Weight Decay

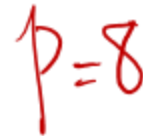


WHY and How to Select λ ?

$$\lambda \rightarrow \infty$$

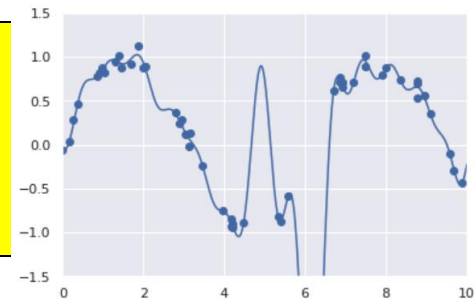
$$\lambda = 0$$

when varying λ ,
how β_j varies.



Overfitting: Can be Handled by Regularization

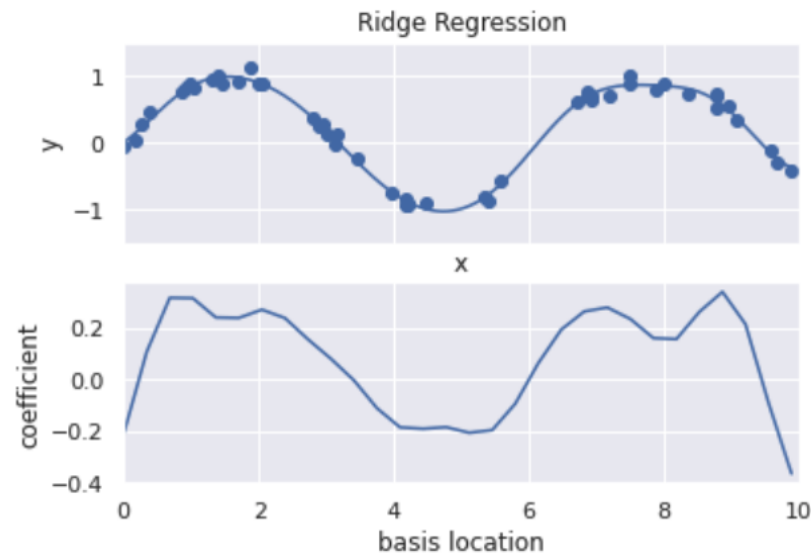
A **regularizer** is an additional criteria to the loss function to make sure that we don't overfit. It's called a **regularizer** since it tries to keep the parameters more normal/regular



code-run:

https://github.com/qiyanjun/2025Fall-UVA-CS-MachineLearningDeep/blob/main/notebook/L7_regularizedRegression_06_Linear_Regression.ipynb

```
from sklearn.linear_model import Ridge
model = make_pipeline(GaussianFeatures(30), Ridge(alpha=0.1))
basis_plot(model, title='Ridge Regression')
```



(Extra) Lasso (least absolute shrinkage and selection operator) / Squared Loss+L1

- The lasso is a shrinkage method like ridge, but acts in a nonlinear manner on the outcome y .
- The lasso is defined by

$$\sum_{i=1}^n (y_i - x_i^T \beta)^2$$

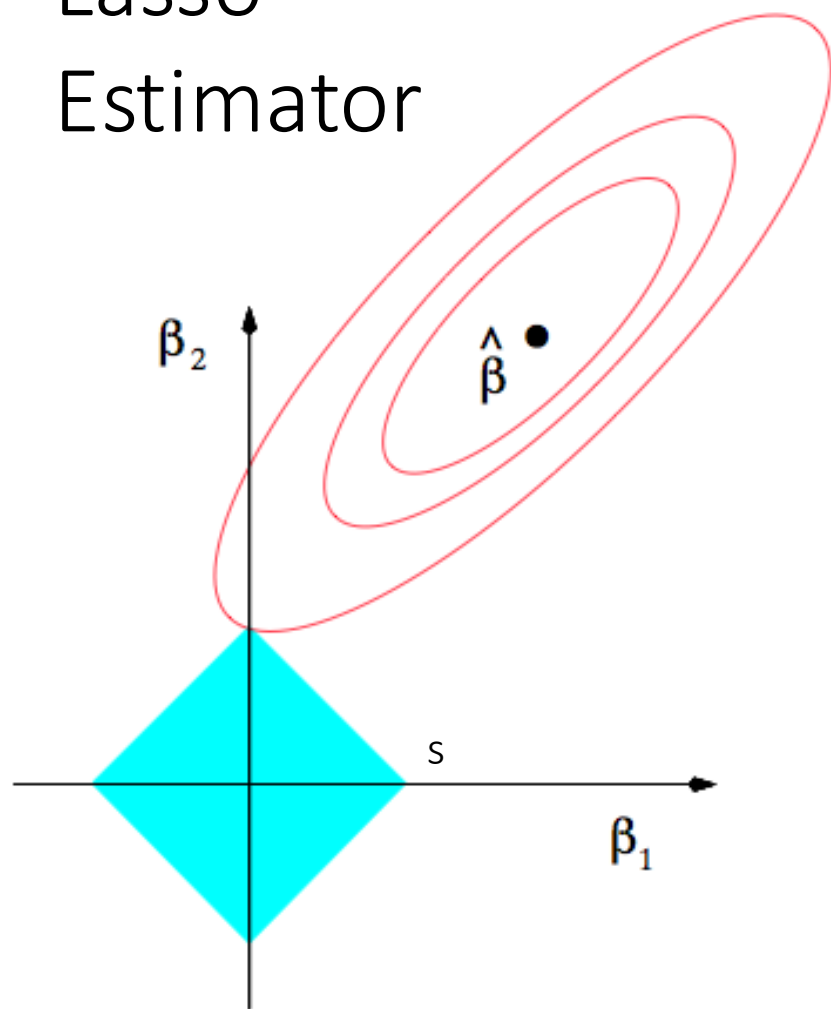

$$\hat{\beta}^{lasso} = \operatorname{argmin}_{\beta} (y - X\beta)^T (y - X\beta)$$

subject to $\sum_j |\beta_j| \leq s$

L1 norm

By convention, the bias/intercept term is typically not regularized.
Here we assume data has been centered ... therefore no bias term

Lasso Estimator



Ridge Regression

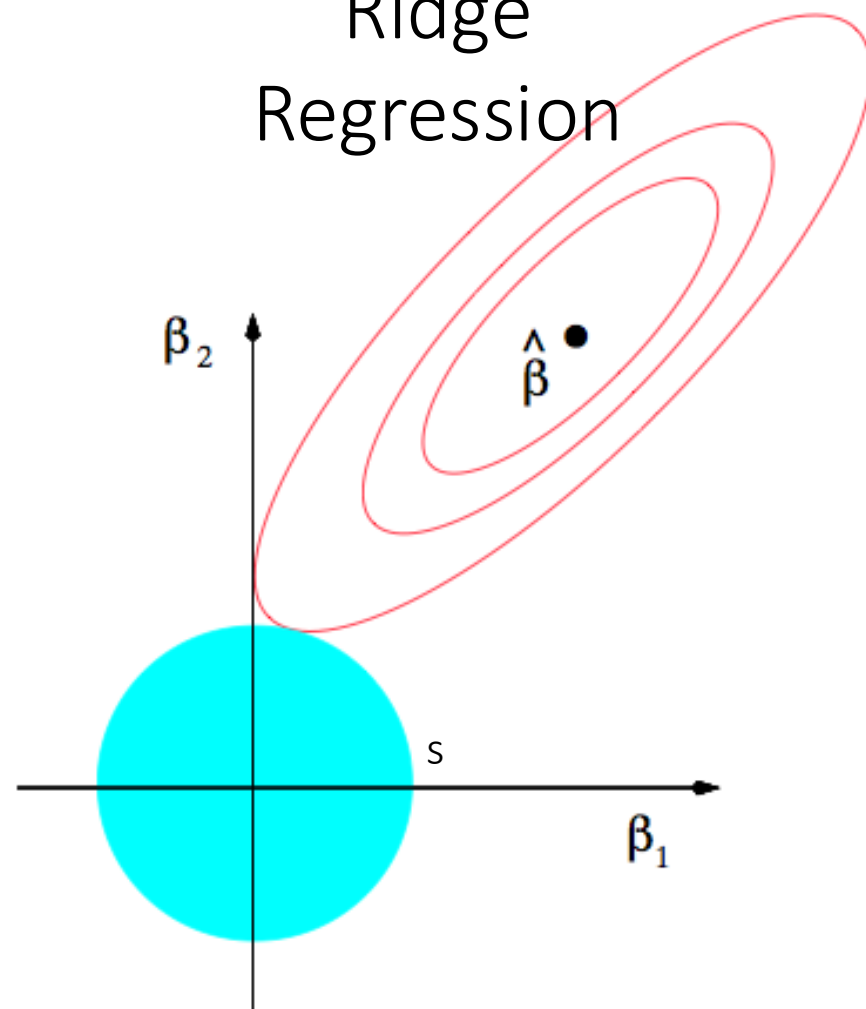
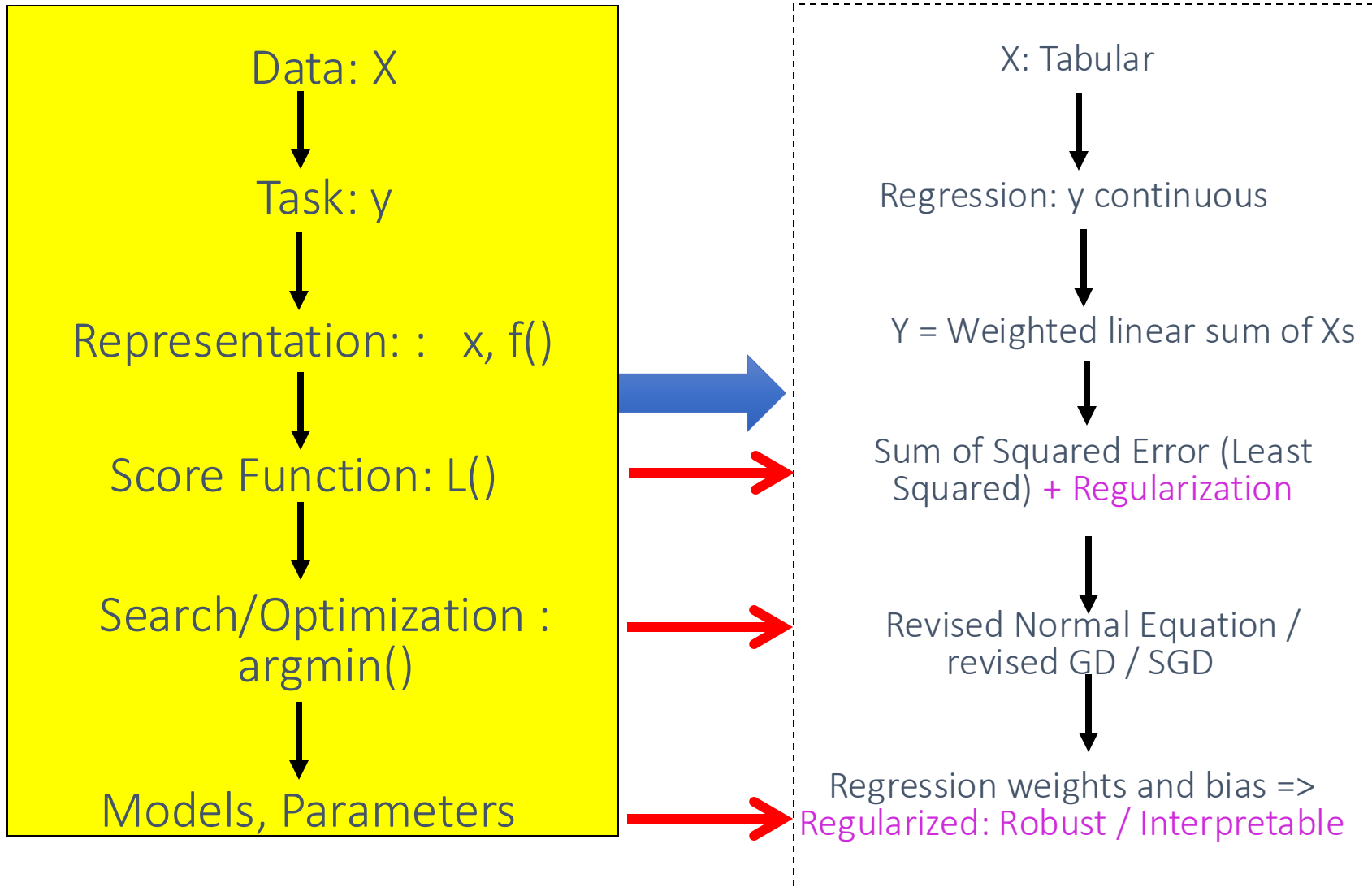


FIGURE 3.11. Estimation picture for the lasso (left) and ridge regression (right). Shown are contours of the error and constraint functions. The solid blue areas are the constraint regions $|\beta_1| + |\beta_2| \leq t$ and $\beta_1^2 + \beta_2^2 \leq t^2$, respectively, while the red ellipses are the contours of the least squares error function.

$$\min J(\beta) = \sum_{i=1}^n \left(Y - \hat{Y} \right)^2 + \lambda \left(\sum_{j=1}^p \beta_j^q \right)^{1/q}$$

Today: Regularized multivariate linear regression



More: A family of shrinkage estimators

$$\beta = \arg \min_{\beta} \sum_{i=1}^N (y_i - x_i^T \beta)^2$$

$$\text{subject to } \sum_j |\beta_j|^q \leq s$$

- for $q \geq 0$, contours of constant value of $\sum_j |\beta_j|^q$ are shown for the case of two inputs.

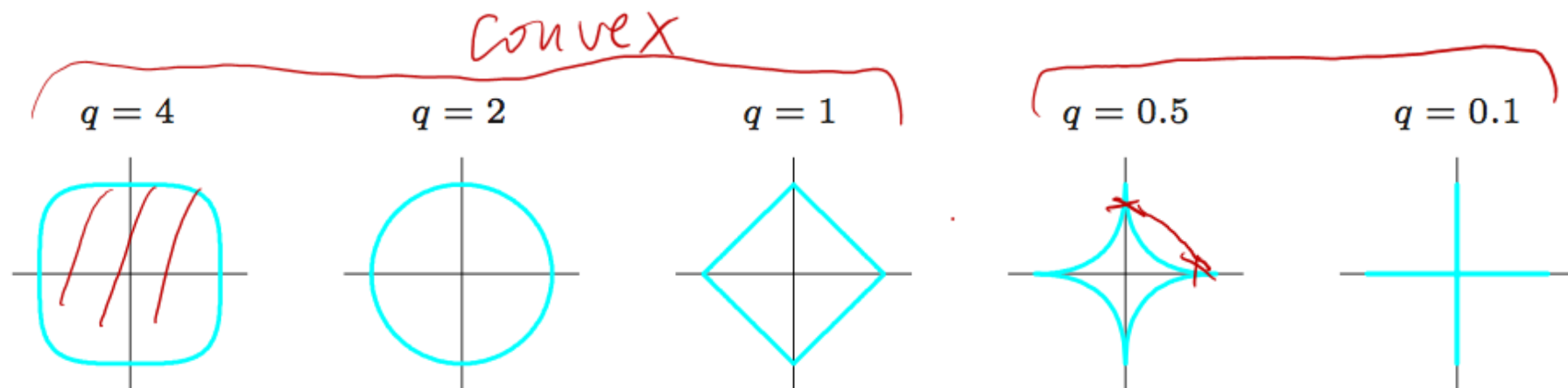
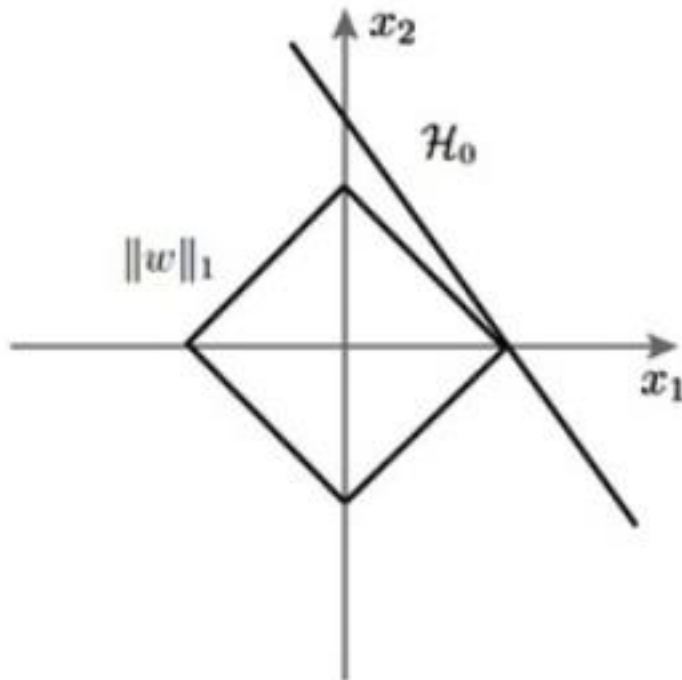
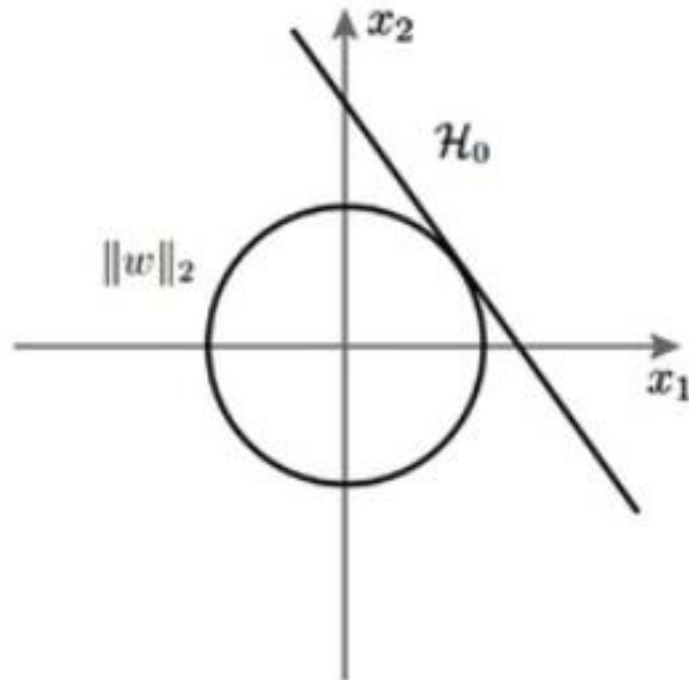


FIGURE 3.12. Contours of constant value of $\sum_j |\beta_j|^q$ for given values of q .

A L1 regularization



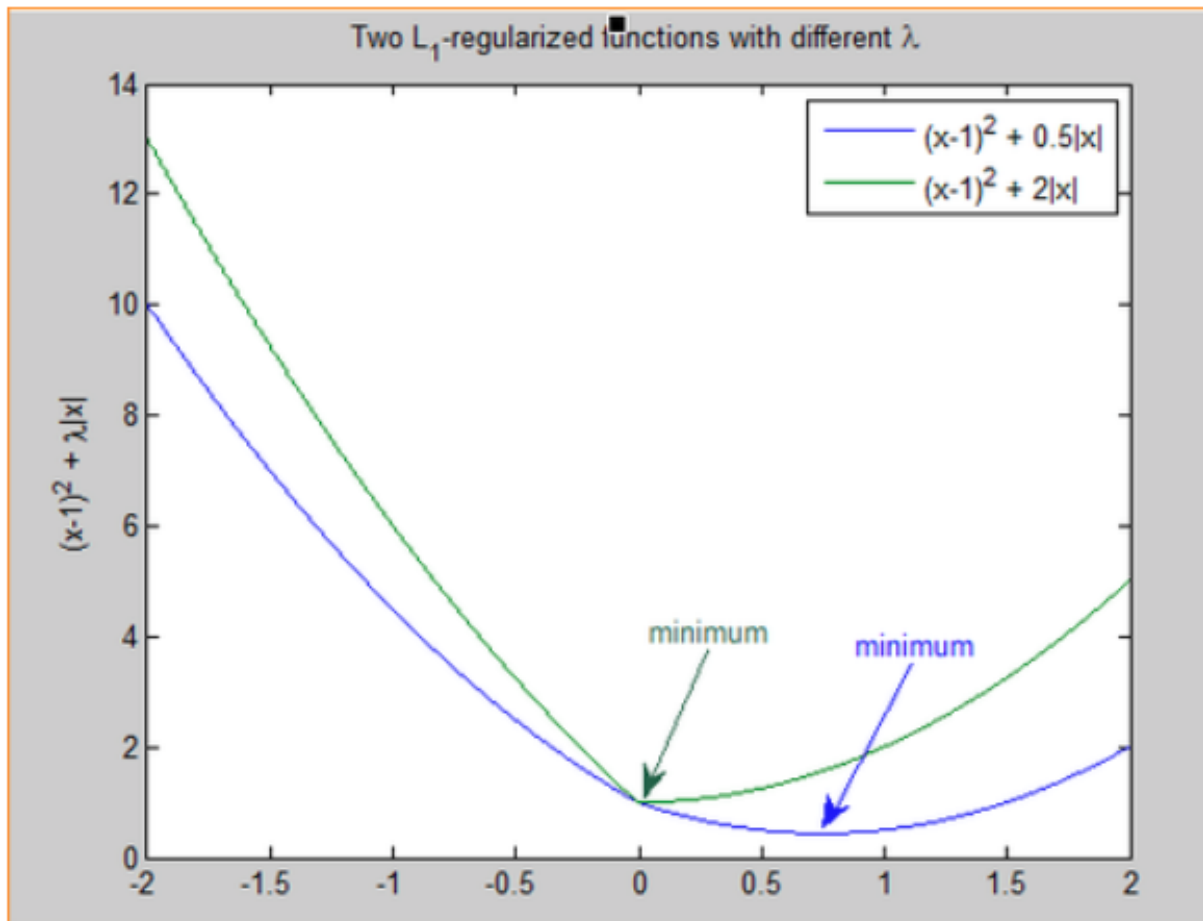
B L2 regularization



due to the nature of L_1 norm, the viable solutions are limited to corners, **which are on a few axis only**
- **in the above case x_1 . Value of $x_2 = 0$.** This means that the solution has eliminated the role of x_2 , leading to sparsity

L_1 -regularized loss function $F(x) = f(x) + \lambda\|x\|_1$ is non-smooth. It's not differentiable at 0. Optimization theory says that the optimum of a function is either the point with 0-derivative or one of the irregularities (corners, kinks, etc.). So, it's possible that the optimal point of F is 0 even if 0 isn't the stationary point of f . In fact, it would be 0 if λ is large enough (stronger regularization effect). Below is a graphical illustration.

<http://www.quora.com/What-is-the-difference-between-L1-and-L2-regularization>



In mathematics, particularly in calculus, a stationary point or critical point of a differentiable function of one variable is a point of the domain of the function where the derivative is zero (equivalently, the slope of the graph at that point is zero).

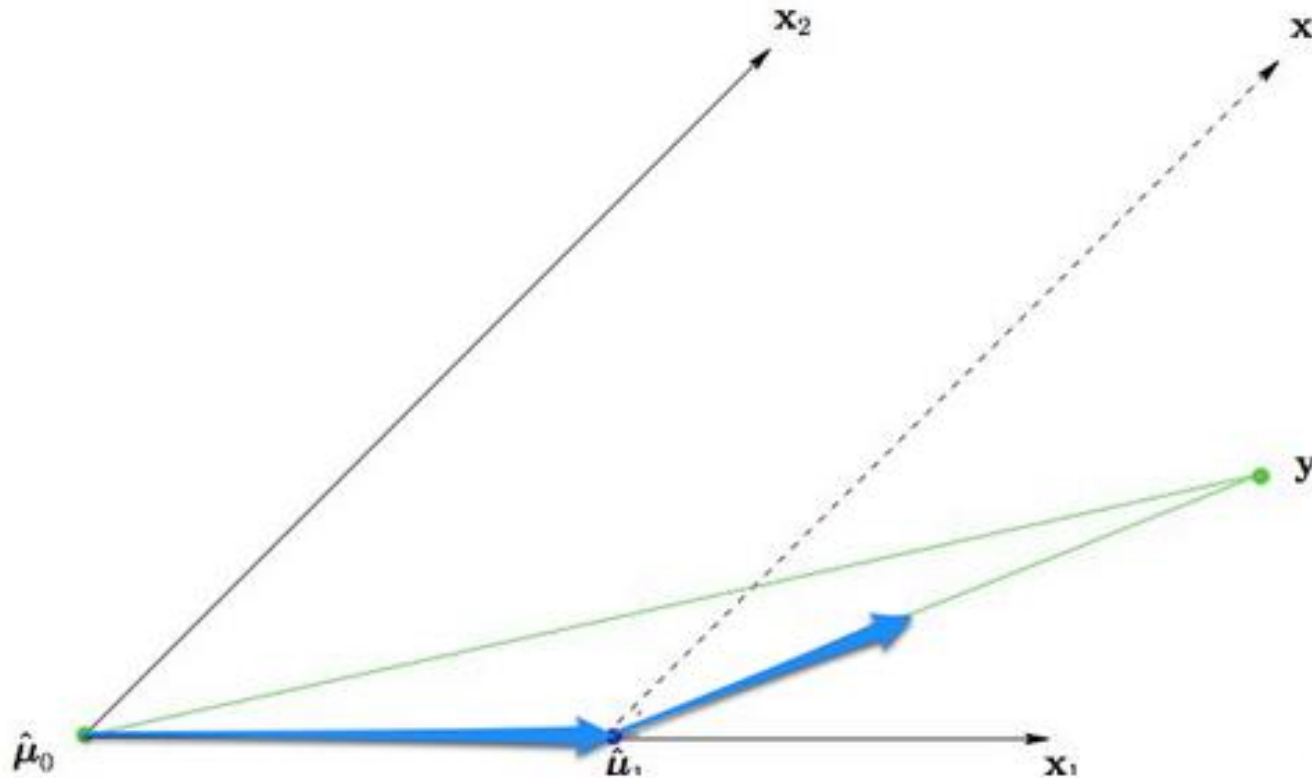
Coordinate descent based Learning of Lasso

1. Initialize β
2. Repeat until converged
3. For $j = 1, 2, \dots, P$ do
 - $a_j = 2 \sum_{i=1}^n x_{ij}^2$
 - $e_j = 2 \sum_{i=1}^n x_{ij} (y_i - x_i^T \beta + x_{ij} \beta_j)$
 - if $e_j < -\lambda$
$$\beta_j = (e_j + \lambda) / a_j$$
 - else if, $e_j > \lambda$
$$\beta_j = (e_j - \lambda) / a_j$$
 - else, soft-thresholding
$$\beta_j = 0$$

Coordinate descent (WIKI) $\rightarrow$ one does line search along one coordinate direction at the current point in each iteration.

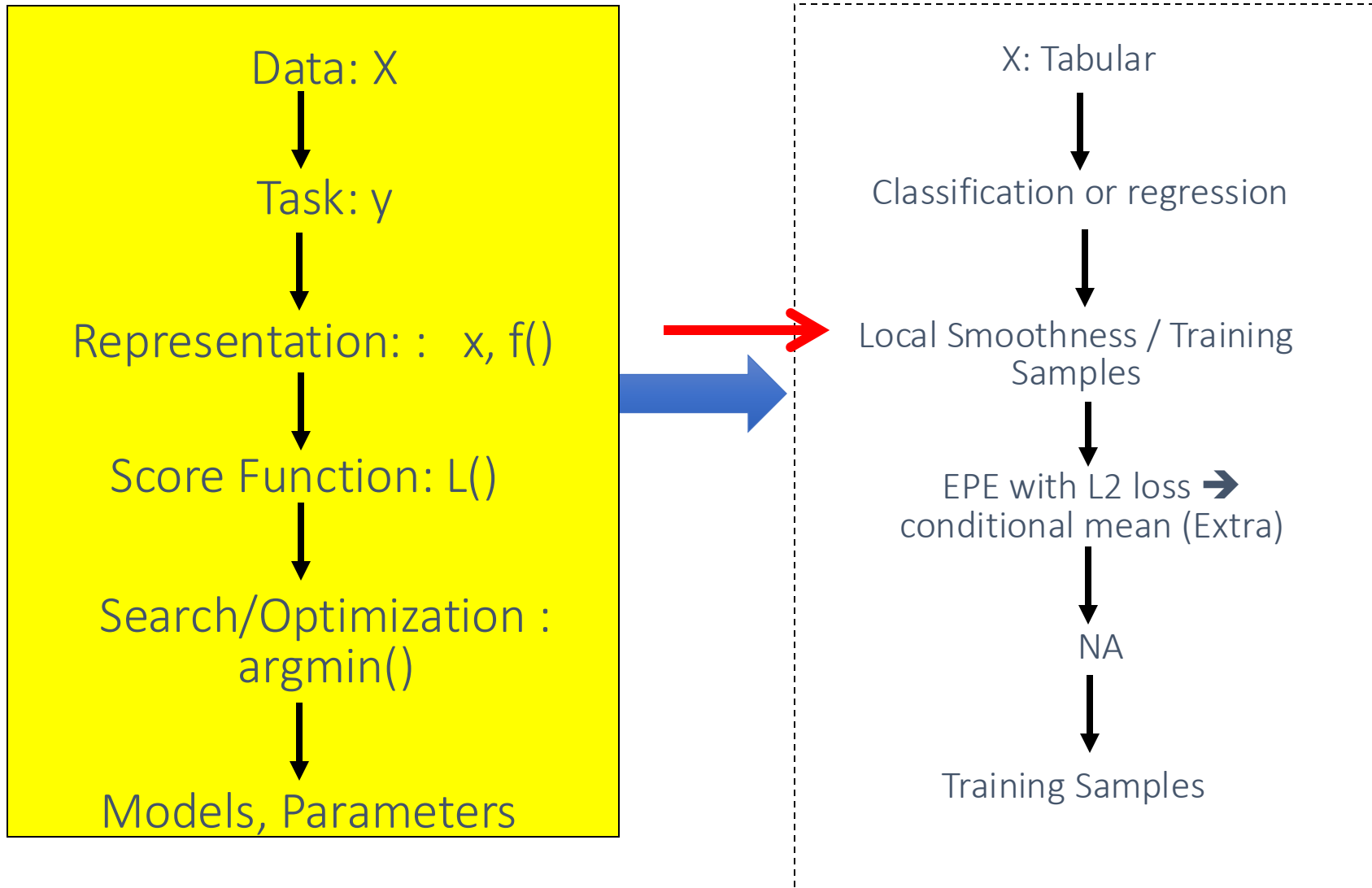
One uses different coordinate directions cyclically throughout the procedure.

Least Angle Regression (LARS) (State-of-the-art LASSO solver)



<http://statweb.stanford.edu/~tibs/ftp/lars.pdf>

L8: K-nearest-neighbor(regressor or classifier)



Code run: https://github.com/qiyanjun/2025Fall-UVA-CS-MachineLearningDeep/blob/main/notebook/L8_Knearest.ipynb

```
[42] # import regressor
      from sklearn.neighbors import KNeighborsRegressor
      # instantiate with K=5
      knn = KNeighborsRegressor(n_neighbors=5)
      # fit with data
      knn.fit(X, y)
```

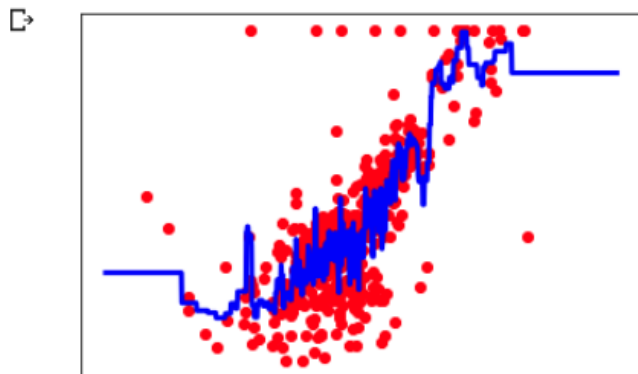
```
↳ KNeighborsRegressor(algorithm='auto', leaf_size=3,
                      metric_params=None, n_jobs=None,
                      weights='uniform')
```

```
###
Xfit = np.linspace(3, 10, 1000).reshape(-1, 1)
yfit = knn.predict(Xfit)

# Plot outputs
plt.scatter(X, y, color='red')
plt.plot(Xfit, yfit, color='blue', linewidth=3)

plt.xticks(())
plt.yticks(())

plt.show()
```



```
[44] # import regressor
      from sklearn.neighbors import KNeighborsRegressor
      # instantiate with K=5
      knn = KNeighborsRegressor(n_neighbors=100)
      # fit with data
      knn.fit(X, y)
```

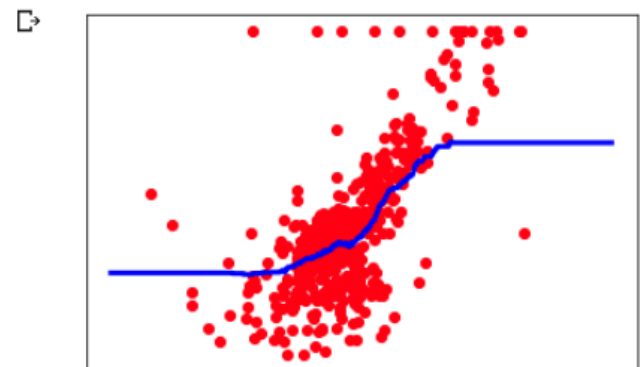
```
↳ KNeighborsRegressor(algorithm='auto', leaf_size=30,
                      metric_params=None, n_jobs=None,
                      weights='uniform')
```

```
###
Xfit = np.linspace(3, 10, 1000).reshape(-1, 1)
yfit = knn.predict(Xfit)

# Plot outputs
plt.scatter(X, y, color='red')
plt.plot(Xfit, yfit, color='blue', linewidth=3)

plt.xticks(())
plt.yticks(())

plt.show()
```



K Nearest neighbor (Testing Mode)

It Needs:

1. The set of stored training samples
2. Distance metric to compute distance between samples
3. The value of k , i.e., the number of nearest neighbors to retrieve

Training Mode:

- (Naïve) version: **DO NOTHING !!!!**

Testing Model: To classify **unknown** sample:

- **Step1**: Compute distance to **all** training records
- **Step2**: Identify **k nearest** neighbors
- **Step3**: Use class labels of nearest neighbors to determine the class label of unknown record (**e.g., by taking majority vote**)

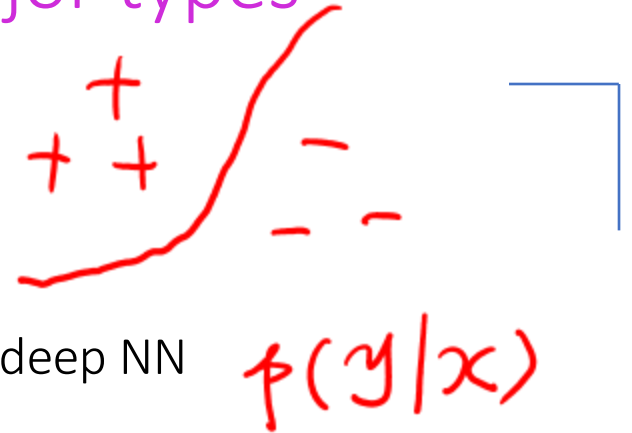
We can divide the large variety of supervised classifiers into roughly three major types

1. Discriminative

directly estimate a decision rule/boundary

e.g., support vector machine, decision tree,

e.g. **logistic regression**, neural networks (NN), deep NN



2. Generative:

build a generative statistical model

e.g., Bayesian networks, **Naïve Bayes classifier**

$$P(x|y=c)$$



3. Instance based classifiers

- Use observation directly (no models)
- e.g. K nearest neighbors

Less
complex

More
complex

① Polynomial Regression

d

$d \downarrow$

$d \uparrow$

② Ridge Reg

λ
 $\lambda > 0$

$\lambda \uparrow$

$\lambda \downarrow$

③ KNN Reg

k

$k \uparrow$

$k \downarrow$

$k \downarrow$

Model Selection for Nearest neighbor classification

- Choosing the value of k:
 - If k is too small, sensitive to noise points
 - If k is too large, neighborhood may include points from other classes

- Bias and variance tradeoff

- A small neighborhood $\rightarrow$ large variance $\rightarrow$ [unreliable] estimation

- A large neighborhood $\rightarrow$ large bias $\rightarrow$ [inaccurate] estimation

overfit

underfit

We aim to make our trained model

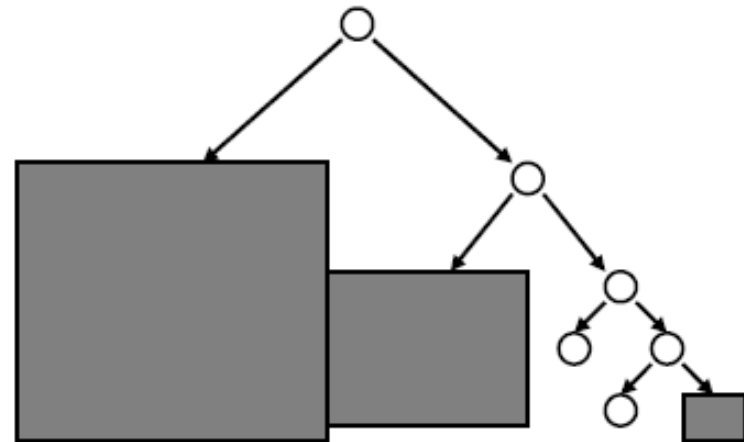
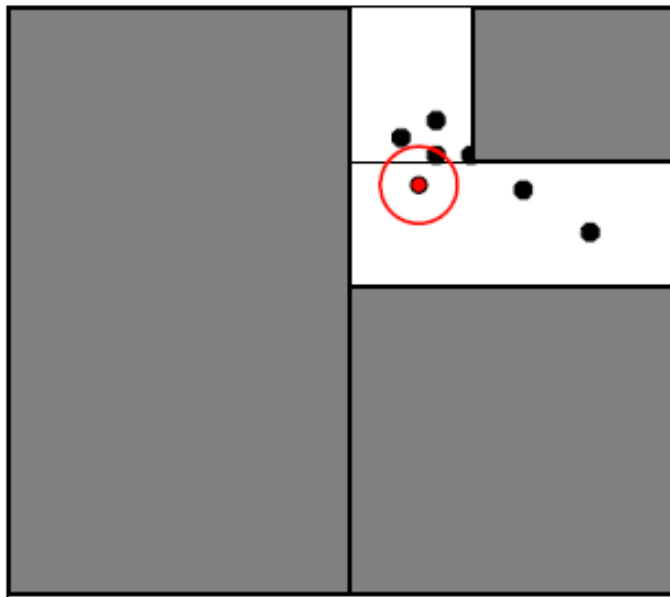
- 1. Generalize Well
- 2. Computational Scalable and Efficient
- 3. Trustworthy: Robust / Interpretable
 - Especially for some domains, this is about trust!

Computational Time Cost

	Train (n)	Test ($m=1$) $\hat{y} = \beta^T x$
Linear Regression	$O(np^2 + p^3)$	$O(p)$
KNN Reg	$O(1)$	$O(np) +$ $O(\text{sort } n-k)$???

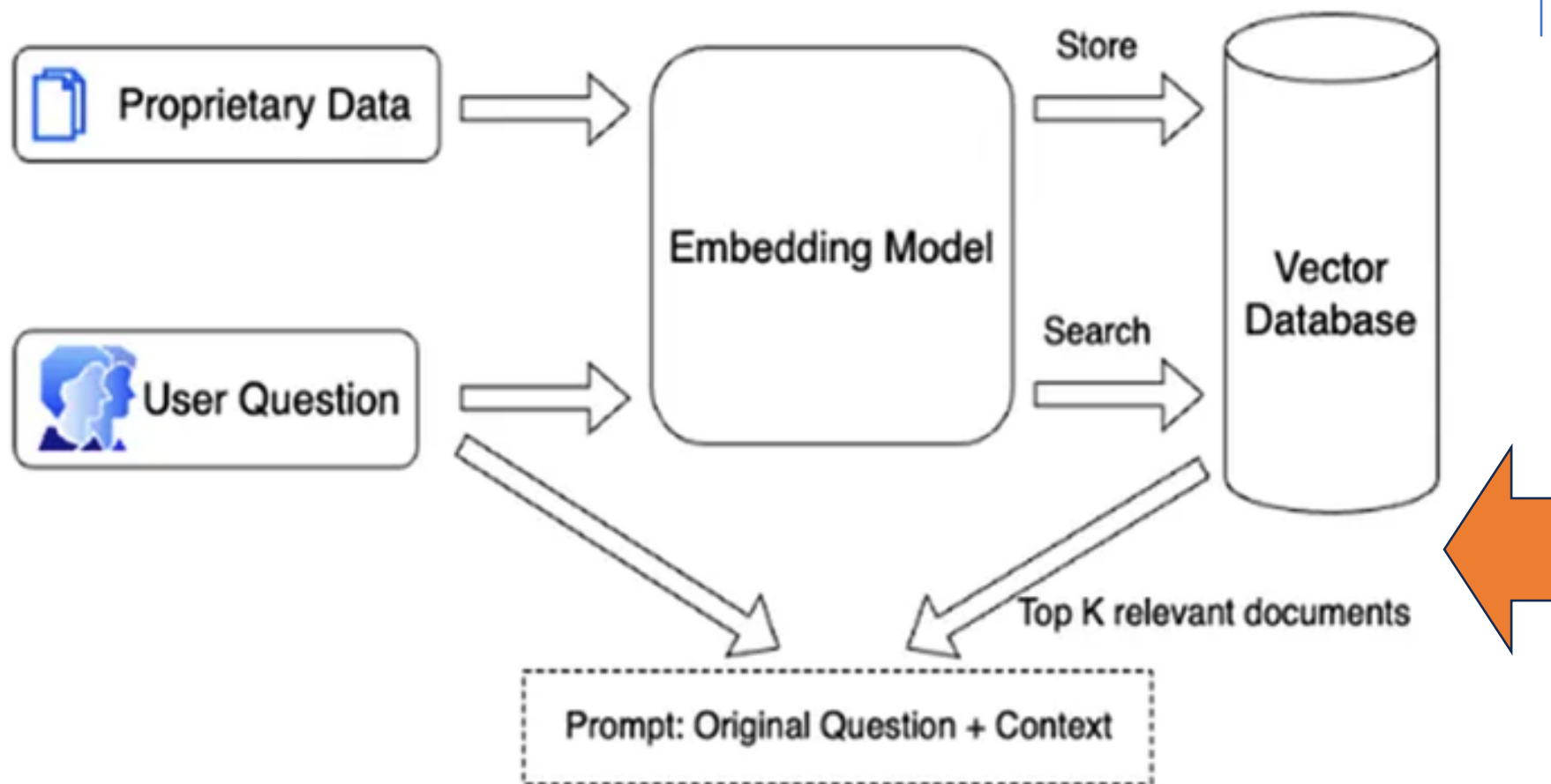
$p = 30,000$
 $n = 20,000$

NN Search by KD Tree



Using the distance bounds and the bounds of the data below each node, we can prune parts of the tree that could NOT include the nearest neighbor.

KNN as the Most critical component in Retrieval Augmented Generation System, e.g.:



09/30/2025 Roadmap

- TA to go over HW1
- One UVA ML club to introduce their setups and projects
- Q5
- Review Q4
- Review QA for L5-L7



10/07

10/07 /2025 Assignments

- HW2 is grading started ..
 - → HW2 key walk-through next Thursday online zoom
- HW3 will get posted by tomorrow
 - Deep NN on Imaging task / Kera / mostly about learning modern DNN library
 - Programming + QA (like calculating marginal prob...)
- Next Tuesday is reading day
 - → we will host makeup-Quiz Q7 next Thursday online
- Course format survey:
 - <https://forms.gle/PkWGMkwHhawqf8QR8>
 - Now go over the results:

Project Process



- Format:
 - Team (1~4 students)
 - Shark Tank alike Screening – https://en.wikipedia.org/wiki/Shark_Tank
 - This week: signup sheet for your team's screening sessions!
 - Next week: Initial project idea collecting!
 - TA Guangzhi will announce the process and signup sheet URL!
- Final deliverables:
 - (1) Code (Github PR to course project repo)
 - (2) Poster presentation class wide (Date: 12/09 TBD)
 - (3) Video Demo (after final exam)

10/07 /2025 Roadmp



Review L5-L8 questions



Quick Review L9-L10



Review Q5

quite disturbing for staying
students around the right after
quiz period



Then Q6

So we will host quiz after review
/ before project screening for all
coming in-person sessions

Questions on L5-L8

Set 1: Bias–Variance, Overfitting, and Model Complexity

- How do bias and variance contribute to generalization error, and how do we find the “sweet spot” without knowing the true distribution?
- What are practical indicators of underfitting vs. overfitting (from graphs, learning curves, or error plots), and how do we fix each?
- How does cross-validation (choice of K) approximate generalization error, and what are the trade-offs (bias vs variance, LOOCV vs k -fold)?
- Why does zero training error often generalize poorly, and how is this linked to variance?
- Would we ever prefer high bias or high variance, and how do we reduce one while controlling the other?

Set 2: Regularization (LASSO, Ridge, Elastic Net, Generalizations)

- What are the key differences between L1 (LASSO), L2 (Ridge), and Elastic Net in terms of sparsity, robustness, computational cost, and when to use each?
- Why do L1 penalties set coefficients to zero, while L2 does not? What happens when $p > n$ or when features are highly correlated?
- How does the choice of λ affect bias–variance, and how do we select it (cross-validation, validation curves)?
- Are there equivalent closed-form solutions for LASSO like Ridge has? Why are L1 and L2 chosen—what about higher-order penalties?
- When is Elastic Net preferable (e.g., grouped correlated features), and can we always default to it?

Questions on L5-L8

Set 3: k-Nearest Neighbors (kNN) and Instance-Based Learning

- How do we pick the best k (odd vs even, weighted vs unweighted, trade-offs with noisy data)?
- What is the computational cost of kNN (sorting term, memory cost), and can it overfit?
- How does the distance metric affect performance, and how are ties handled in classification?
- What are the advantages/disadvantages of kNN vs gradient descent or regularized linear models?
- In practice, how large must the dataset be to offset outliers, and is kNN more effective for regression or classification?

Set 4: Maximum Likelihood Estimation (MLE) and Probability Foundations

- Why do we usually maximize the log-likelihood instead of the likelihood itself, and how does this connect to squared error in linear regression?
- How does MLE extend from discrete distributions (e.g., coin flips) to continuous (e.g., Gaussians)?
- Why is the MLE for Bernoulli just the sample proportion, and what happens with small samples or noisy data?
- What makes MLE consistent and efficient, and are there situations where maximum likelihood may not yield the most “ideal” parameter?
- How does the bias–variance decomposition change with different loss functions (e.g., 0–1 loss, Laplace errors)?

10/07 /2025 Roadmp



Review L5-L8 questions



Quick Review L9-L10



Review Q5

quite disturbing for staying
students around the right after
quiz period



Then Q6

So we will host quiz after review
/ before project screening for all
coming in-person sessions

Lecture 10: Maximum Likelihood Estimation (MLE)

➔ Probability Review

- The big picture
- Events and Event spaces
- Random variables
- Joint probability, Marginalization, conditioning, chain rule, Bayes Rule, law of total probability, etc.
- Structural properties, e.g., Independence, conditional independence
- Maximum Likelihood Estimation

If hard to directly estimate from data, most likely we can estimate

- 1. Joint probability

- Use Chain Rule

$$P(A, B) = P(B) P(A|B)$$

- 2. Marginal probability

- Use the total law of probability

$$P(B) = P(B, A) + P(B, \sim A)$$
$$\parallel$$
$$P(B, A \cup \sim A) //$$

- 3. Conditional probability

- Use the Bayes Rule

$$P(A|B)$$
$$P(B|A) = \frac{P(A, B)}{P(A)} = \frac{P(A|B) P(B)}{P(A)}$$

One Example

Assume we have a dark box with 3 red balls and 1 blue ball. That is, we have the set $\{r, r, r, b\}$. What is the probability of drawing 2 red balls in the first 2 tries?

$$P(B_1 = r, B_2 = r) = \underbrace{P(B_1 = r)}_{\frac{3}{4}} P(B_2 = r | B_1 = r) = \frac{1}{2}$$

$$P(B_2 = r) = P(B_1 = r, B_2 = r) + P(B_1 = b, B_2 = r)$$

$$P(B_1 = r | B_2 = r) = \frac{P(B_1 = r, B_2 = r)}{P(B_2 = r)}$$

MLE idea is to

- ✓ assume a particular **model with unknown parameters**, θ
- ✓ we can then define the probability of observing a given event conditional on a particular set of parameters. $P(Z_i|\theta)$
- ✓ We have observed **a set of outcomes** in the real world.
- ✓ It is then possible to choose a set of parameters which are most likely to have produced the observed results.

$$\hat{\theta} = \underset{\theta}{\operatorname{argmax}} P(Z_1 \dots Z_n | \theta)$$


This is maximum likelihood.

In most cases this scorer is **both consistent and efficient**.

$$\log(L(\theta)) = \sum_{i=1}^n \log(P(Z_i | \theta))$$


It is often convenient to work with the Log of the likelihood function.

Deriving the Maximum Likelihood Estimate for Bernoulli

$$\begin{aligned}\log(L(p)) &= \log \left[\prod_{i=1}^n p^{z_i} (1-p)^{1-z_i} \right] \\ &= \sum_{i=1}^n (z_i \log p + (1-z_i) \log(1-p)) \\ &= \log p \sum_{i=1}^n z_i + \log(1-p) \sum_{i=1}^n (1-z_i) \\ &= x \log p + (n-x) \log(1-p)\end{aligned}$$

Deriving the Maximum Likelihood Estimate for Bernoulli

$$\underset{p}{\operatorname{argmax}} \{-l(p)\} = \underset{p}{\operatorname{argmin}} \{-x \log(p) - (n-x) \log(1-p)\}$$

$$\frac{dl(p)}{dp} = -\frac{x}{p} - \frac{-(n-x)}{1-p} = 0$$

$$0 = -x + pn$$

$$0 = -\frac{x}{p} + \frac{n-x}{1-p}$$

Minimize the negative log-likelihood

→ MLE parameter estimation

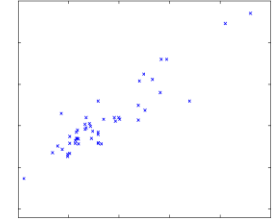
$$0 = \frac{-x(1-p) + p(n-x)}{p(1-p)}$$

$$0 = -x + px + pn - px$$

$$\hat{p} = \frac{x}{n}$$

i.e. Relative frequency of a binary event

DETOUR: Probabilistic Interpretation of Linear Regression



- Let us assume that the target variable and the inputs are related by the equation:

$$y_i = \theta^T \mathbf{x}_i + \varepsilon_i$$

RV $\varepsilon \sim N(0, \sigma^2)$

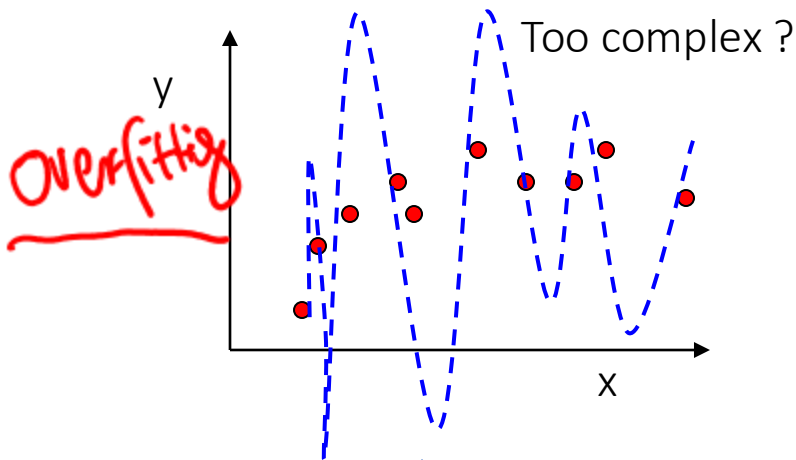
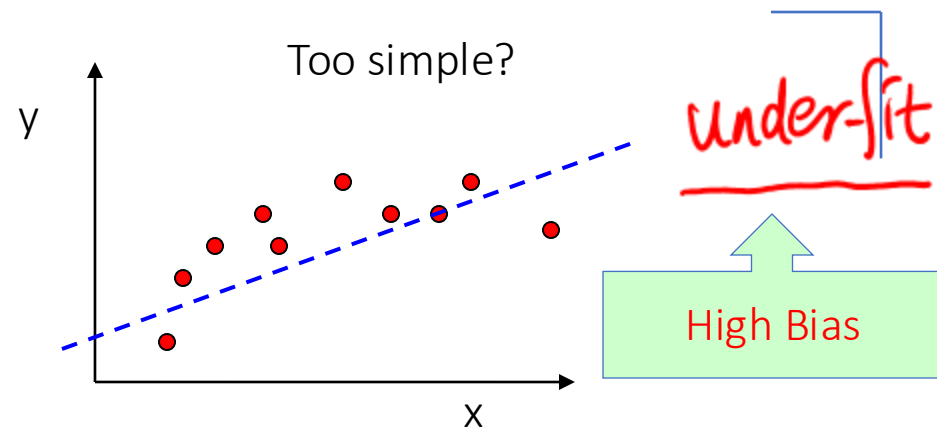
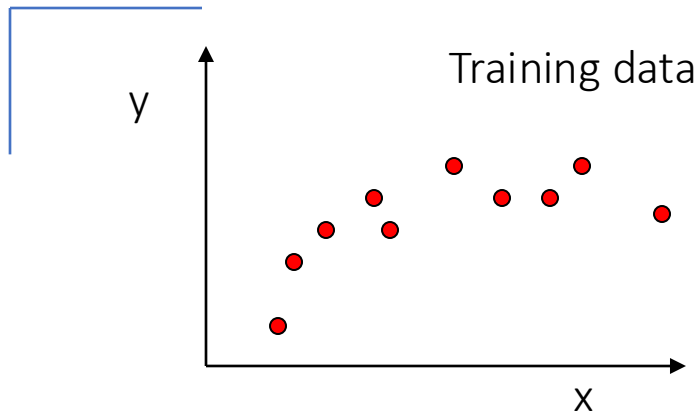
where ε is an error term of unmodeled effects or random noise₂

- Now assume that ε follows a Gaussian $N(0, \sigma)$, then we have:

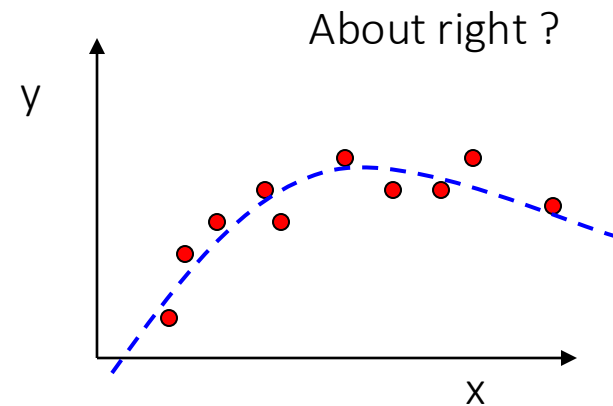
$$p(y_i | x_i; \theta) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - \theta^T \mathbf{x}_i)^2}{2\sigma^2}\right)$$

RV $y|x;\theta \sim N(\theta^T x, \sigma)$

L9: Complexity / Goodness of Fit / Generalization



High Variance



What ultimately matters: GENERALIZATION

Statistical Decision Theory (Extra)

- Random input vector: X
- Random output variable: Y
- Joint distribution: $\Pr(X, Y)$
- Loss function $L(Y, f(X))$

$$P(t_1, t_2): \begin{cases} H & H \\ H & T \\ T & H \\ T & T \end{cases}$$

$$\Rightarrow D = \begin{pmatrix} (\vec{x}_1, y_1) \\ \vdots \\ (\vec{x}_n, y_n) \end{pmatrix}$$

- Expected prediction error (EPE):

$$\text{EPE}(f) = \mathbb{E}(L(Y, f(X))) = \int L(y, f(x)) \Pr(dx, dy)$$

$$\text{e.g.} = \int (y - f(x))^2 \Pr(dx, dy)$$

e.g. Squared error loss (also called L2 loss)

One way to define generalization: by considering the joint population distribution

Decomposition of EPE

- When additive error model: $Y = \underline{f}(X) + \epsilon, \epsilon \sim (0, \sigma^2)$
- Notations
 - Output random variable: Y
 - True function: $f \rightarrow \text{true}$
 - Prediction estimator: $\hat{f} \rightarrow D \rightarrow \hat{f}$

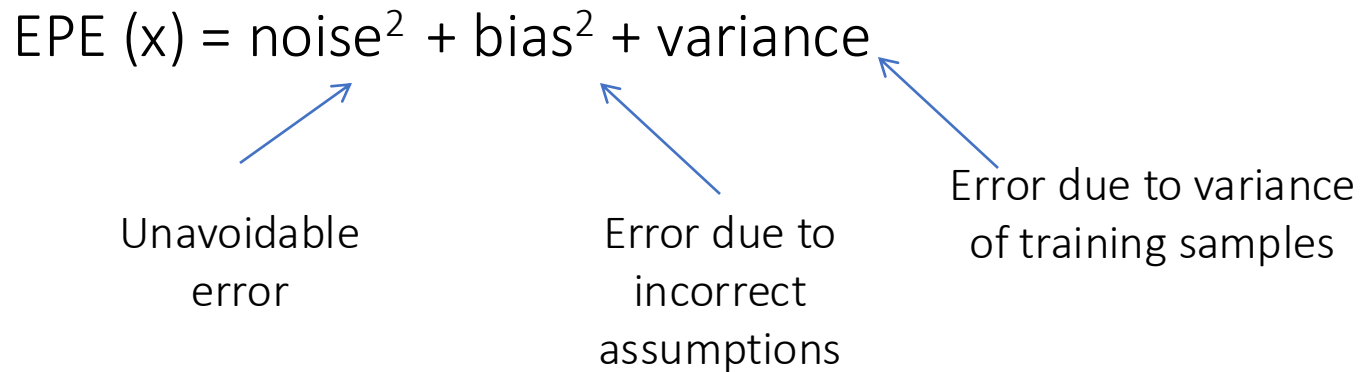
$$\begin{aligned} EPE(x) &= E[(Y - \hat{f})^2 | X = x] \\ &= E[((Y - f) + (f - \hat{f}))^2 | X = x] \\ &= \underbrace{E[(Y - f)^2 | X = x]}_{\epsilon} + \underbrace{E[(f - \hat{f})^2 | X = x]}_{\text{Bayes Error}} \\ &= \sigma^2 + \underbrace{Var(\hat{f}) + Bias^2(\hat{f})}_{\text{Bayes Error}} \end{aligned}$$

Irreducible / Bayes error

Bias-Variance Trade-off for EPE:

$$\text{EPE}(x) = \text{noise}^2 + \text{bias}^2 + \text{variance}$$

Unavoidable
error



The diagram illustrates the Bias-Variance Trade-off for Expected Prediction Error (EPE). The equation $\text{EPE}(x) = \text{noise}^2 + \text{bias}^2 + \text{variance}$ is centered at the top. Three blue arrows point from descriptive text labels below to the terms in the equation: one from 'Unavoidable error' to 'noise²', one from 'Error due to incorrect assumptions' to 'bias²', and one from 'Error due to variance of training samples' to 'variance'.

Error due to
incorrect
assumptions

Error due to variance
of training samples

BIAS AND VARIANCE TRADE-OFF for Parameter Estimation

- θ : true value (normally unknown)
- $\hat{\theta}$: estimator
- $\bar{\theta} := E[\hat{\theta}]$ (mean, i.e. expectation of the estimator)

- Bias $E[(\bar{\theta} - \theta)^2]$

- measures accuracy or quality of the estimator
- low bias implies on average we will accurately estimate true parameter from training data

- Variance $E[(\hat{\theta} - \bar{\theta})^2]$

- Measures precision or specificity of the estimator
- Low variance implies the estimator does not change much as the training set varies

Model “bias” & Model “variance”

- Middle RED:
 - TRUE function
- Error due to bias:
 - How far off in general from the middle red

$$E[(\bar{\theta} - \theta)^2]$$

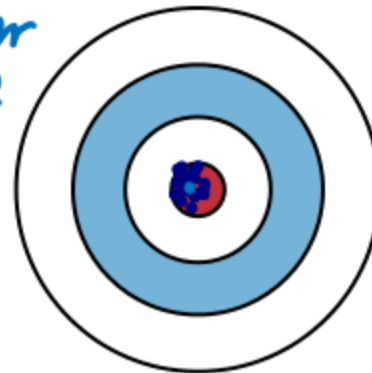
- Error due to variance:
 - How wildly the blue points spread

$$E[(\hat{\theta} - \bar{\theta})^2]$$

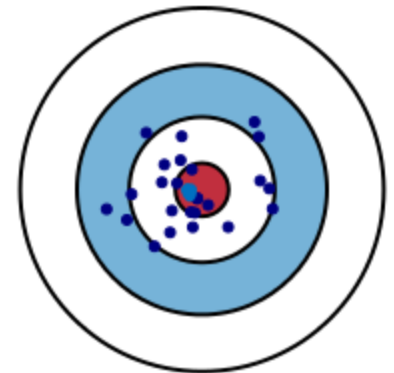
red
vs.
center
blue

Low Bias

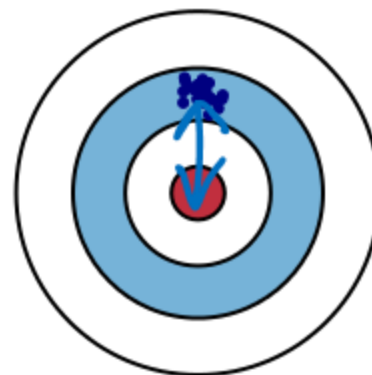
Low Variance



High Variance



High Bias



need to make assumptions that are able to generalize

- Underfitting: model is too “simple” to represent all the relevant characteristics
 - High bias and low variance
 - High training error and high test error
- Overfitting: model is too “complex” and fits irrelevant characteristics (noise) in the data
 - Low bias and high variance
 - Low training error and high test error

Bias Variance Tradeoff

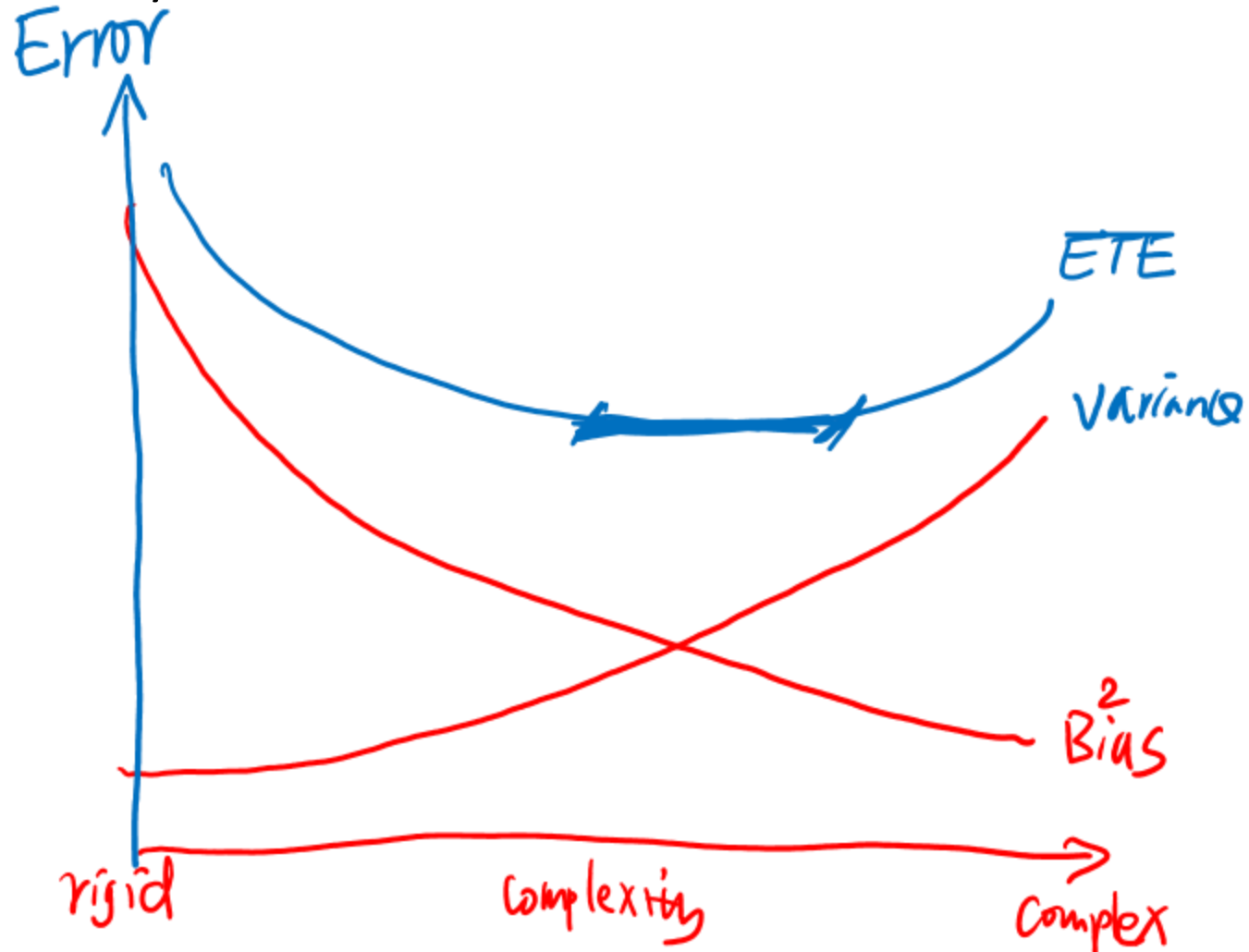
- (1) Randomness of Training Sets
- (2) Training error can always be reduced when increasing model complexity
- (3) Randomness in the Testing Error!!!
- (4) Cross Validation Error as good approximation for Expected Test error -- good appx of generalization

Review:

One important Control of Bias Variance Tradeoff

➔ Model Complexity

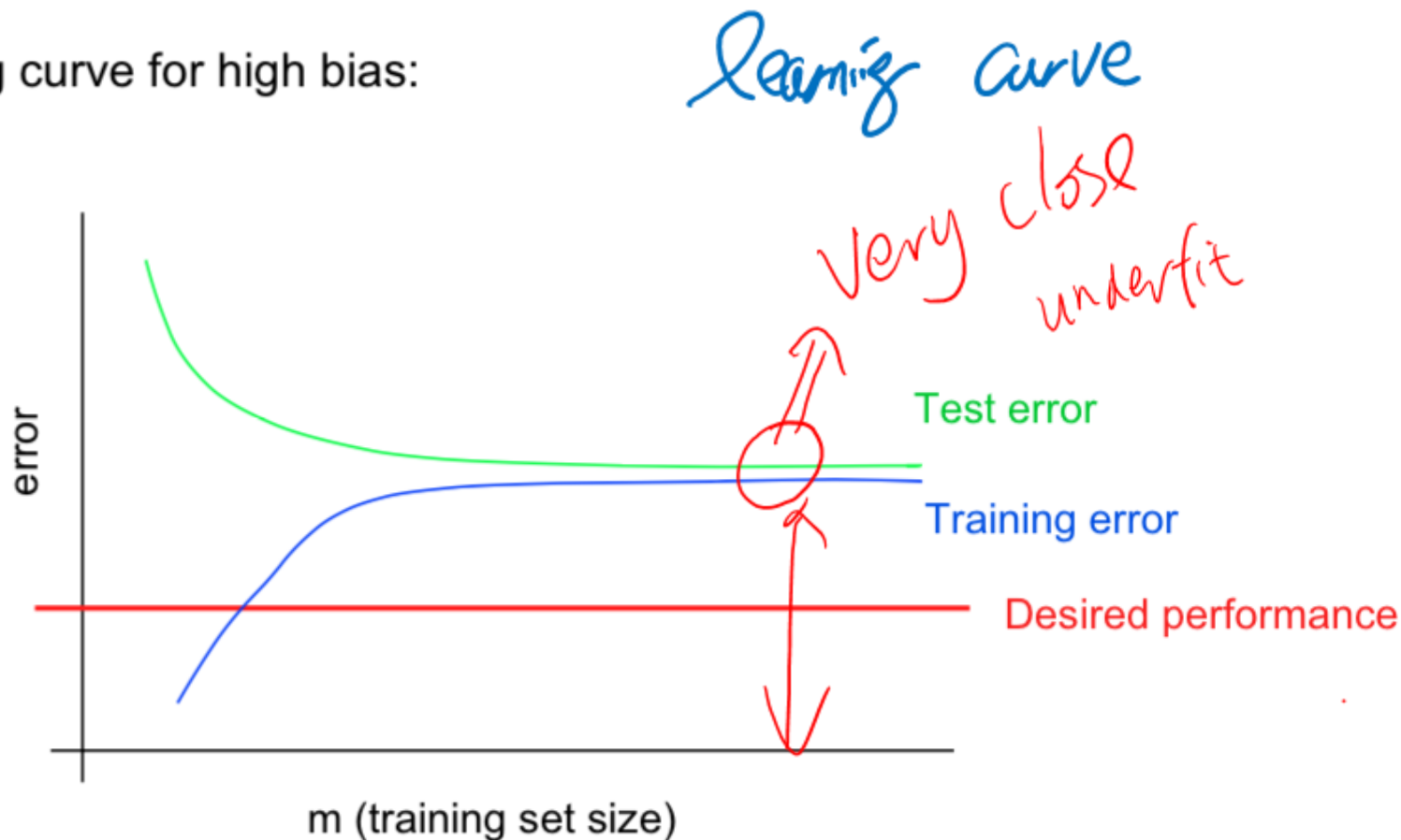
- bias decrease with model gets more complex;
- Variance increase with bigger model capacity
- Sum of $\text{Bias}^2 + \text{Variance}$



Another important Control of Bias Variance Tradeoff

➔ Training Size (Extra)

Typical learning curve for high bias:



- Even training error is unacceptably high.
- Small gap between training and test error.

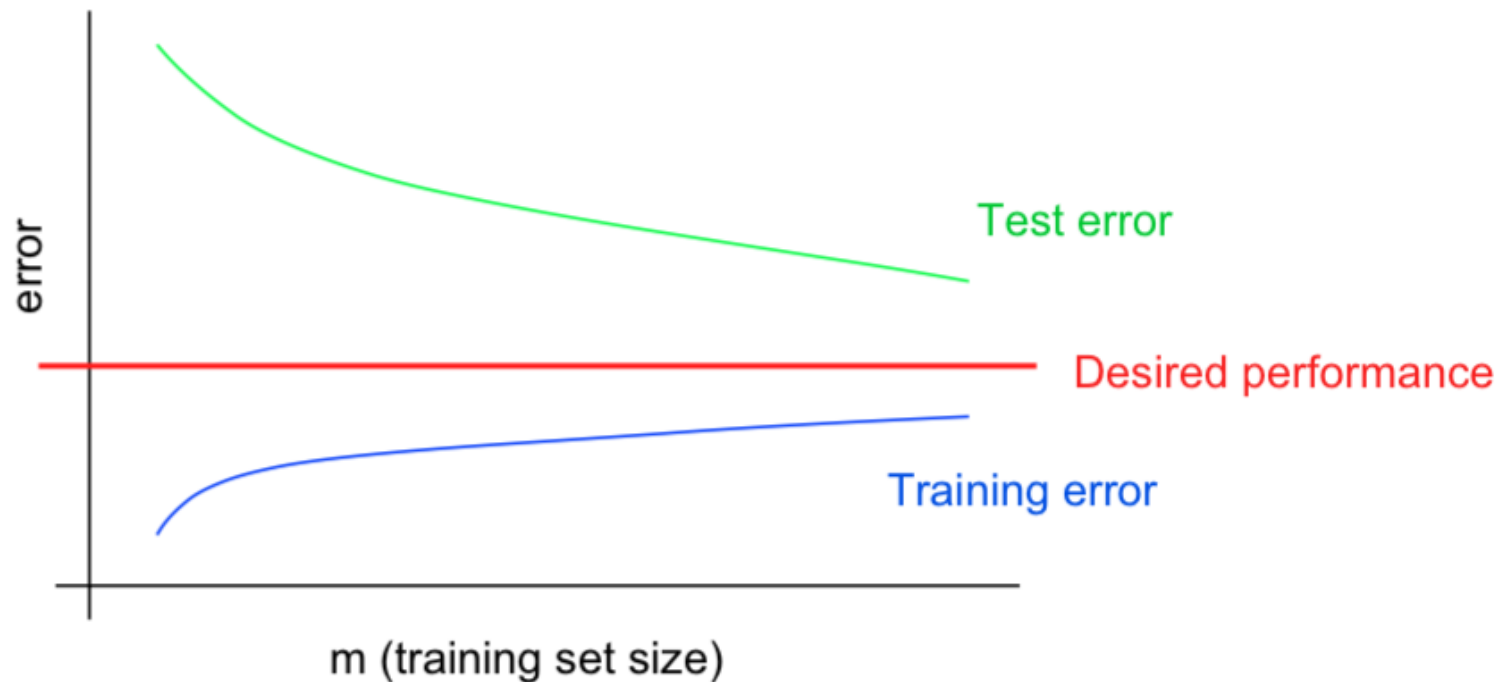
High training error and high test error

Another important Control of Bias Variance Tradeoff

➔ Training Size (Extra)

Typical learning curve for high variance:

learning curve



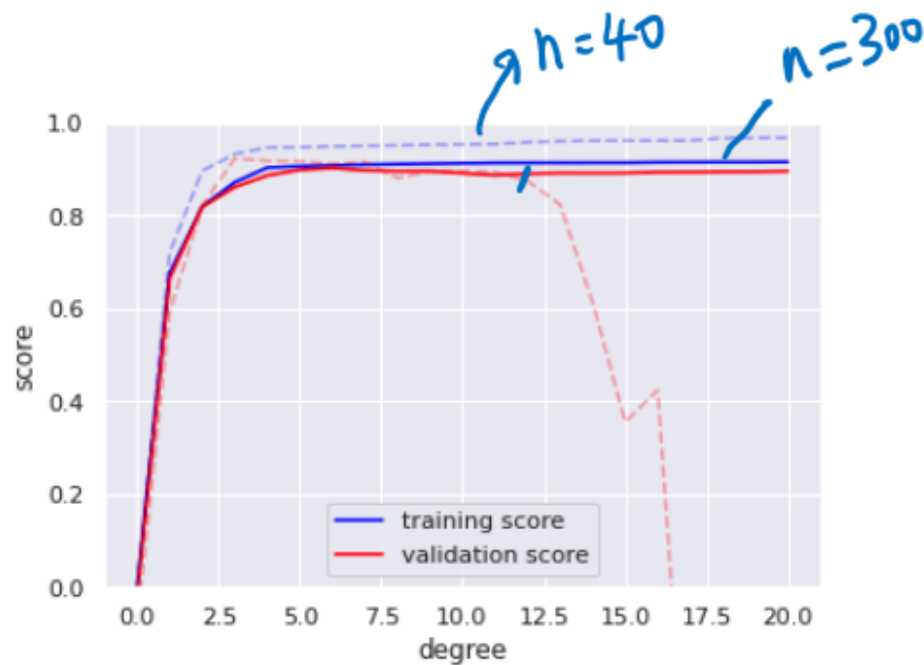
How to reduce Model High Variance?

- Choose a simpler classifier
- Regularize the parameters
- Get more training data
- Try smaller set of features *feature selection* $n < p$
- Try feature engineering
- Try multiple models and then use all as ensemble

Take Away : Three types of plots

- (1) Sanity check (S)GD type Optimization
 - Train / Vali Loss vs. Epochs to help you
 - https://scikit-learn.org/stable/auto_examples/linear_model/plot_sgd_early_stopping.html#sphx-glr-auto-examples-linear-model-plot-sgd-early-stopping-py
- (2) Sanity check hyperparameter tuning (validation curve)
 - Train / Vali Loss vs. hyperparameter Values
 - `from sklearn.model_selection import validation_curve`
- (3) Sanity check if your current model overfits or underfits
 - Train / Vali Loss vs. Varying Size of Training (learning curve)
 - https://scikit-learn.org/stable/auto_examples/model_selection/plot_learning_curve.html#sphx-glr-auto-examples-model-selection-plot-learning-curve-py

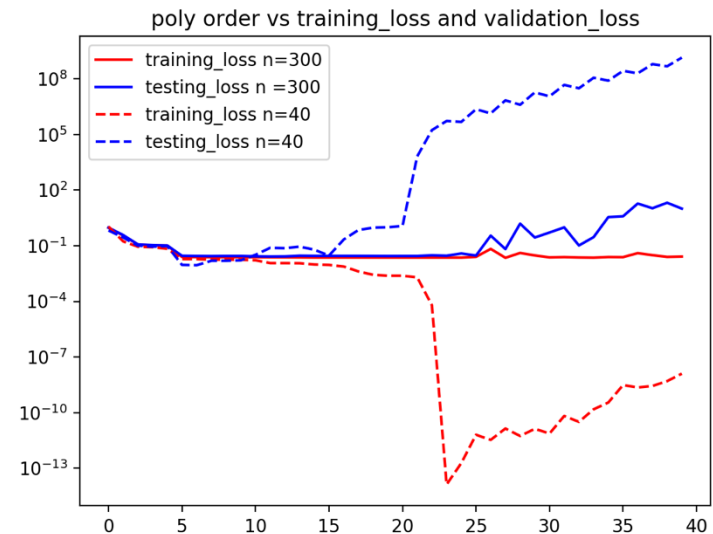
I will Code run: <https://github.com/qiyanjun/2025Fall-UVA-CS-MachineLearningDeep/blob/main/notebook/L9-LearningCurves.ipynb>



(1) Validation curve

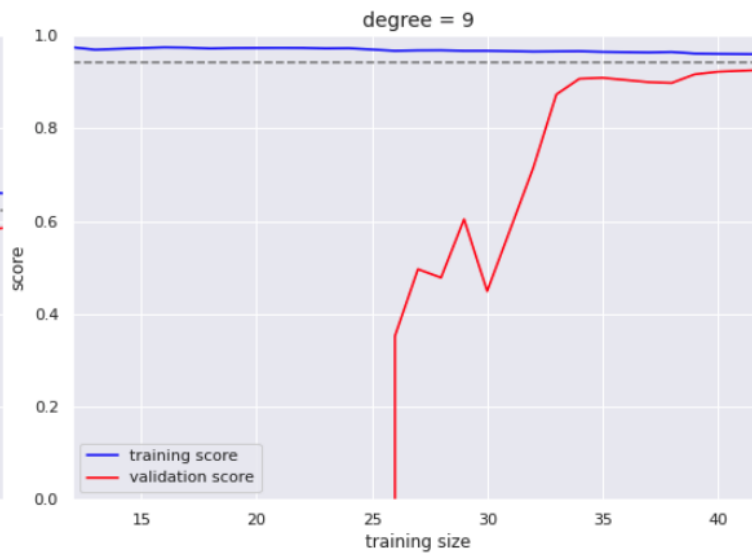
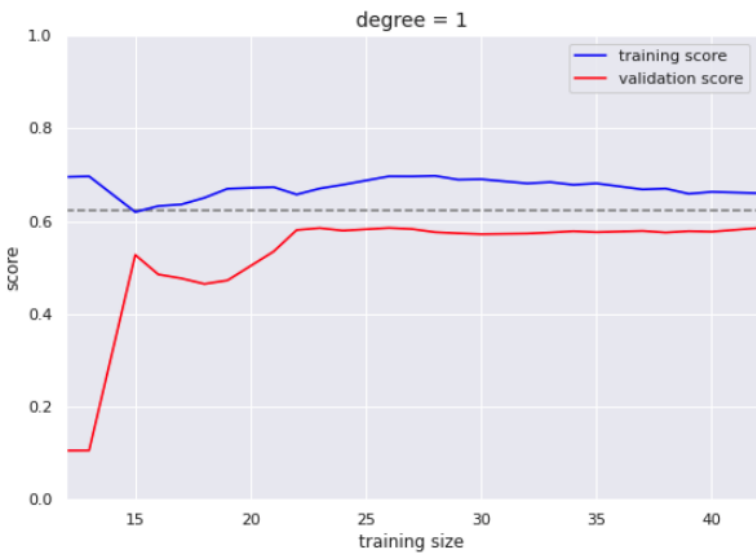
By scikitlearn Validation_curve function
(normalize all metrics to positive range)

https://scikit-learn.org/stable/modules/model_evaluation.html#the-scoring-parameter-defining-model-evaluation-rules

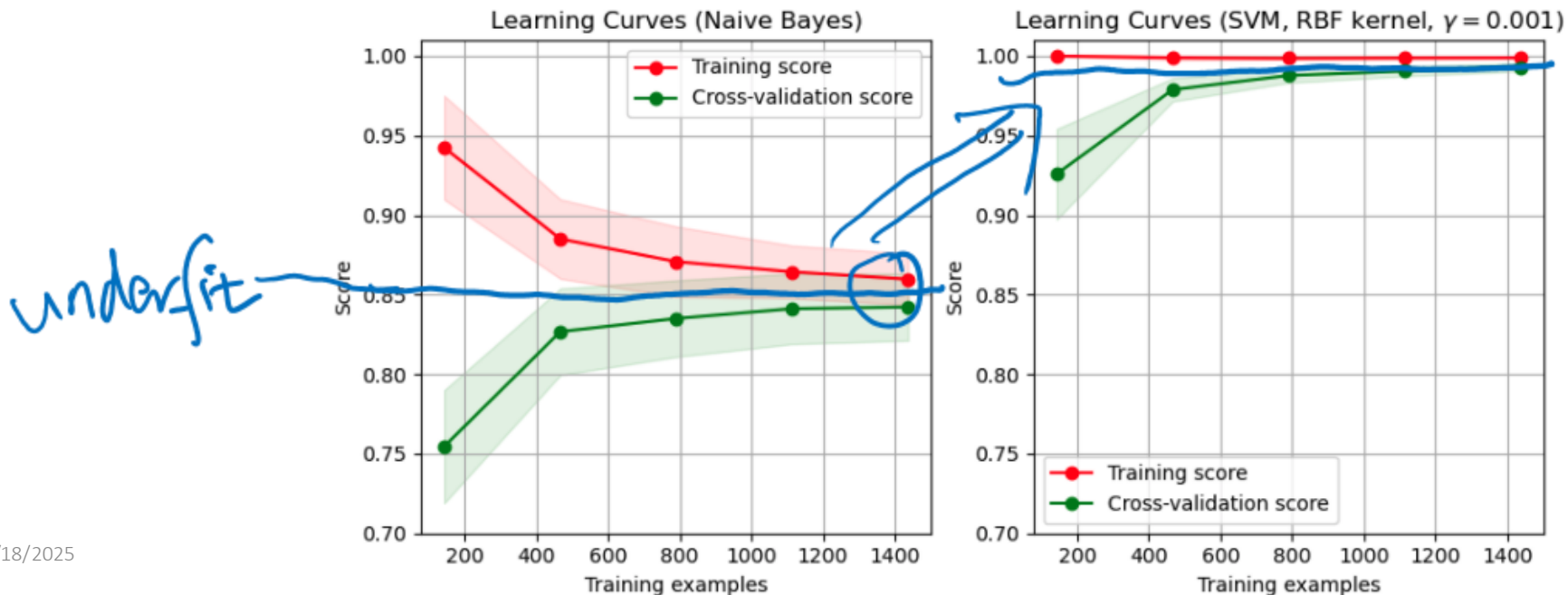


(1) Validation curve

By our HW2 (more close to
modern deep learning
library style)



(1) Learning Curves for polynomial regression (up) and classification (down) / by scikitlearn



```
X2, y2 = make_data(200)
```

```
degree = np.arange(200)
```

```
train_score2, val_score2 = validation_curve(PolynomialReg  
                                             'polynomial
```

```
plt.plot(degree, np.median(train_score2, 1), color='blue'  
plt.plot(degree, np.median(val_score2, 1), color='red',  
plt.legend(loc='lower center')  
plt.ylim(0, 1)  
plt.xlabel('degree')  
plt.ylabel('score');
```

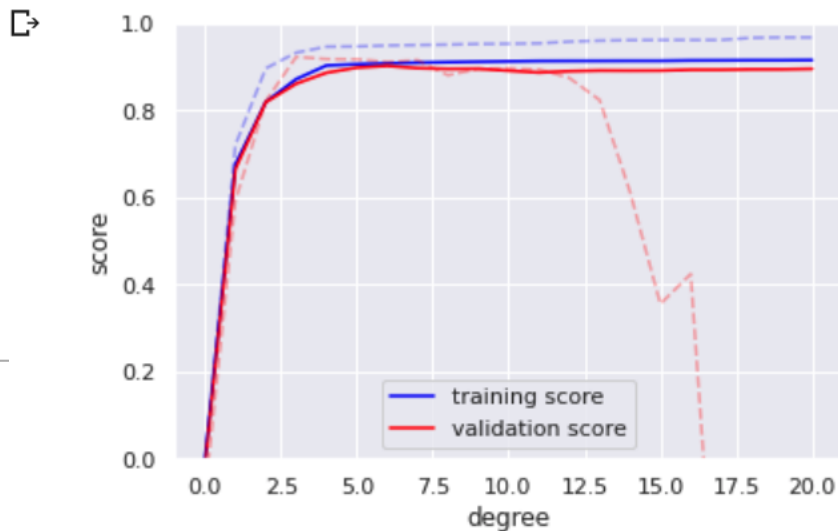


```
X2, y2 = make_data(200)
```

```
degree = np.arange(21)
```

```
train_score2, val_score2 = validation_curve(PolynomialReg  
                                             'polynomialfe
```

```
plt.plot(degree, np.median(train_score2, 1), color='blue'  
plt.plot(degree, np.median(val_score2, 1), color='red', 1  
plt.plot(degree, np.median(train_score, 1), color='blue',  
plt.plot(degree, np.median(val_score, 1), color='red', al  
plt.legend(loc='lower center')  
plt.ylim(0, 1)  
plt.xlabel('degree')  
plt.ylabel('score');
```



Interesting Relation between

- the right range of model complexity
- the number of training points

Is the bias-variance trade off dependent on the number of samples? (EXTRA)

$n \uparrow$
variance $\downarrow$

In the usual application of linear regression, your coefficient estimators are unbiased so sample size is irrelevant. But more generally, you can have bias that is a function of sample size as in the case of the variance estimator obtained from applying the population variance formula to a sample (sum of squares divided by n)....

... the bias and variance for an estimator are generally a decreasing function of training size n . Dealing with this is a core topic in nonparametric statistics. For nonparametric methods with tuning parameters a very standard practice is to theoretically derive rates of convergence (as sample size goes to infinity) of the bias and variance as a function of the tuning parameter, and then you find the optimal (in terms of MSE) rate of convergence of the tuning parameter by balancing the rates of the bias and variance. Then you get asymptotic results of your estimator with the tuning parameter converging at that particular rate. Ideally you also provide a data-based method of choosing the tuning parameter (since simply setting the tuning parameter to some fixed function of sample size could have poor finite sample performance), and then show that the tuning parameter chosen this way attains the optimal rate.



10/16

Agenda

- Going over HW2 Solution
- A tutorial talk on [Huggingface.co](https://huggingface.co)

Course Content Plan → Regarding Tasks

JTD

☒ ~~Regression (supervised)~~

☒ ~~Learning theory~~

☐ Classification (supervised)

☐ Unsupervised models

☐ Graphical models

☐ Reinforcement Learning

Y is a continuous

About $f()$

Y is a discrete

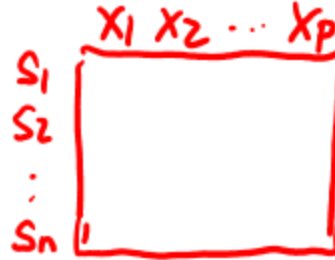
NO Y

About interactions among $Y, X_1, \dots, X_p$

Learn to Interact with environment

Course Content Plan → Regarding Data

- Tabular / Matrix



- 2D Grid Structured: Imaging



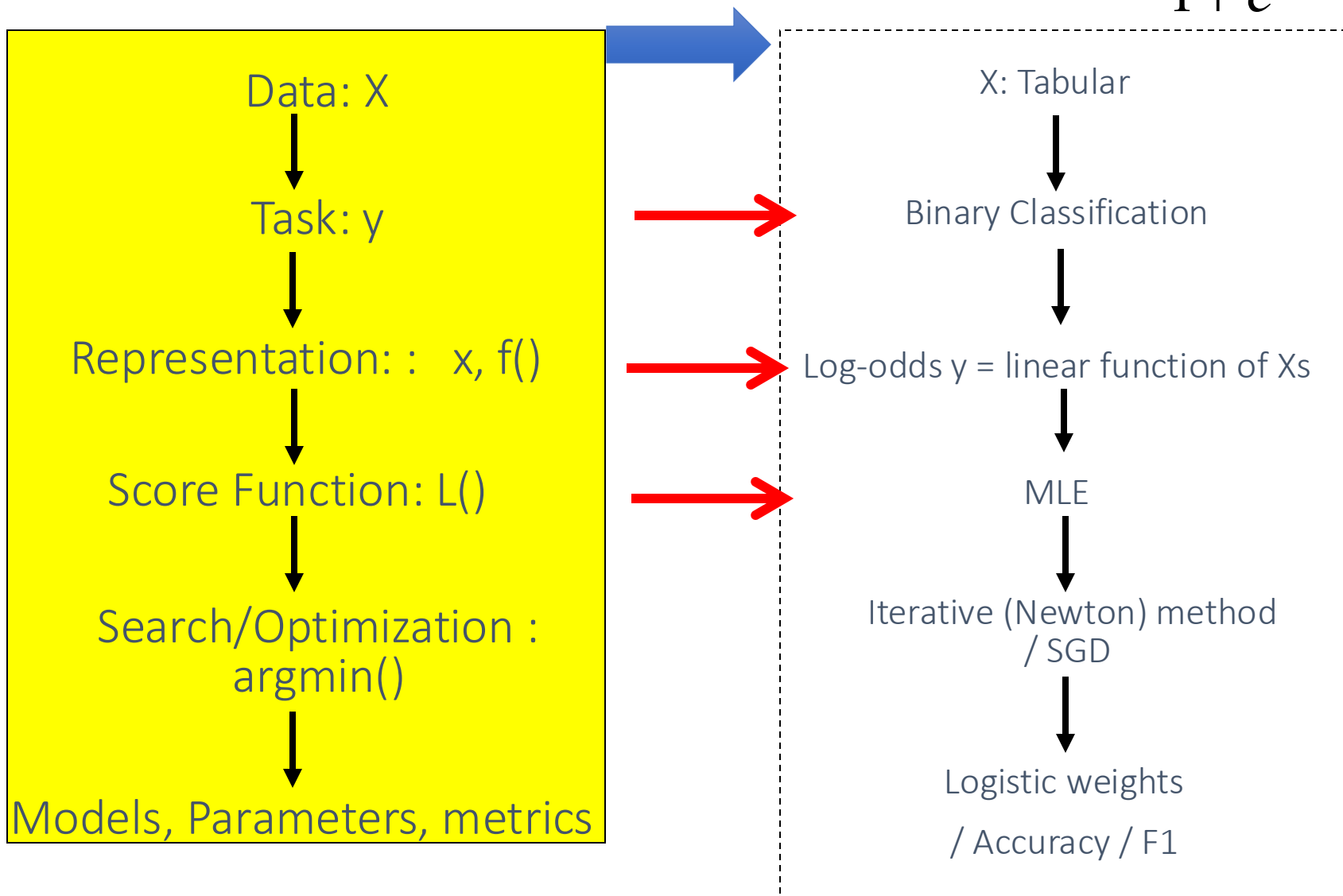
- 1D Sequential Structured: Text

- Graph Structured (Relational)

- Set Structured / 3D /

Today: Logistic Regression Classifier

$$P(y = 1|x) = \frac{e^{\beta_0 + \beta^T x}}{1 + e^{\beta_0 + \beta^T x}}$$



Bayes Classifiers – Predict via MAP Rule

Task: Classify a new instance X : $X = \langle X_1, X_2, \dots, X_p \rangle$

based on:

$$c_{MAP} = \operatorname{argmax}_{c_j \in C} P(c_j \mid x_1, x_2, \dots, x_p)$$



MAP Rule

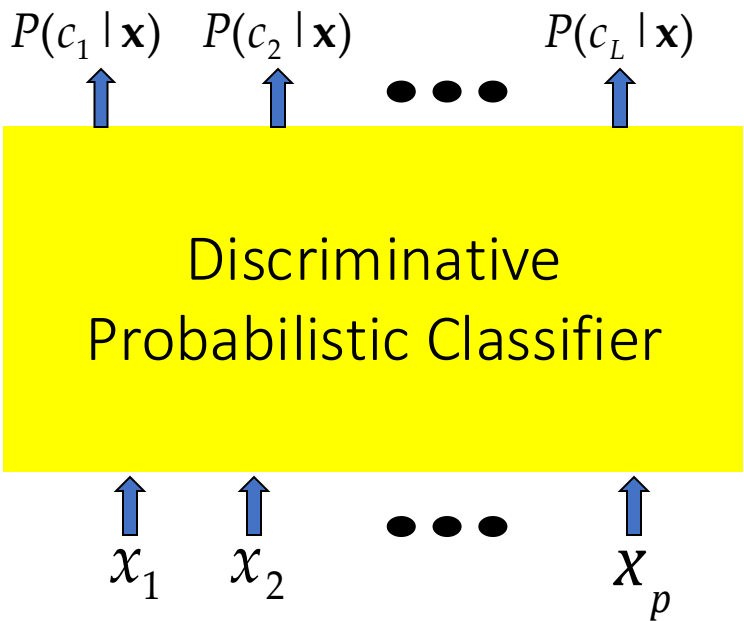
MAP = Maximum A posteriori Probability

Our Whole Section 2:

$X \rightarrow C : \underbrace{P(C|X)}_{f(x)}$

- Discriminative Classifiers

$$\arg \max_{c \in C} P(c | \mathbf{X}), C = \{c_1, \dots, c_L\}$$

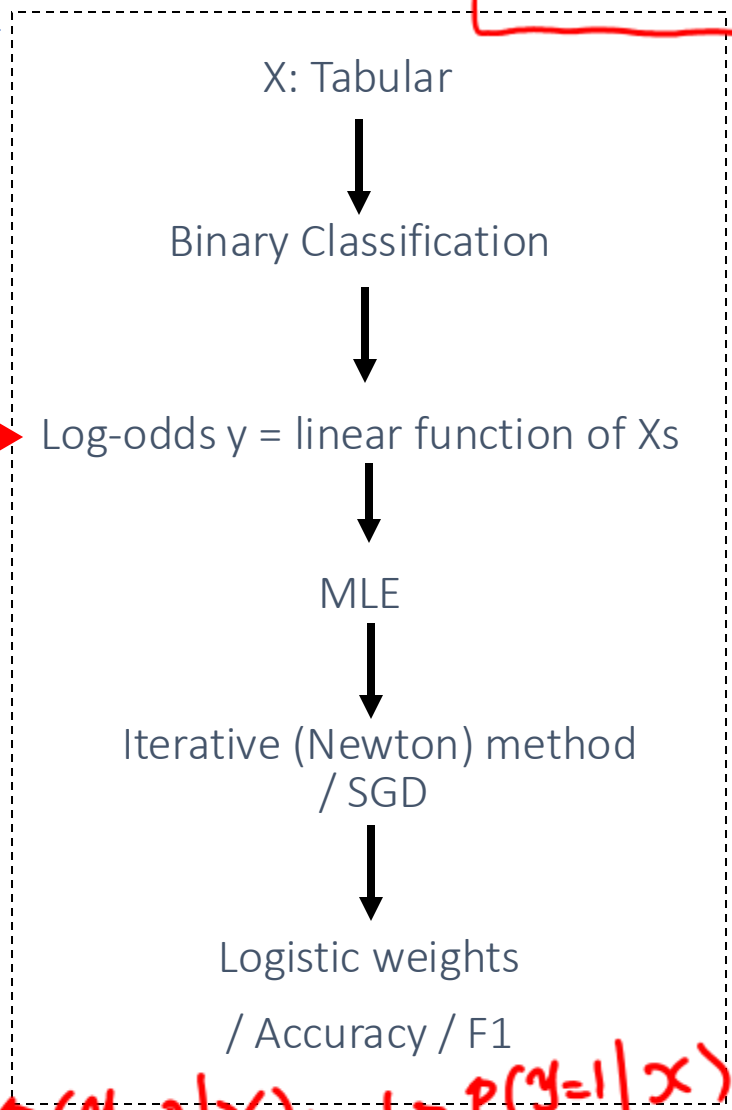
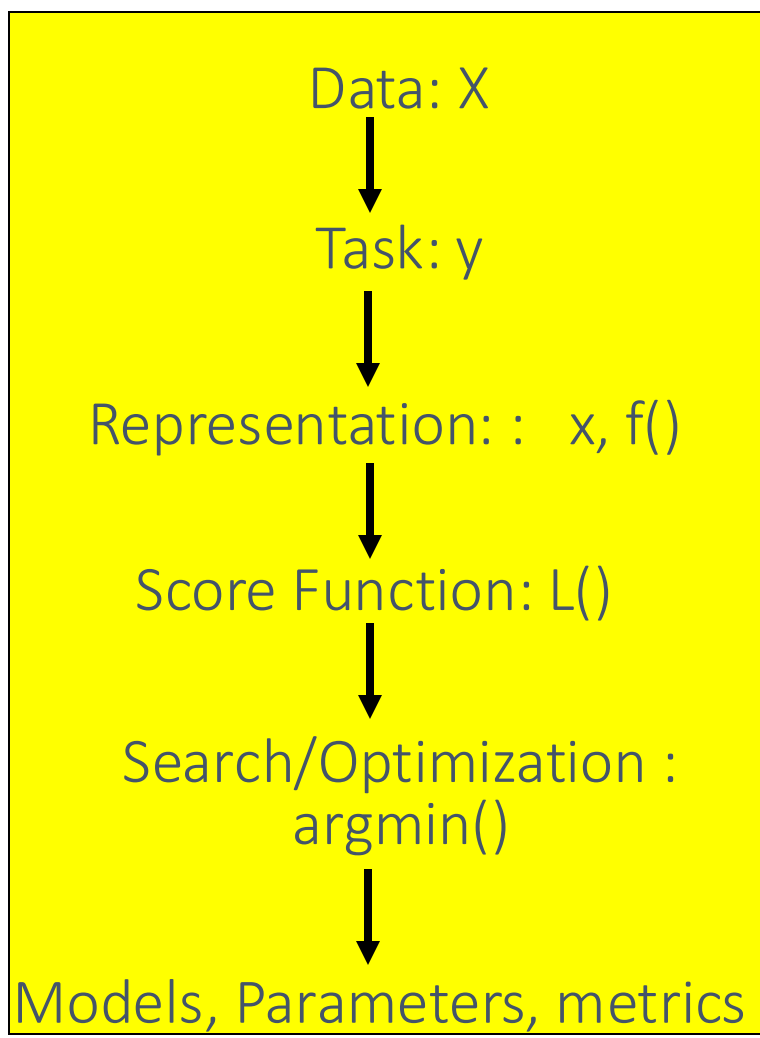


$$\underbrace{P(C|\vec{x})}_{\vec{x}}$$

$$\mathbf{x} = (x_1, x_2, \dots, x_p)$$

Today: Logistic Regression Classifier

$$P(y = 1|x) = \frac{e^{\beta_0 + \beta^T x}}{1 + e^{\beta_0 + \beta^T x}}$$



$$P(y=0|x) = 1 - P(y=1|x)$$

Logistic Regression $p(y|x)$

$$\log \left[\frac{P(\text{Head}|x)}{P(\text{Tail}|x)} \right]$$

$$\ln \left[\frac{P(y|x)}{1 - P(y|x)} \right] = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p$$

Logit



$P(\text{Head})$

$p(c|x)$

$$P(y|x) = \frac{e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p}}{1 + e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p}} = \frac{1}{1 + e^{-(\beta_0 + \beta^T X)}}$$

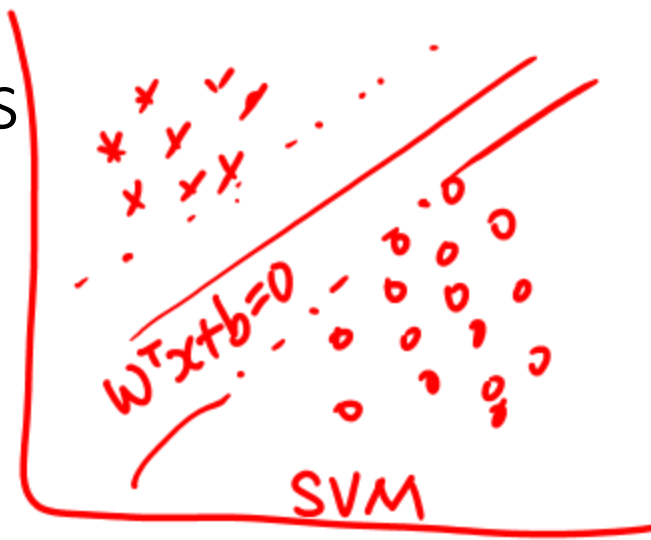
View IV: Logistic Regression models a linear classification boundary!

$$y = \{H, T\}$$

$$\underset{y \in \{0,1\}}{\operatorname{argmax}} p(y|x)$$

$$\frac{p(y=0|x) = p(y=1|x)}{\log\left(\frac{p(y=1|x)}{p(y=0|x)}\right) = \beta^T x = \log(1) = 0} \Rightarrow \frac{p(y=1|x)}{p(y=0|x)} = 1$$

Decision Boundary



Summary: MLE for Logistic Regression Training

Let's fit the logistic regression model for $K=2$, i.e., number of classes is 2

Training set: $(x_i, y_i), i=1, \dots, N$

(conditional)
Log-likelihood:

How?

For Bernoulli distribution

$$p(y | x)^y (1 - p)^{1-y}$$

$$\begin{aligned} l(\beta) &= \sum_{i=1}^N \{\log \Pr(Y = y_i | X = x_i)\} \\ &= \sum_{i=1}^N y_i \log(\Pr(Y = 1 | X = x_i)) + (1 - y_i) \log(\Pr(Y = 0 | X = x_i)) \\ &= \sum_{i=1}^N \left(y_i \log \frac{\exp(\beta^T x_i)}{1 + \exp(\beta^T x_i)} + (1 - y_i) \log \frac{1}{1 + \exp(\beta^T x_i)} \right) \\ &= \sum_{i=1}^N (y_i \beta^T x_i - \log(1 + \exp(\beta^T x_i))) \end{aligned}$$

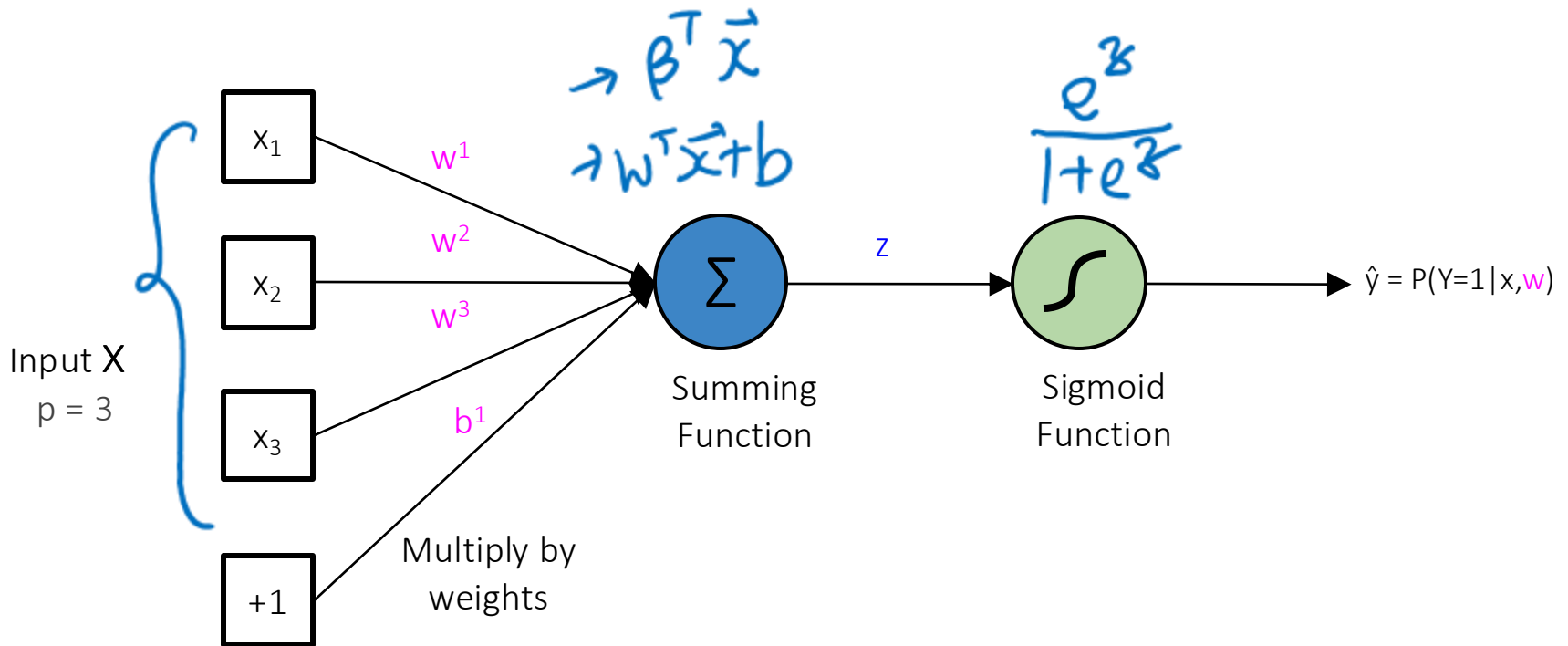
x_i are $(p+1)$ -dimensional input vector with leading entry 1

β is a $(p+1)$ -dimensional vector

We want to **maximize** the log-likelihood in order to estimate β

See Extra Slides How to use Newton-Raphson optimization

One “Neuron”: Block View of Logistic Regression



$$z = \mathbf{w}^T \cdot \mathbf{X} + b$$

$$y = \text{sigmoid}(z) = \frac{e^z}{1 + e^z}$$



10/21

Agenda

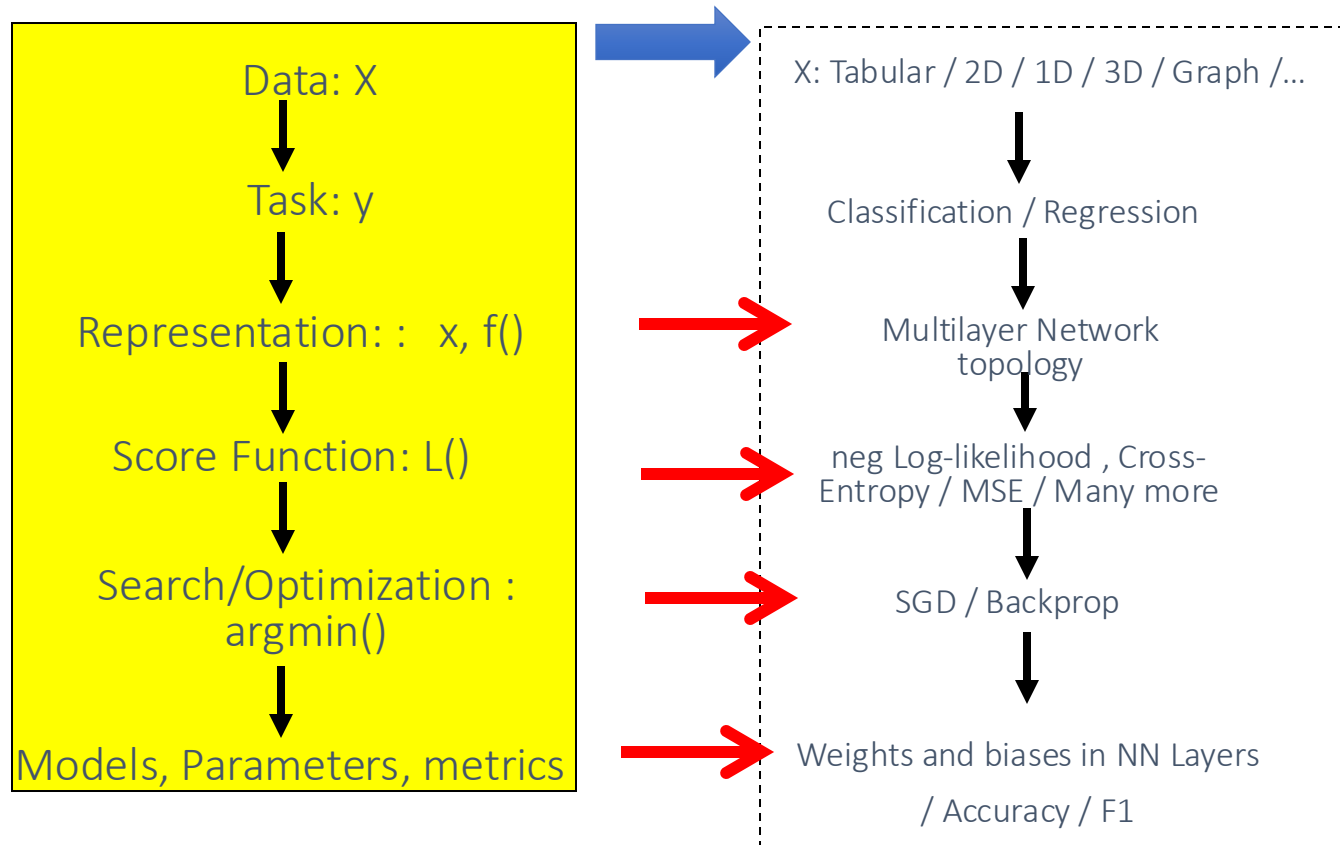
- HW3 is due
- Please select your Project's Shark Tank Sessions ASAP
- Today:
 - Review MLP / DNN / CNN / PCA / Word Embedding, Transformer
 - Quiz 8 Today

Takeaway: Logistic Regression Classifier

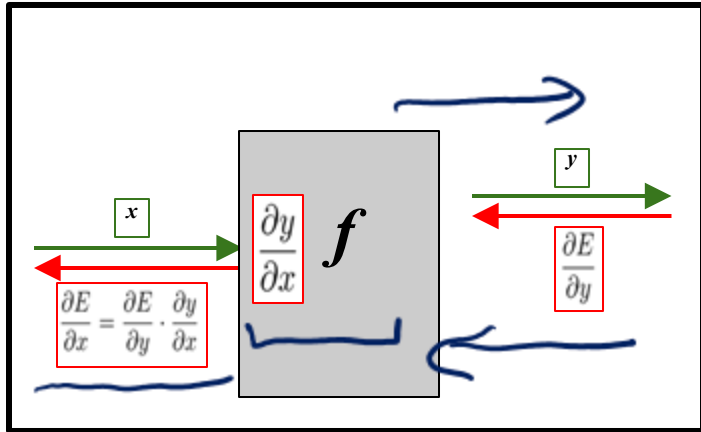
- View I: $\text{logit}(y)$ as linear of X s
- View II: model Y as Bernoulli with $p(y=1 | x)$ as $p(\text{Head})$
- View III: S" shape function compress to $[0,1]$
- View IV: models a linear classification boundary!
- View V: Two stages: summation + sigmoid

Lecture 12: Neural Network (NN) and More: BackProp

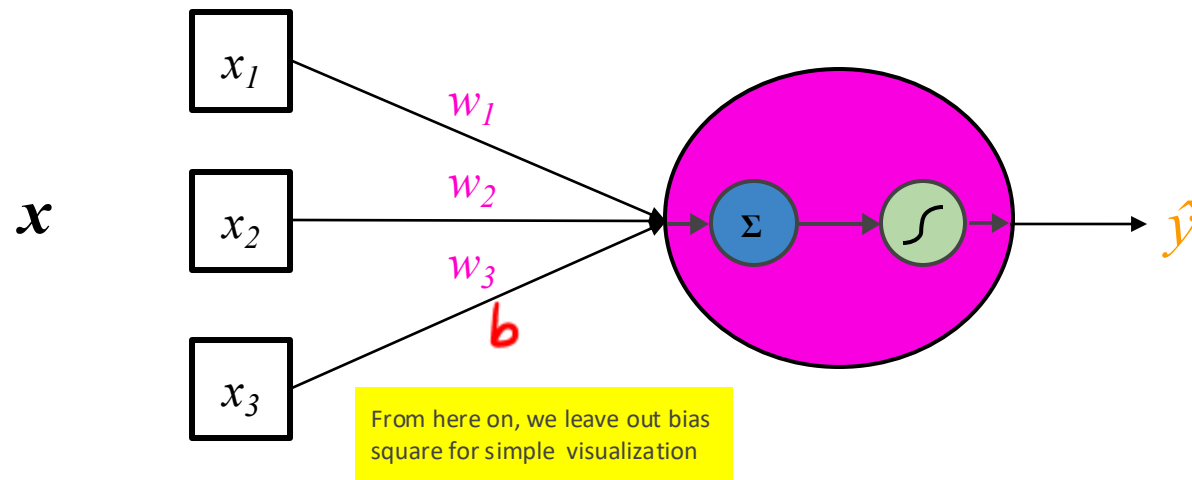
Today: Basic Neural Network Models



Building Deep Neural Nets

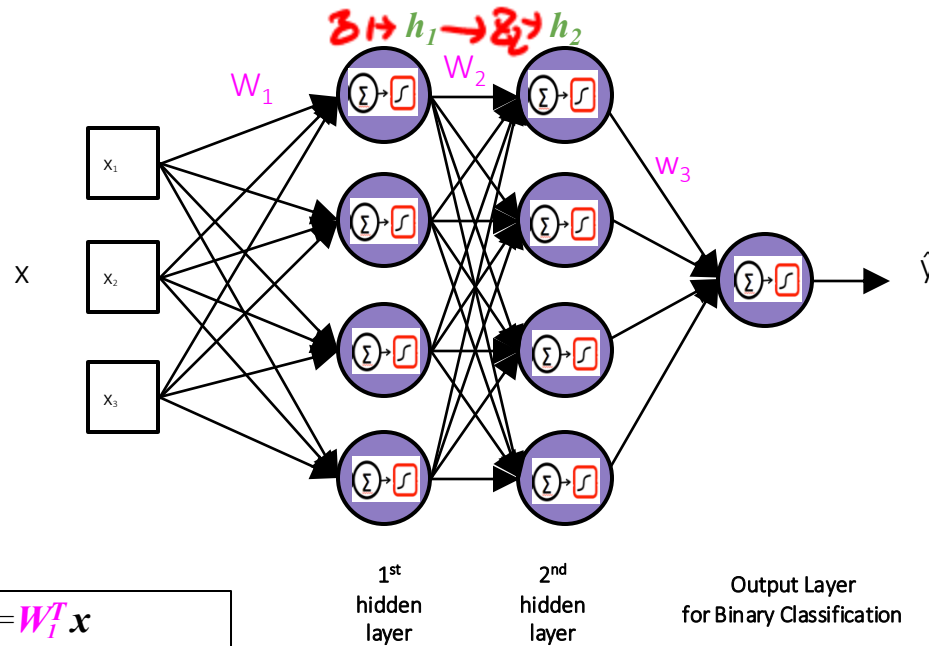


Neuron Representation



The linear transformation and nonlinearity together is typically considered a single neuron

Multi-Layer Perceptron (MLP)- (Feed-Forward NN)

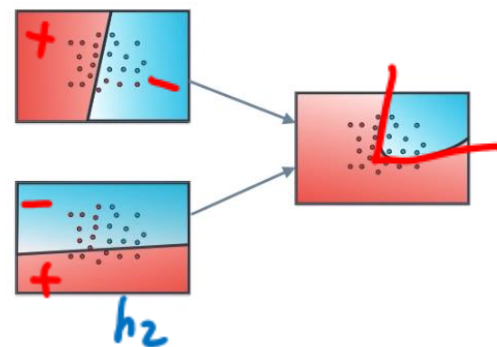
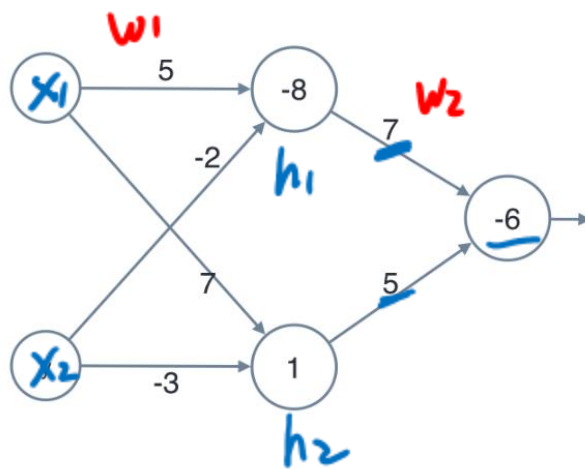
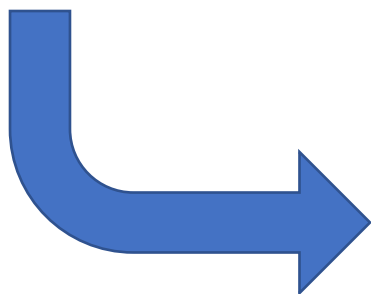
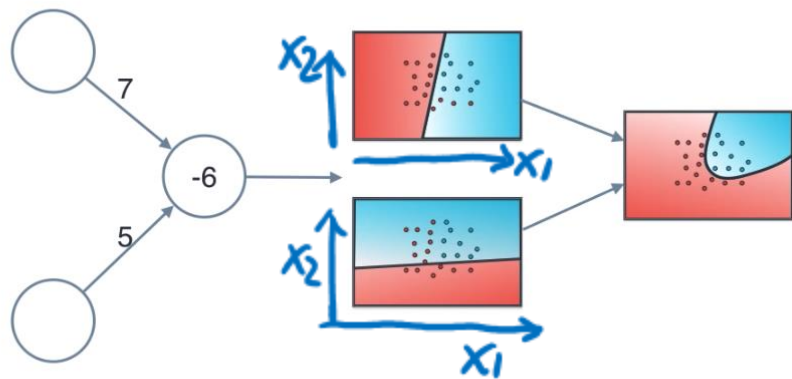
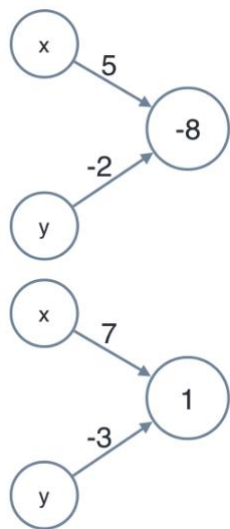


hidden layer 1 output

hidden layer 2 output

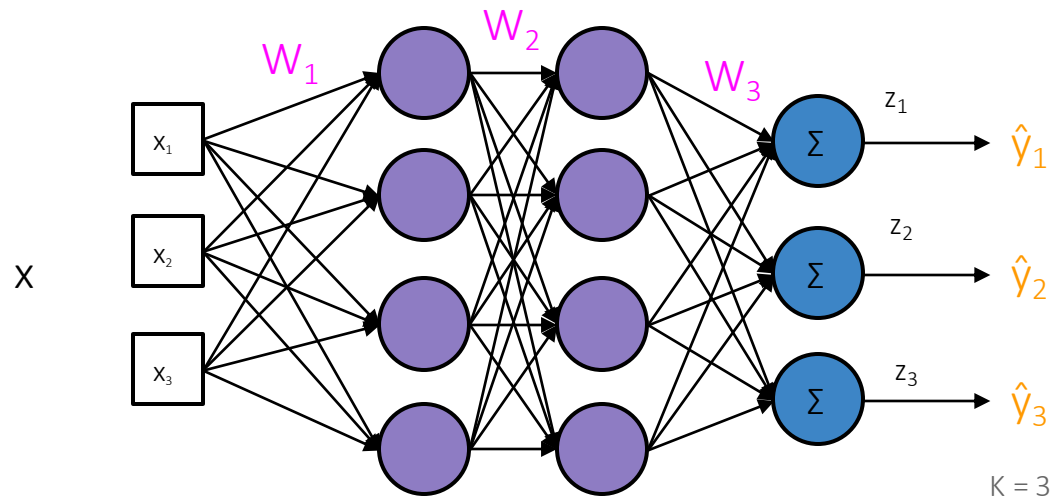
$$\begin{aligned} z_1 &= W_1^T x \\ h_1 &= \text{sigmoid}(z_1) \\ z_2 &= W_2^T h_1 \\ h_2 &= \text{sigmoid}(z_2) \\ z_3 &= W_3^T h_2 \\ \hat{y} &= \text{sigmoid}(z_3) \end{aligned}$$

$$x \xrightarrow{w_1} z_1 \rightarrow h_1 \xrightarrow{w_2} z_2 \rightarrow h_2 \xrightarrow{w_3} \hat{y}$$



Recap: Multi-Class Classification Loss

Cross Entropy Loss



$$\hat{y}_i = \frac{e^{z_i}}{\sum_j e^{z_j}} = P(y_i = 1 \mid x)$$

“Softmax” function.
Normalizing function which
converts each class output to
a probability.

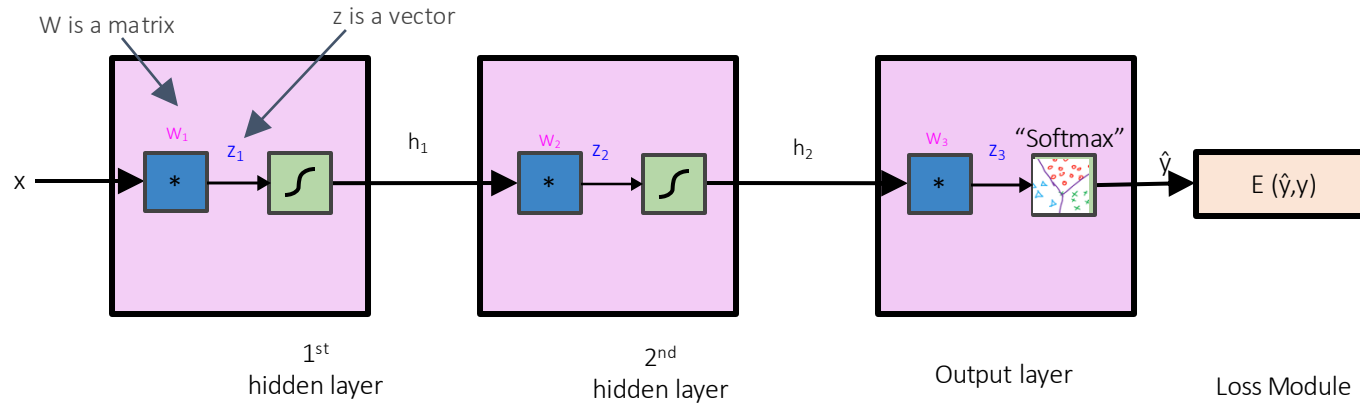
$$\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0.1 \\ 0.7 \\ 0.2 \end{pmatrix}$$

y $\hat{y}$

$$E = \text{loss} = - \sum_{j=1 \dots K} y_j \log \hat{y}_j$$

“0” for all except true class

e.g., “Block View” of multi-layered multi-class NN



$$x \xrightarrow{W_1} h_1 \xrightarrow{W_2} h_2 \xrightarrow{W_3} \hat{y} \rightarrow E(\hat{y}, y)$$

Extra

$\text{argmin}_w \{f_4(f_3(f_2(f_1(\dots))))\}$

f_1	$z_1 = x_1 w_1 + x_2 w_3 + b_1$ $z_2 = x_1 w_2 + x_2 w_4 + b_2$
f_2	$h_1 = \frac{\exp(z_1)}{1 + \exp(z_1)}$ $h_2 = \frac{\exp(z_2)}{1 + \exp(z_2)}$
f_3	$\hat{y} = h_1 w_5 + h_2 w_6 + b_3$
f_4	$E = (y - \hat{y})^2$

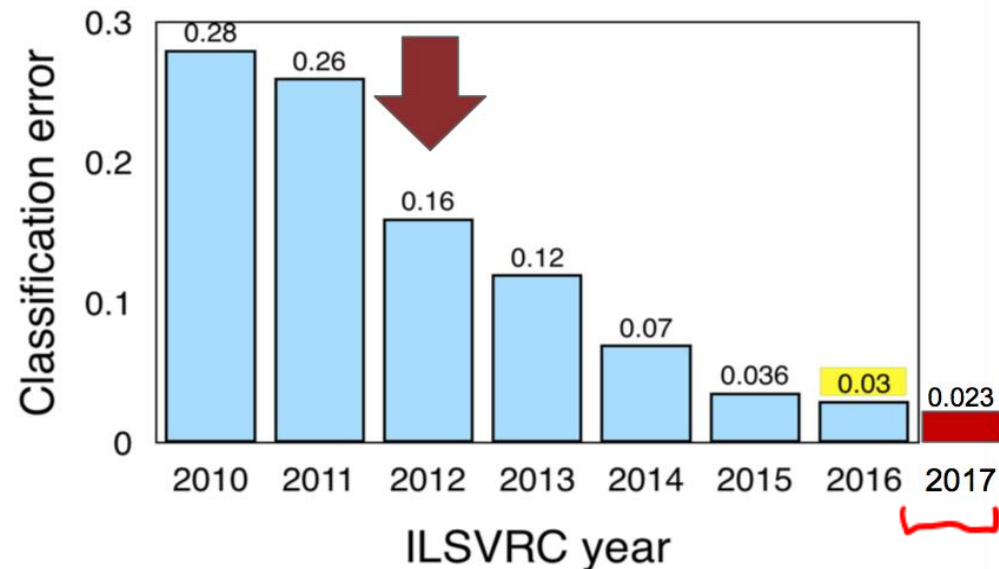
Input	Output	Local Gradients = $\frac{\partial \text{Output}}{\partial \text{Input}}$
$x_1, x_2, w_1, \dots$	z_1, z_2	
z_1, z_2	h_1, h_2	$\frac{\partial h_1}{\partial z_1} = h_1(1-h_1)$
w_5, h_1, h_2	$\hat{y}$	$\frac{\partial \hat{y}}{\partial h_1} = w_5$
$\hat{y}$	loss E	$\frac{\partial E}{\partial \hat{y}} = -2(y - \hat{y})$

$$\begin{aligned}
 \frac{\partial E}{\partial w_1} &= \frac{\partial E}{\partial w_1} = \frac{\partial f_4}{\partial f_3} \frac{\partial f_3}{\partial f_2} \frac{\partial f_2}{\partial f_1} \frac{\partial f_1}{\partial w_1} \\
 &= -2(y - \hat{y}) \frac{\partial (h_1 w_5 + h_2 w_6 + b_3)}{\partial w_1} \\
 &= -2(y - \hat{y}) (w_5 \frac{\partial h_1}{\partial w_1} + w_6 \frac{\partial h_2}{\partial w_1}) \\
 &= -2(y - \hat{y}) w_5 \frac{\partial h_1}{\partial w_1} = -2(y - \hat{y}) w_5 \frac{\partial h_1}{\partial z_1} \frac{\partial z_1}{\partial w_1} \\
 &= \underbrace{-2(y - \hat{y})}_{f_4 \text{ local}} \underbrace{w_5}_{f_3 \text{ local}} \underbrace{h_1(1-h_1)}_{f_2 \text{ local}} \underbrace{x_1}_{\frac{\partial f_1}{\partial w_1}}
 \end{aligned}$$

ImageNet Challenge



- 2010-11: hand-crafted computer vision pipelines
- 2012-2016: ConvNets
 - 2012: AlexNet
 - major deep learning success
 - 2013: ZFNet
 - improvements over AlexNet
 - 2014
 - VGGNet: deeper, simpler
 - InceptionNet: deeper, faster
 - 2015
 - ResNet: even deeper
 - 2016
 - ensembled networks
 - 2017
 - Squeeze and Excitation Network



Lecture 13: Supervised Image Classification and Convolutional Neural Networks

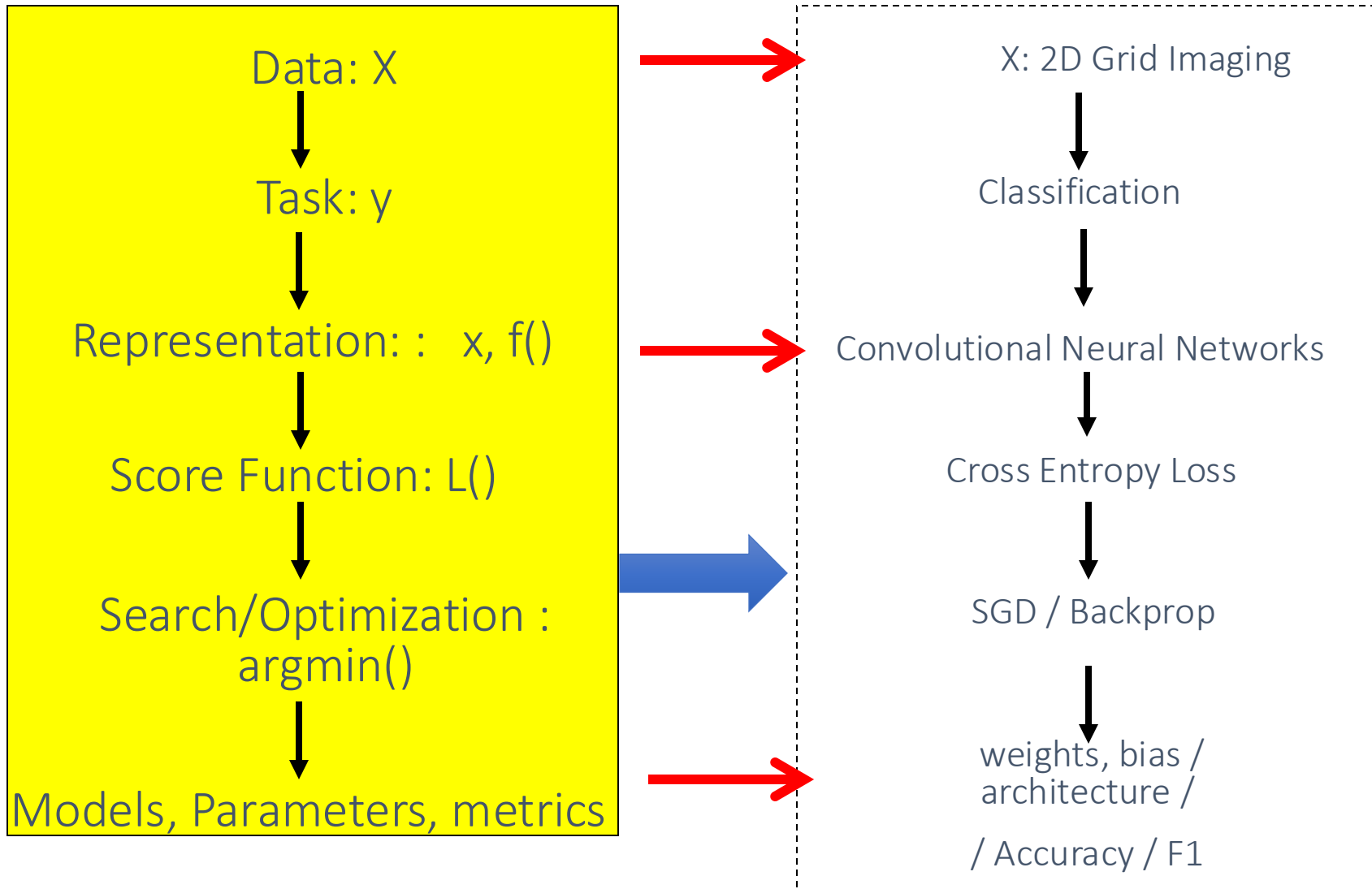
Metric	Formula	Interpretation
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$	Overall performance of model
Precision	$\frac{TP}{TP + FP}$	How accurate the positive predictions are
Recall Sensitivity	$\frac{TP}{TP + FN}$	Coverage of actual positive sample
Specificity	$\frac{TN}{TN + FP}$	Coverage of actual negative sample
F1 score	$\frac{2TP}{2TP + FP + FN}$	

	actual	
	+	−
predicted+	<i>TP</i>	<i>FP</i>
predicted−	<i>FN</i>	<i>TN</i>

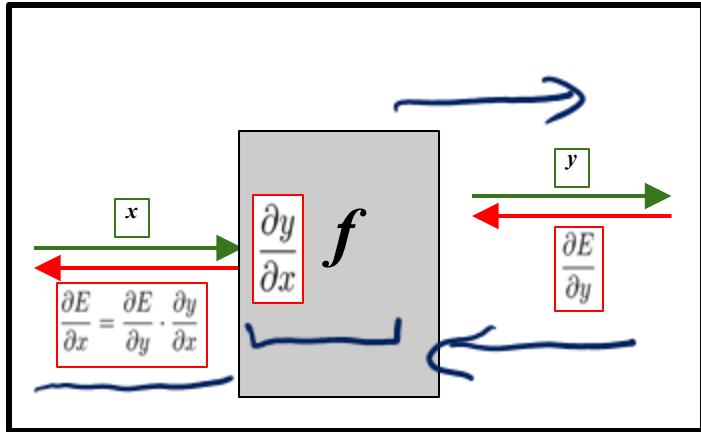
ed classes

Lecture 13: Supervised Image Classification and Convolutional Neural Networks

Today: Convolutional Network Models on 2D Grid / Image

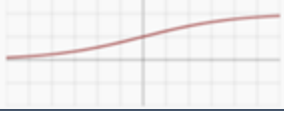
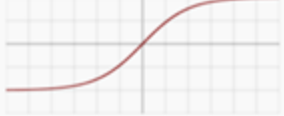


Building Deep Neural Nets



E.g., Many Possible Nonlinearity Functions

(aka transfer or activation functions)

Name	Plot	Equation	Derivative (w.r.t x)
Binary step		$f(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$	$f'(x) = \begin{cases} 0 & \text{for } x \neq 0 \\ ? & \text{for } x = 0 \end{cases}$
Logistic (a.k.a Soft step)		$f(x) = \frac{1}{1 + e^{-x}}$	$f'(x) = f(x)(1 - f(x))$
TanH		$f(x) = \tanh(x) = \frac{2}{1 + e^{-2x}} - 1$	$f'(x) = 1 - f(x)^2$
Rectifier (ReLU) ^[9]		$f(x) = \begin{cases} 0 & \text{for } x < 0 \\ x & \text{for } x \geq 0 \end{cases}$	$f'(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$

usually works best in practice

Important Block: Convolutional Neural Networks (CNN)

- Prof. Yann LeCun invented CNN in 1998
- First NN successfully trained with many layers



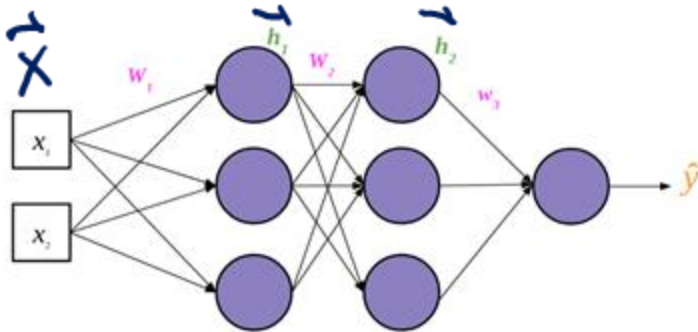
The bird occupies a local area and looks the same in different parts of an image.
We should construct neural nets which exploit these properties!

Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, Gradient-based learning applied to document recognition, Proceedings of the IEEE 86(11): 2278–2324, 1998.

TF Keras Sample Code

<https://www.kaggle.com/code/shawon10/covid-19-diagnosis-from-images-using-densenet121>

Pytorch Sample Code



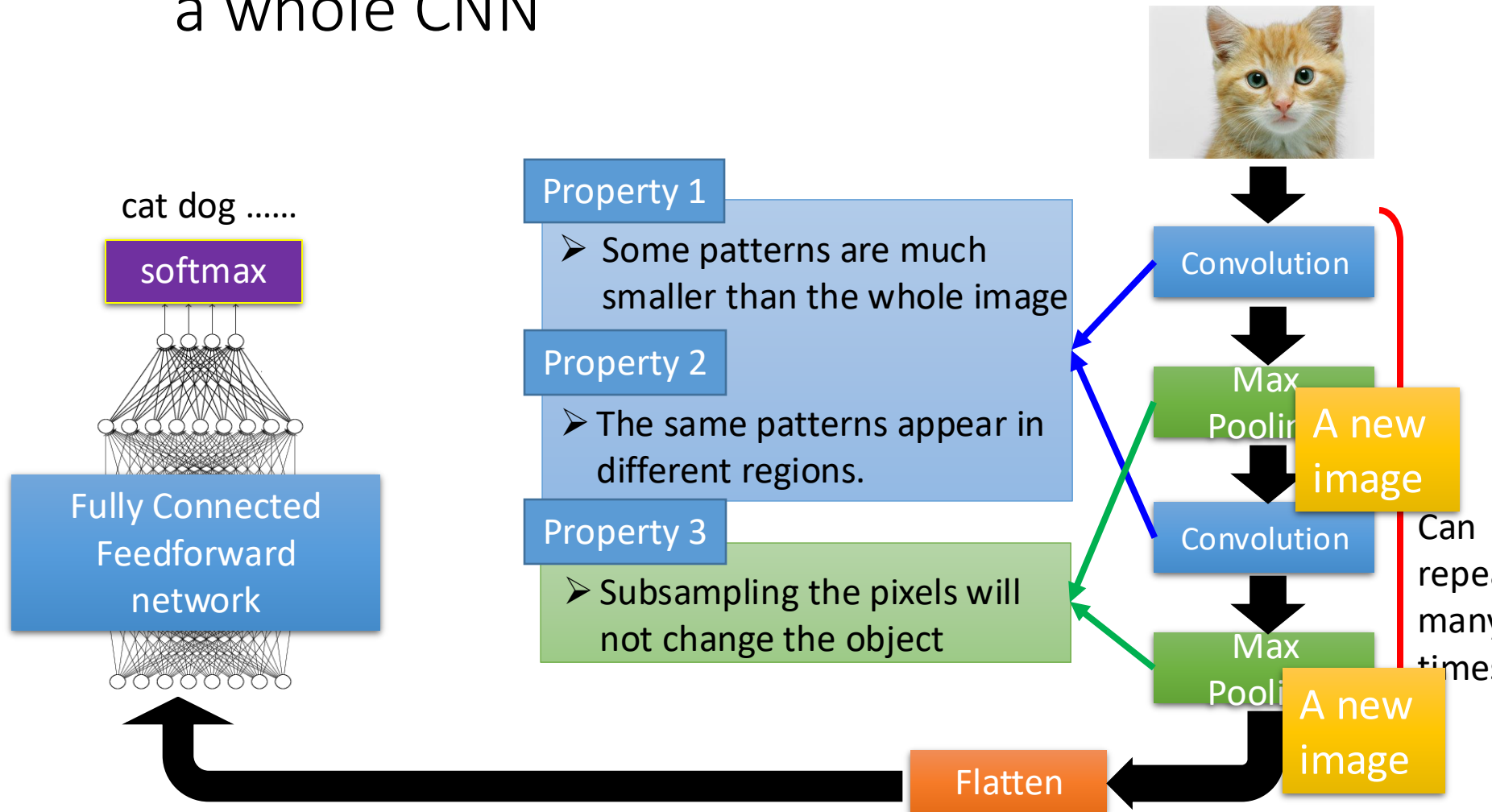
```
import torch.nn as nn
import torch.nn.functional as F

class ThreeLayerNet(torch.nn.Module):
    def __init__(self, d_in, d_hidden, d_out):
        super().__init__()
        self.W1 = nn.Linear(d_in, d_hidden)
        self.W2 = nn.Linear(d_hidden, d_hidden)
        self.w3 = nn.Linear(d_hidden, d_out)
        self.nonlinear = nn.Sigmoid()

    def forward(self, x):
        h1 = self.nonlinear(self.W1(x))
        h2 = self.nonlinear(self.W2(h1))
        y_hat = self.nonlinear(self.w3(h2))
        return y_hat

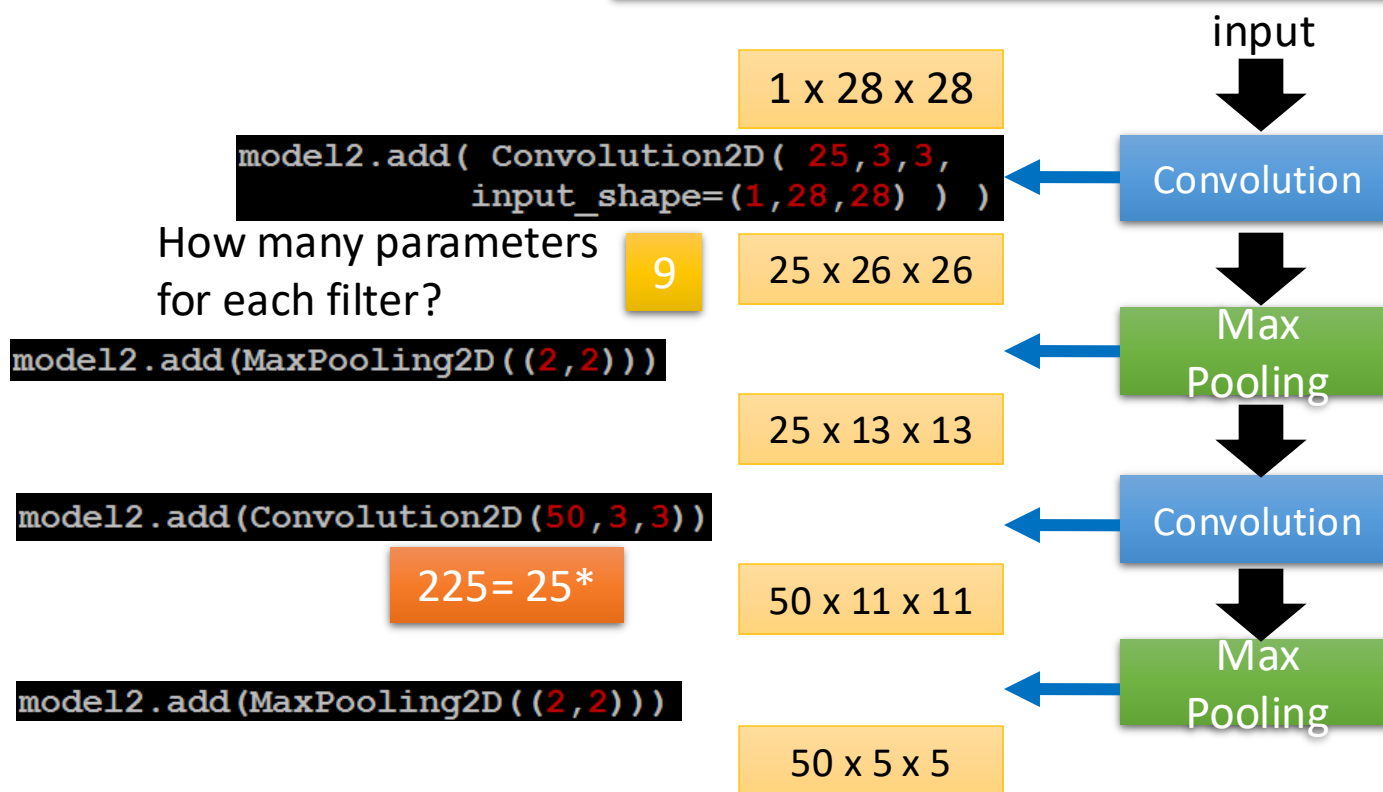
model = ThreeLayerNet(2, 3, 1)
```

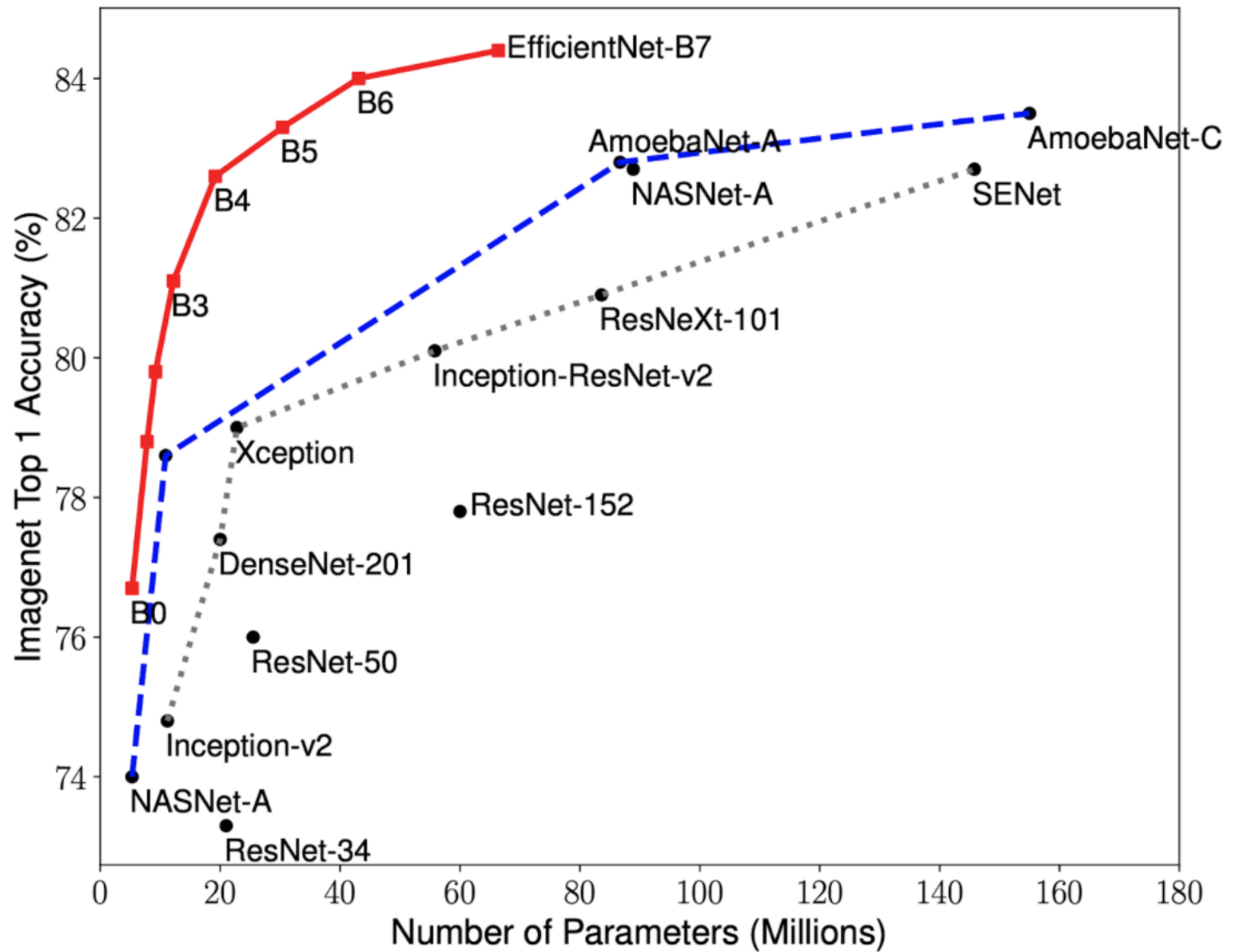
a whole CNN



CNN in Keras

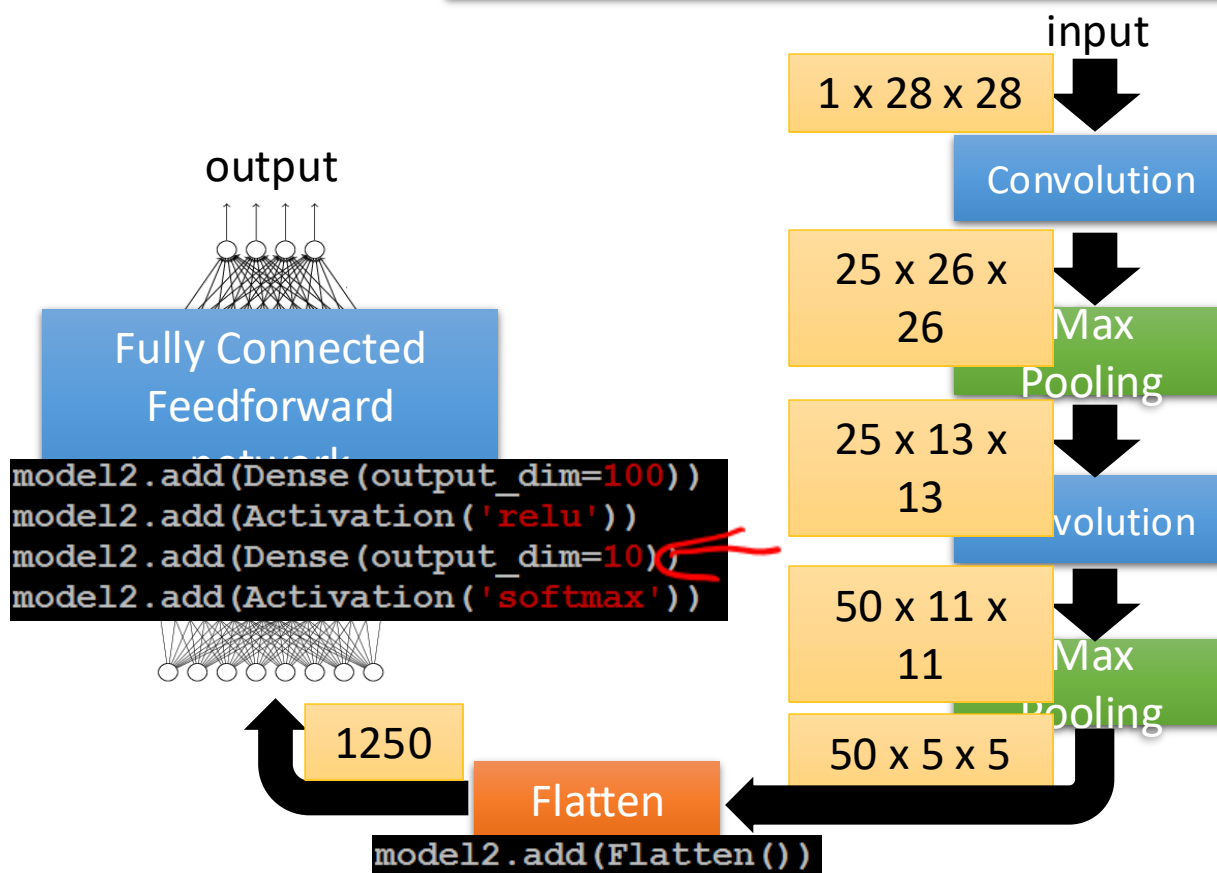
*network structure and input format
(vector -> 3-D tensor)*





CNN in Keras

Only modified the *network structure* and *input format (vector -> 3-D tensor)*



Lecture 14: Dimension Reduction

Today: Dimensionality Reduction (Two Ways)

Feature selection: chooses a subset of the **original** features.



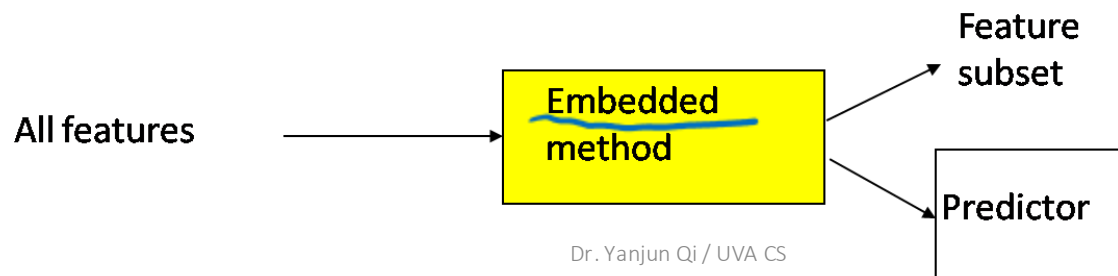
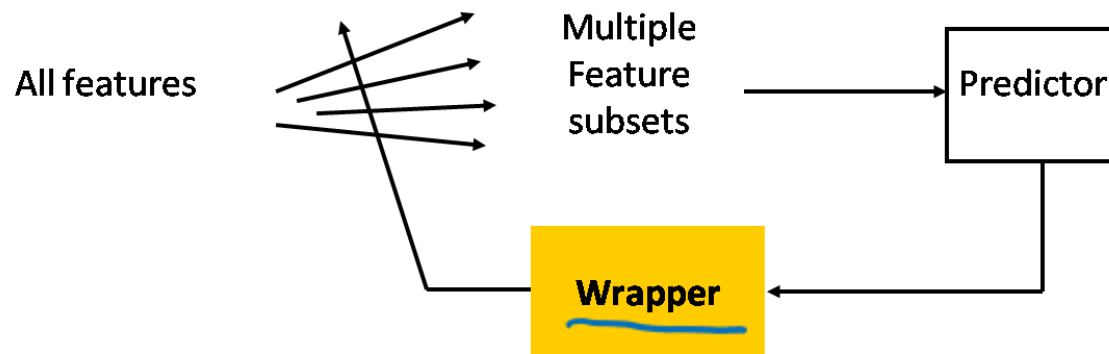
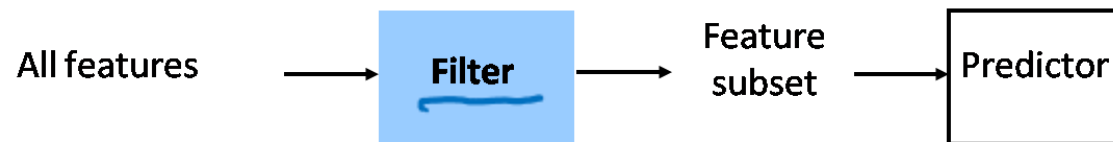
$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix} \longrightarrow \mathbf{x}' = \begin{bmatrix} x_{k1} \\ x_{k2} \\ \vdots \\ x_{kK} \end{bmatrix}$$

$K \ll N$

Summary: Feature Selection

=> filters vs. wrappers vs. embedding

- Main goal: rank subsets of useful features



(I) Filtering : (many choices)

Method		X			Y			Comments
Name	Formula	B	M	C	B	M	C	
Bayesian accuracy	Eq. 3.1	+	s		+	s		Theoretically the golden standard, rescaled Bayesian relevance Eq. 3.2.
Balanced accuracy	Eq. 3.4	+	s		+	s		Average of sensitivity and specificity; used for unbalanced dataset, same as AUC for binary targets.
Bi-normal separation	Eq. 3.5	+	s		+	s		Used in information retrieval.
F-measure ✓	Eq. 3.7	+	s		+	s		Harmonic of recall and precision, popular in information retrieval.
Odds ratio ✓	Eq. 3.6	+	s		+	s		Popular in information retrieval.
Means separation	Eq. 3.10	+	i	+	+			Based on two class means, related to Fisher's criterion.
T-statistics	Eq. 3.11	+	i	+	+			Based also on the means separation.
Pearson correlation ✓	Eq. 3.9	+	i	+	+	i	+	Linear correlation, significance test Eq. 3.12, or a permutation test.
Group correlation ✓	Eq. 3.13	+	i	+	+	i	+	Pearson's coefficient for subset of features.
χ^2 → ✓	Eq. 3.8	+	s		+	s		Results depend on the number of samples m .
Relief	Eq. 3.15	+	s	+	+	s	+	Family of methods, the formula is for a simplified version ReliefX, captures local correlations and feature interactions.
Separability Split Value	Eq. 3.41	+	s	+	+	s		Decision tree index.
Kolmogorov distance	Eq. 3.16	+	s	+	+	s	+	Difference between joint and product probabilities.
Bayesian measure	Eq. 3.16	+	s	+	+	s	+	Same as Vajda entropy Eq. 3.23 and Gini Eq. 3.39.
Kullback-Leibler divergence	Eq. 3.20	+	s	+	+	s	+	Equivalent to mutual information.
Jeffreys-Matusita distance	Eq. 3.22	+	s	+	+	s	+	Rarely used but worth trying.
Value Difference Metric	Eq. 3.22	+	s		+	s		Used for symbolic data in similarity-based methods, and symbolic feature-feature correlations.
Mutual Information ✓	Eq. 3.29	+	s	+	+	s	+	Equivalent to information gain Eq. 3.30.
Information Gain Ratio ✓	Eq. 3.32	+	s	+	+	s	+	Information gain divided by feature entropy, stable evaluation.
Symmetrical Uncertainty	Eq. 3.35	+	s	+	+	s	+	Low bias for multivalued features.
J-measure	Eq. 3.36	+	s	+	+	s	+	Measures information provided by a logical rule.
Weight of evidence	Eq. 3.37	+	s	+	+	s	+	So far rarely used.
MDL 11/18/2025	Eq. 3.38	+	s		+	s		Low bias for multivalued features.

Dr. Yanjun Qi / UVA CS

Guyon-Elisseeff, JMLR 2004; Springer 2006

102

Guyon-Elisseff, JMLR 2004;

Springer 2006

(2) Wrapper : Feature Subset Selection

Wrapper Methods

- Learner is considered a black-box
- Interface of the black-box is used to **score subsets** of variables **according to the predictive power** of the learner when using the subsets.
- Results vary for different learners

(b). Search: even more search strategies for selecting feature subset

$$p \longrightarrow 2^p \text{ feature subsets}$$

- **Forward selection** or **backward elimination**.
- **Beam search**: keep k best path at each step.
- **GSFS**: generalized sequential forward selection – when $(n-k)$ features are left try all subsets of g features. More trainings at each step, but fewer steps.
- **PTA(l, r)**: plus l , take away r – at each step, run SFS l times then SBS r times.
- **Floating search**: One step of SFS (resp. SBS), then SBS (resp. SFS) as long as we find better subsets than those of the same size obtained so far.

(3) Embedded

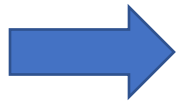
- Embedding approach:

uses a **predictor to build** a (single) model with a subset of features that are internally selected.

Lasso
elasticNet

Today: Dimensionality Reduction (Two Ways)

Feature extraction: finds a set of **new** features (i.e., through some mapping $f()$) from the **existing** features.



The mapping $f()$
could be linear or
non-linear

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix} \xrightarrow{f()} \mathbf{h} = \begin{bmatrix} h_1 \\ h_2 \\ \vdots \\ h_K \end{bmatrix}$$

$K \ll N$

Feature selection: chooses a subset of the **original** features.

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix} \longrightarrow \mathbf{x}' = \begin{bmatrix} x_{k1} \\ x_{k2} \\ \vdots \\ x_{kK} \end{bmatrix}$$

$K \ll N$

Feature Extraction (linear or nonlinear)

- **Linear** combinations are particularly attractive because they are simpler to compute and analytically tractable.
- Given $\mathbf{x} \in \mathbb{R}^p$, find an $N \times K$ matrix $\mathbf{U}$ such that:

$$\mathbf{y} = \mathbf{U}^T \mathbf{x} \in \mathbb{R}^K \text{ where } K < P$$

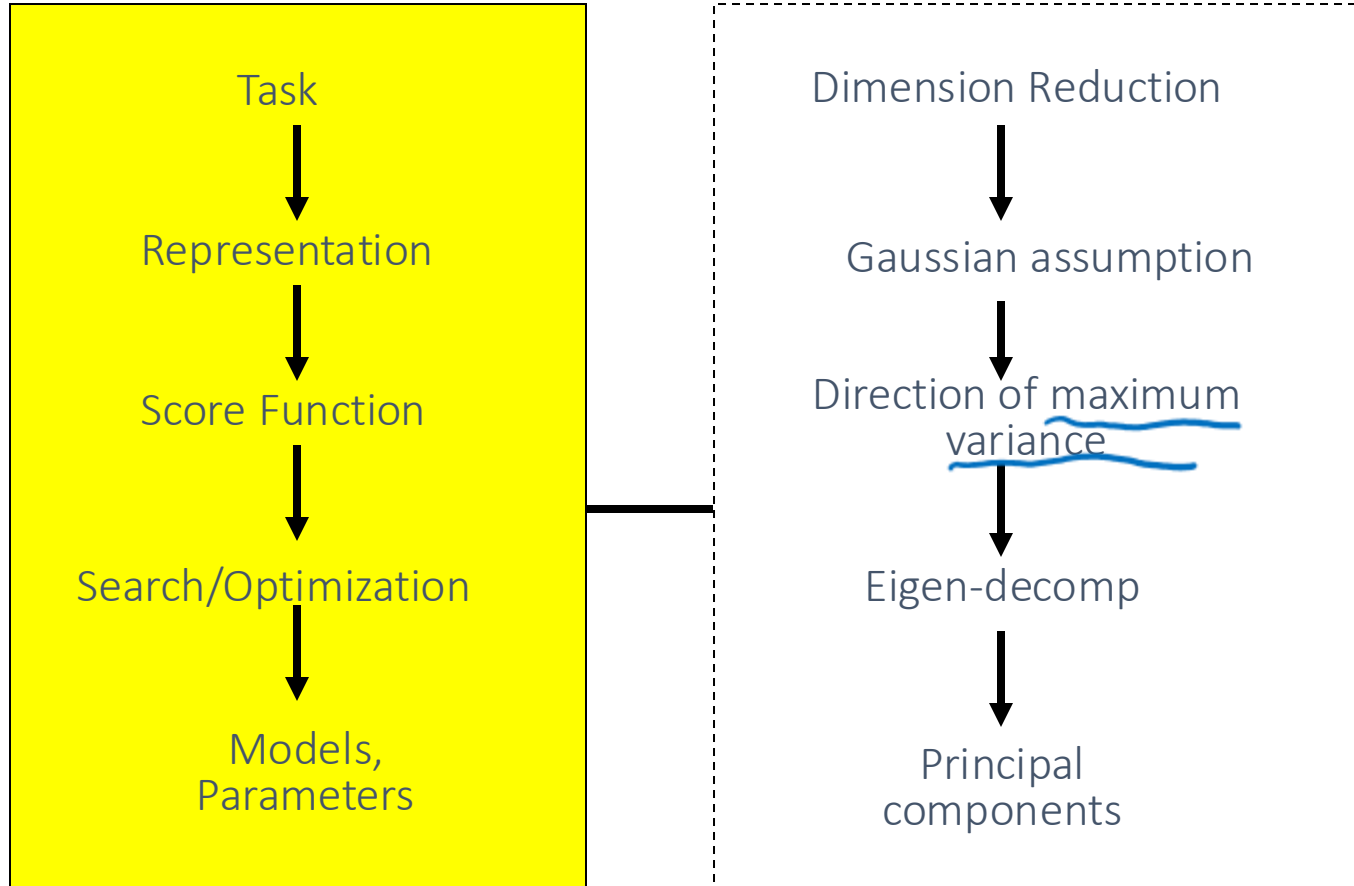
This is a
projection
from the N -
dimensional
space to a K -
dimensional
space.

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix} \xrightarrow{\mathbf{U}^T} \mathbf{h} = \begin{bmatrix} h_1 \\ h_2 \\ \vdots \\ h_K \end{bmatrix}$$

Feature Extraction (cont'd)

- Commonly used **linear** feature extraction methods:
 - **Principal Components Analysis (PCA)**: Seeks a projection that **preserves** as much **information** in the data as possible.
 - **Linear Discriminant Analysis (LDA)**: Seeks a projection that **best discriminates** the data.
- Recent **nonlinear** feature extraction methods:
 - Like Word Embedding / Autoencoder / ...

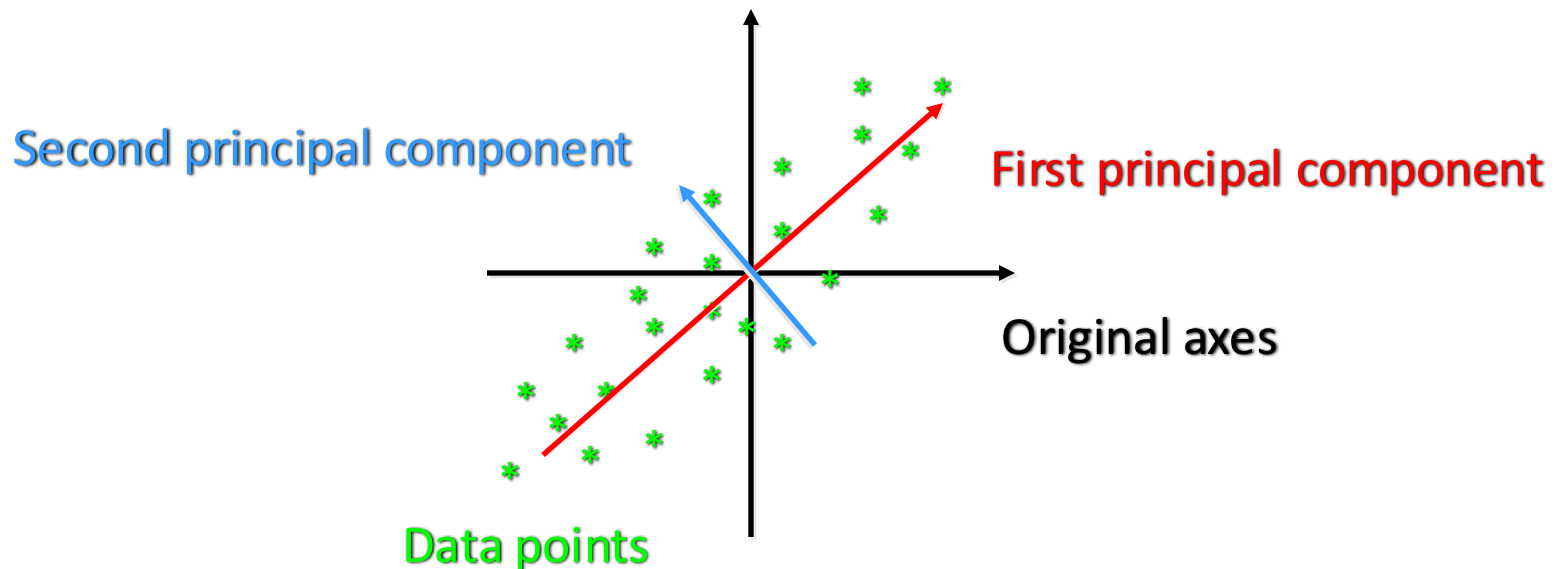
Principal Component Analysis



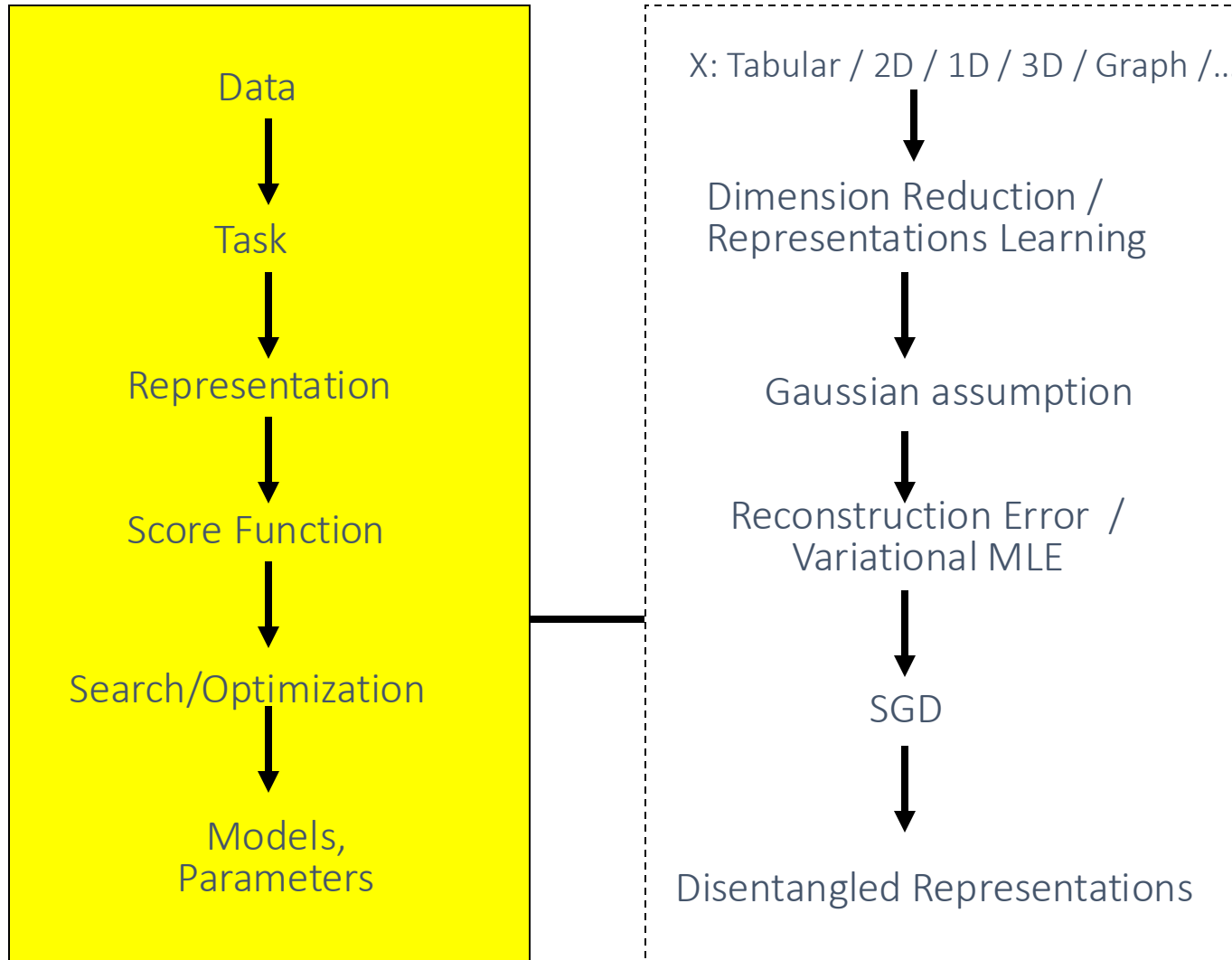
How does PCA work? Explaining Variance

- Each PC always explains some proportion of the total variance in the data. Between them they explain everything
 - PC1 always explains the most
 - PC2 is the next highest etc. etc.

$\mathcal{D} \rightarrow \text{PCs}$

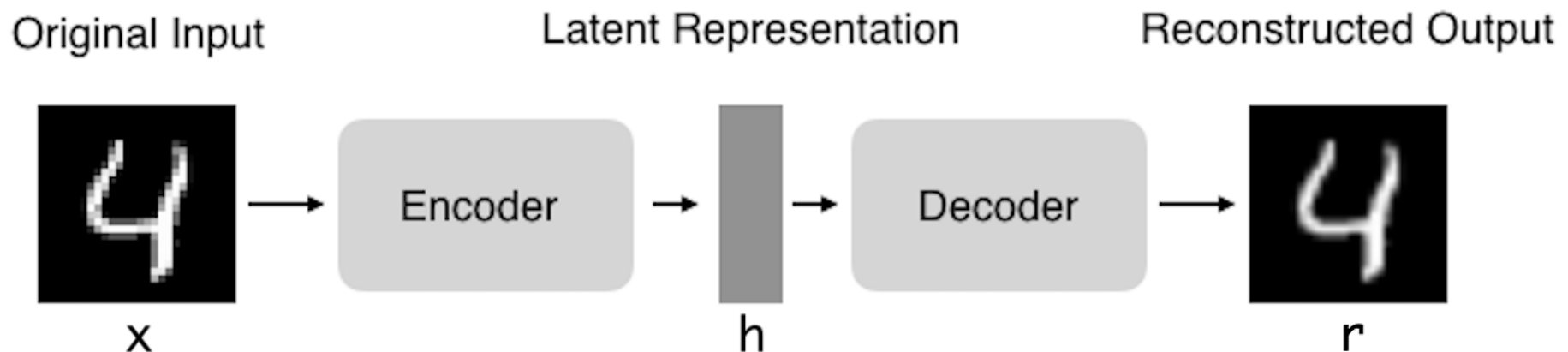


Auto Encoder

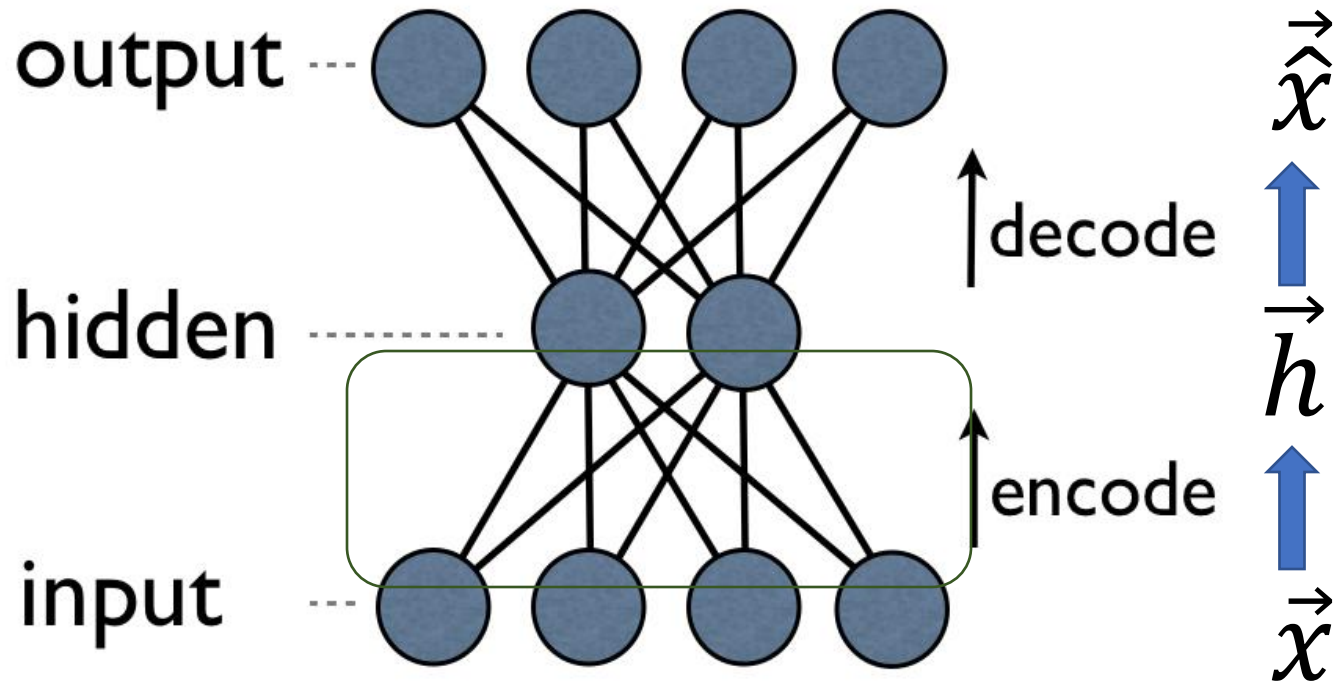


Autoencoders: structure

- Encoder: compress input into a latent-space of usually smaller dimension. $h = f(x)$
- Decoder: reconstruct input from the latent space. $r = g(f(x))$ with r as close to x as possible



an auto-encoder-decoder is trained to reproduce the input



Minimize diff

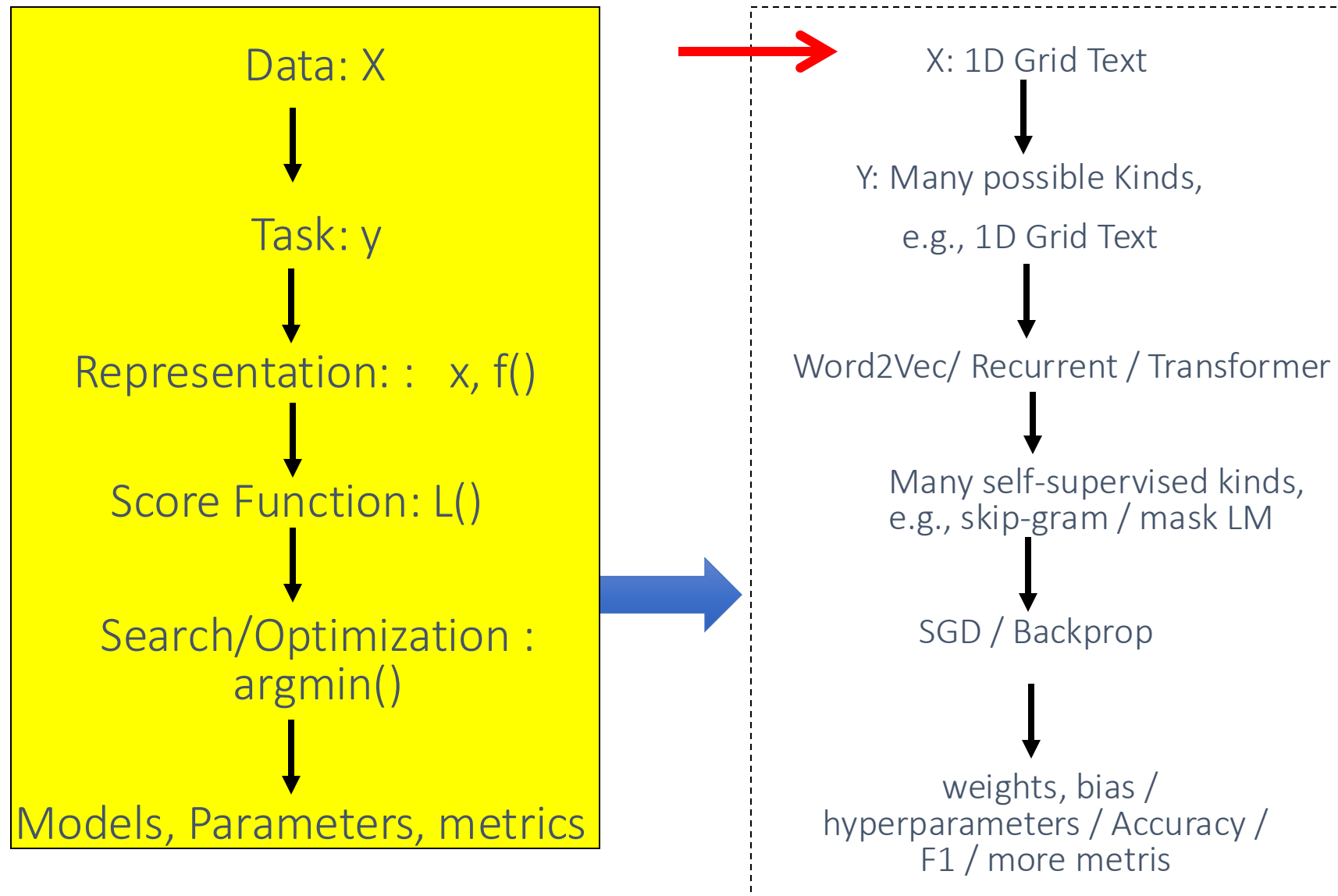
$$|\hat{\vec{x}} - \vec{x}|$$

Reconstruction Loss: force the 'hidden layer' units to become good / reliable feature detectors

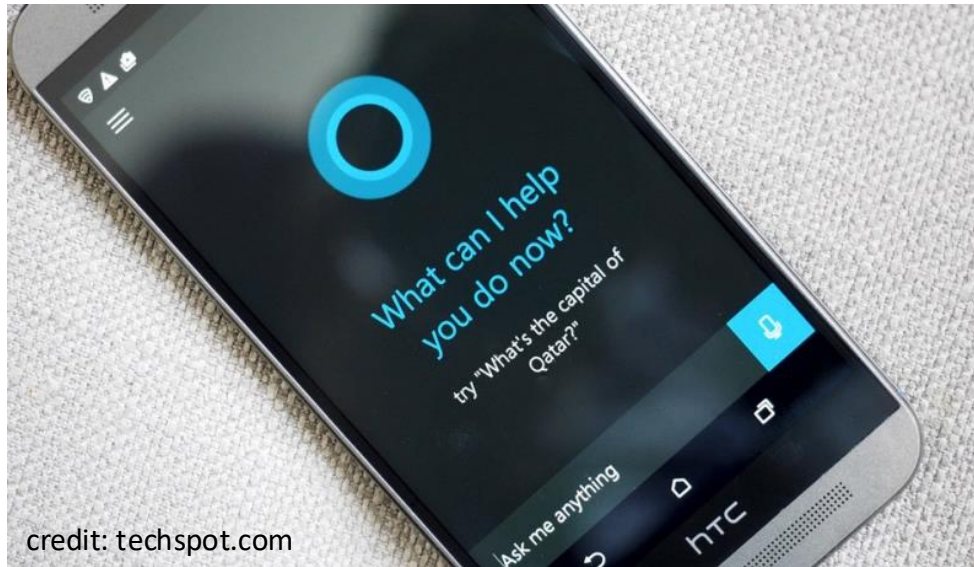


10/28

Today: Neural Network Models on 1D Grid / Language Data

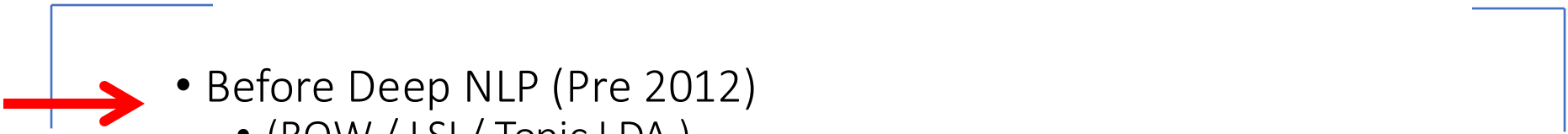


Deep Learning for natural language processing, e.g., Digital personal assistant



- Semantic parsing – understand tasks
- Entity linking – “my wife” = “Kellie” in the phone book

f() on natural language

- 
- Before Deep NLP (Pre 2012)
 - (BOW / LSI / Topic LDA)
 - Word2Vec (2013-2016)
 - (GloVe/ FastText)
 - Recurrent NN (2014-2016)
 - LSTM
 - Seq2Seq
 - Attention / Self-Attention (2016 – now)
 - Attention
 - Transformer (self-attention, attention only)
 - BERT / XLNet/ GPT-x / T5 ...

Recap: The bag of words representation

$$f(\text{Table}) = C$$

great	2
love	2
recommend	1
laugh	1
happy	1
...	...

How to Represent A Word in DNN: Feature Extraction / Embedding

Solution:

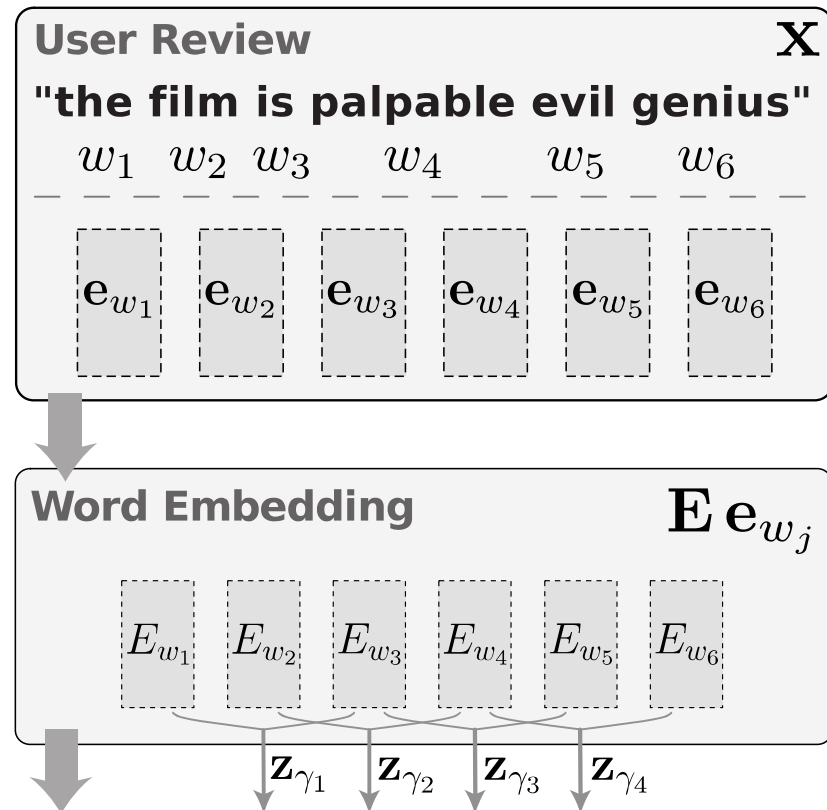
Distributional Word Embedding Vectors

- GloVe (Global Vectors)
 - Pennington et al., 2014
 - Skip-gram
- fasttext
 - Bojanowski et al., 2017

However, Natural language is

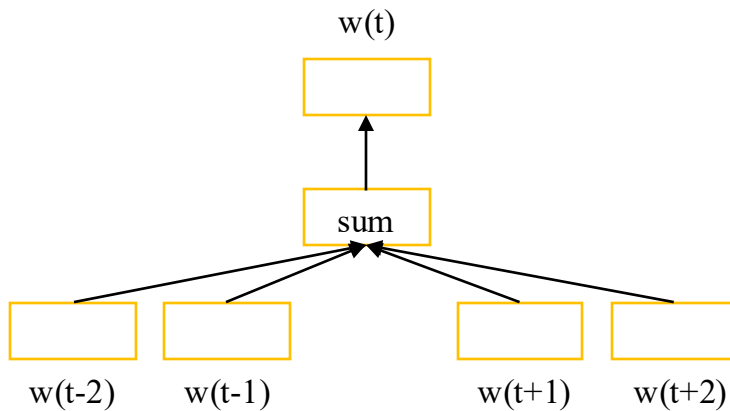
- Variable-length
- Composition of multiple words
- Word meaning is contextual

- Elmo
 - Peters, 2018
- BERT
 - Devlin et al., 2018

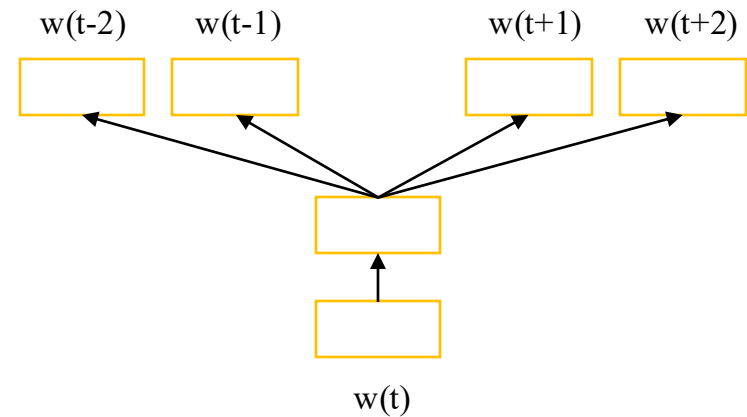


Word2vec: CBOW / SkipGram (Basic Word2Vec)

- Distributed representations of words and phrases and their compositionality (NIPS 2013, Mikolov et al.)
- CBOW
 - predict the input tokens based on context tokens
- SkipGram
 - predict context tokens based on input tokens



CBOW

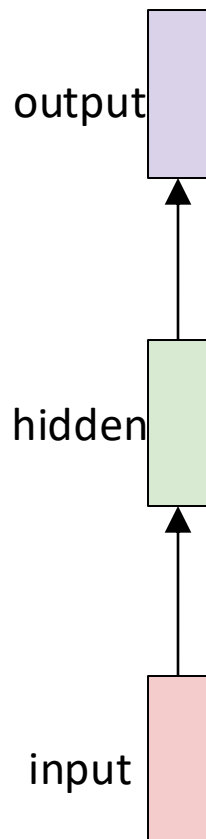


SkipGram

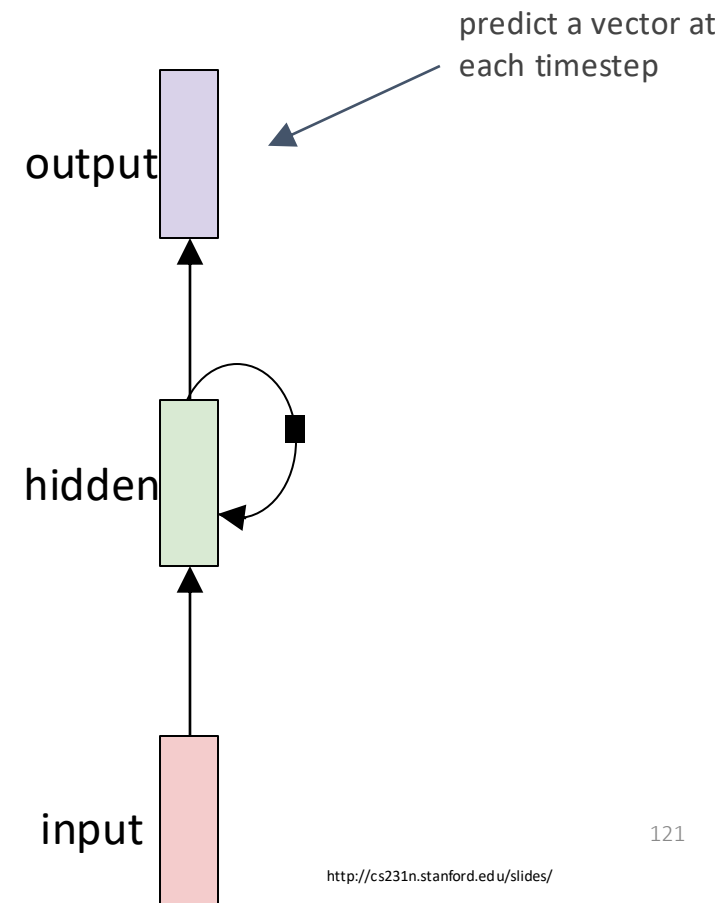
RNN models dynamic temporal dependency

- Make **fully**-connected layer model **each unit recurrently**
- Units form a **directed chain graph** along a sequence
- Each unit uses **recent history** and current input in modeling

**Traditional “Feed Forward”
Neural Network**



Recurrent Neural Network

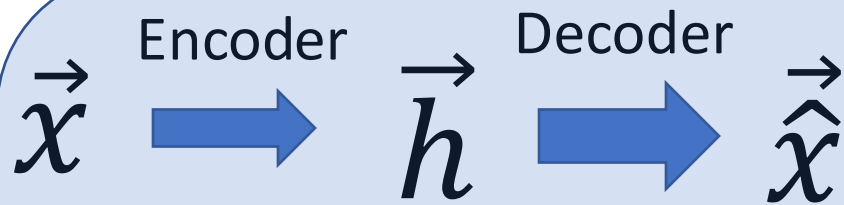


Seq2Seq for NLP Sequence-to-Sequence Generation Tasks

Given source sentences, learn an optimal model to automatically generate accurate and diversified target sentences that look like human generated sentences.

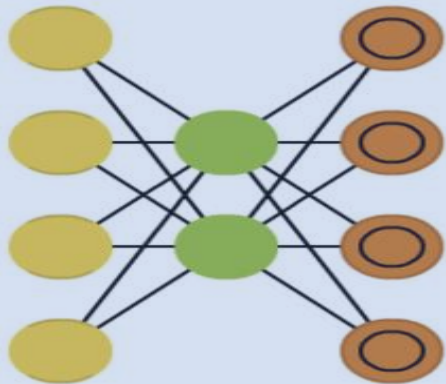


- **Paraphrase generation:** “How did Trump win the election?” → “How did Trump become president?”
- **Dialogue generation:** “You know French?” → “Sure do ... my Mom's from Canada”
- **Question answering:** “What was the name of the 1937 treaty?” → “Bald Eagle Protection Act”
- **Style Transfer:** “Just a dum funny question hahahaha” → “Just a senseless , funny question.”



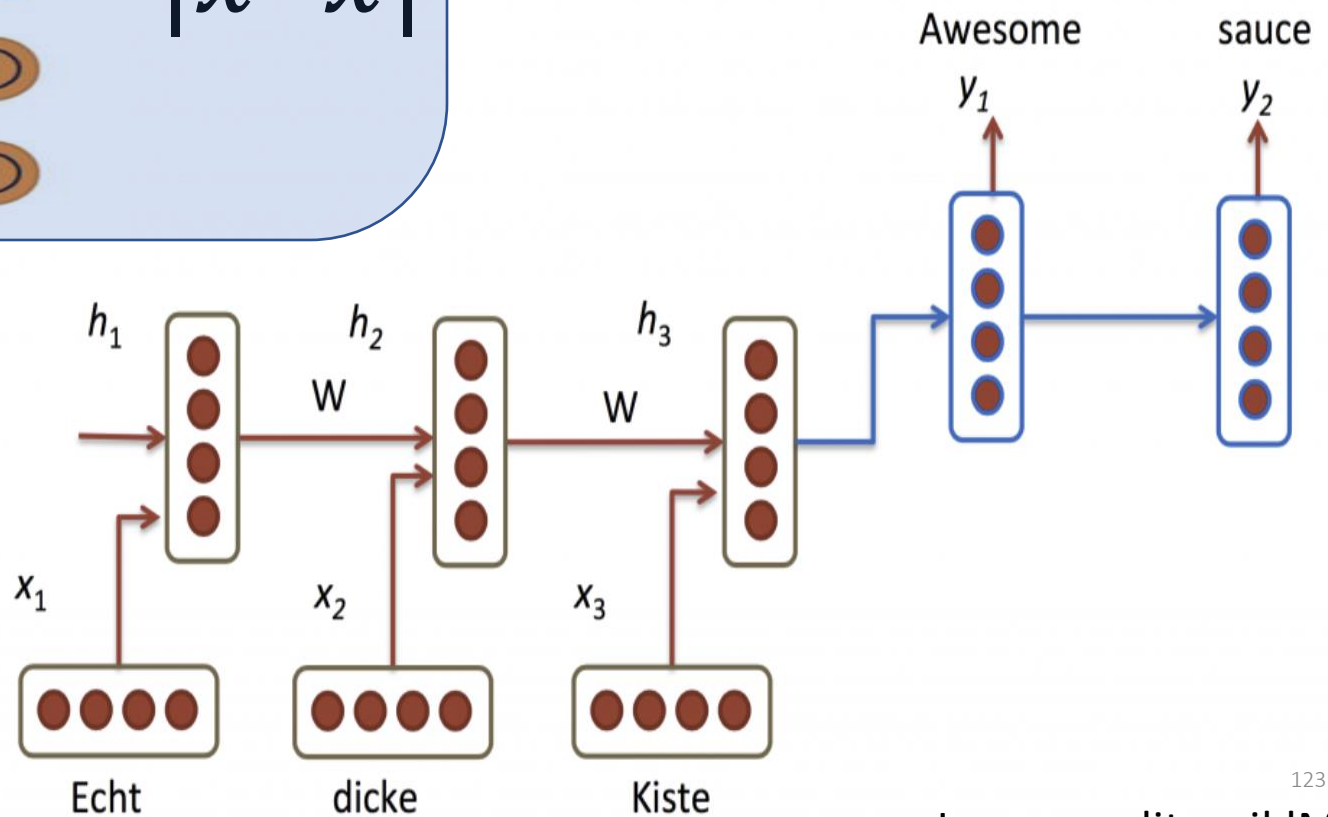
Seq2Seq

Auto Encoder (AE)

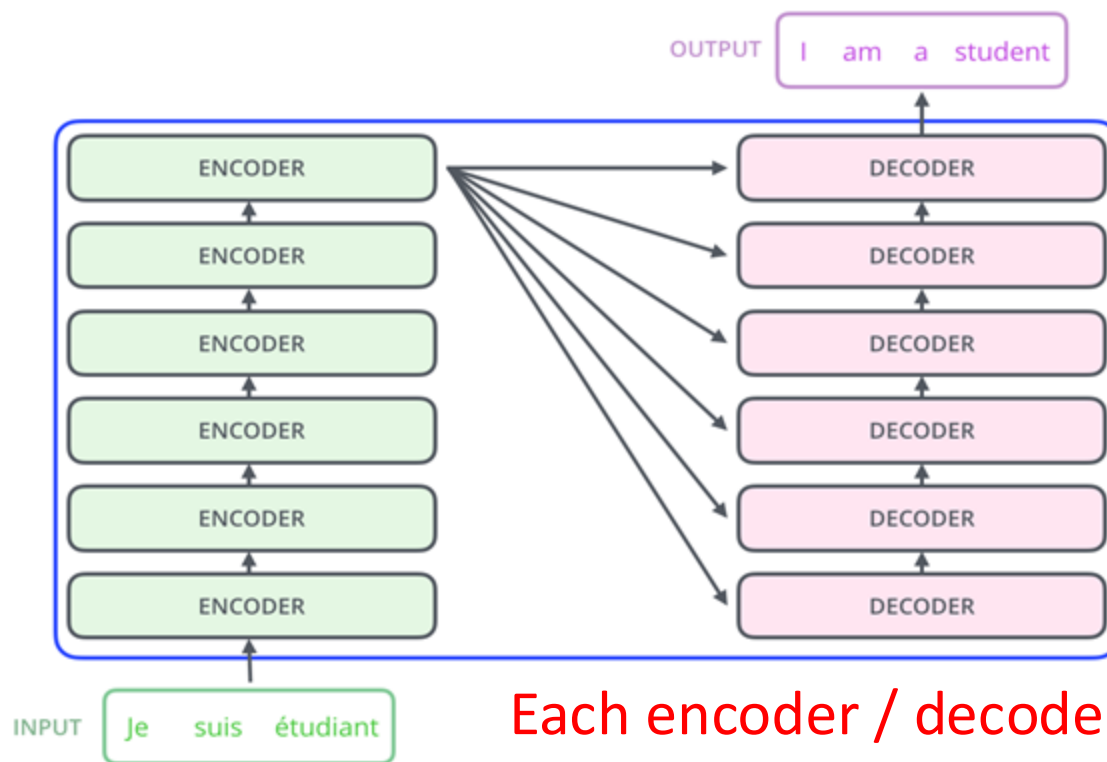


Minimize diff

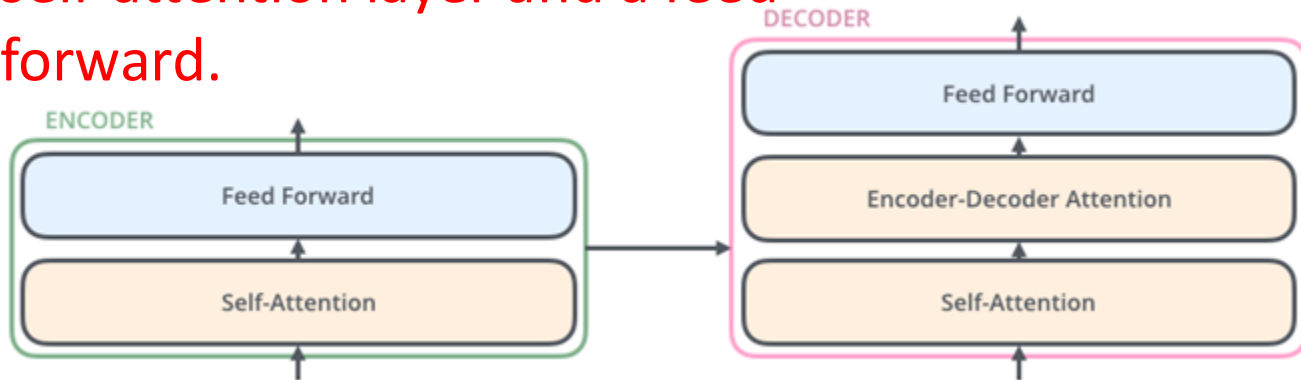
$$|\hat{\vec{x}} - \vec{x}|$$



Original Transformer is Seq2Seq model

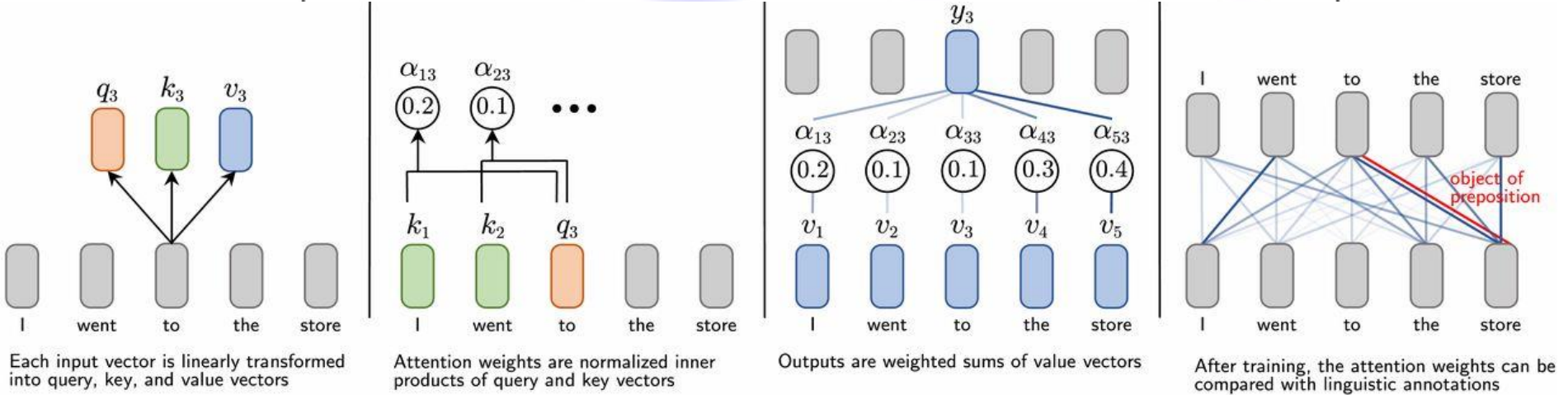


Each encoder / decoder layer has a self-attention layer and a feed forward.

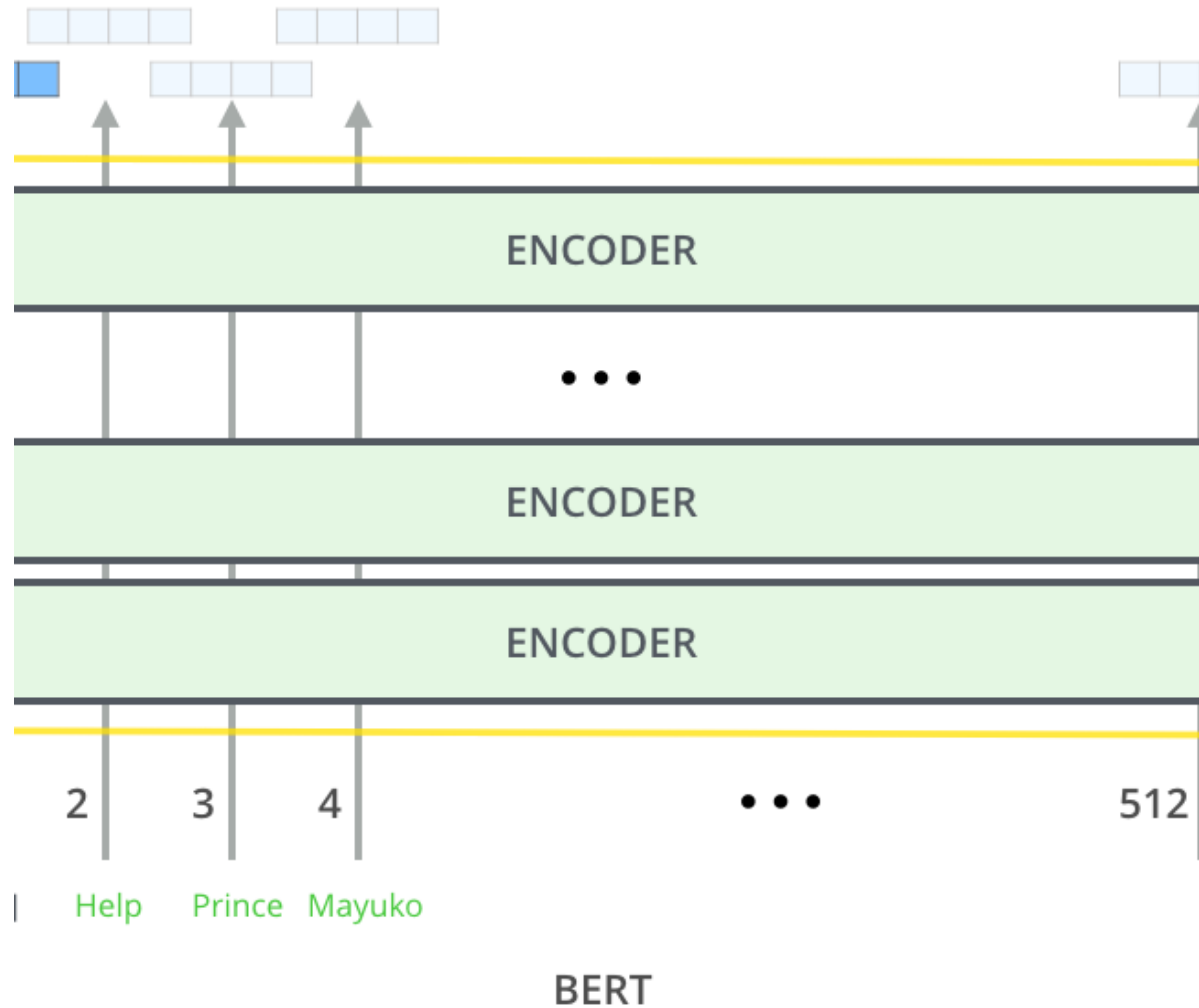


Self-attention creates attention layers mapping from a sequence to itself.

The FBI is chasing a criminal on the run .
The FBI is chasing a criminal on the run .
The FBI is chasing a criminal on the run .
The FBI is chasing a criminal on the run .
The FBI is chasing a criminal on the run .
The FBI is chasing a criminal on the run .
The FBI is chasing a criminal on the run .
The FBI is chasing a criminal on the run .
The FBI is chasing a criminal on the run .



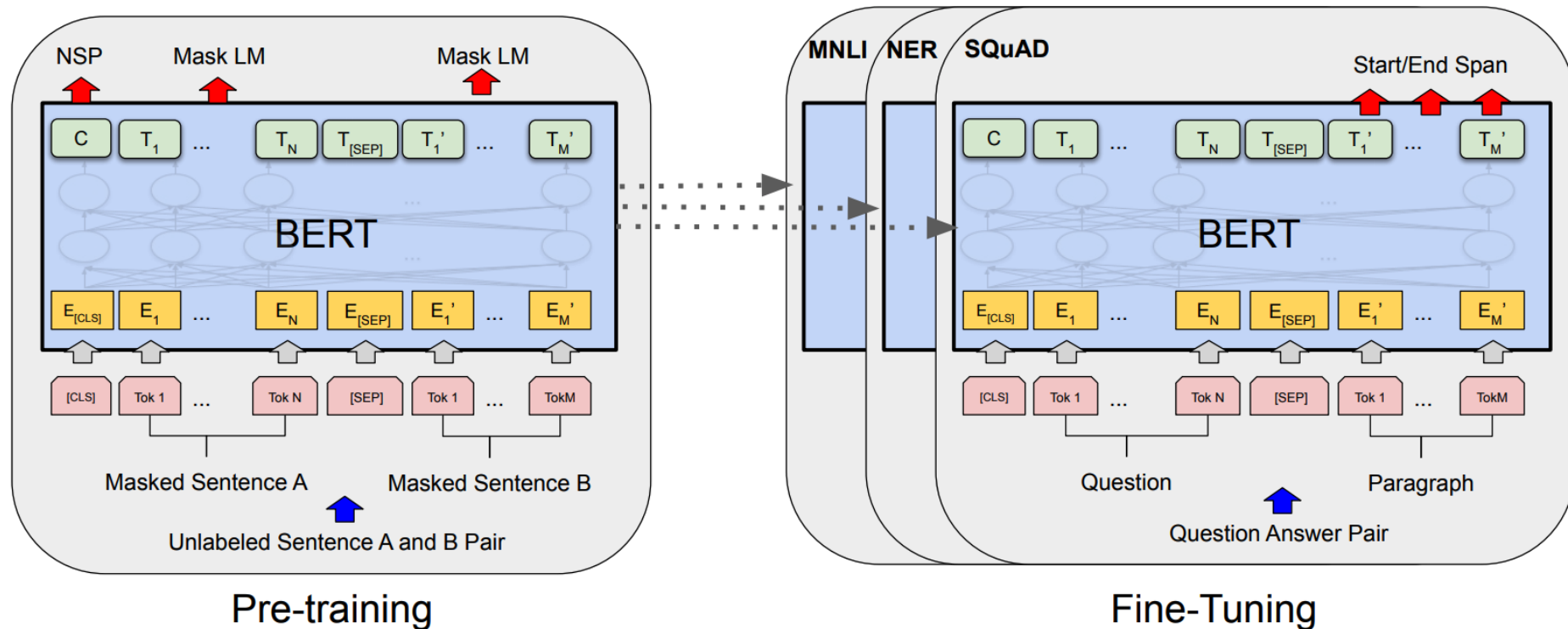
BERT: Bidirectional Encoder Representations from Transformers.



BERT's architecture is just a transformer's encoder stack.

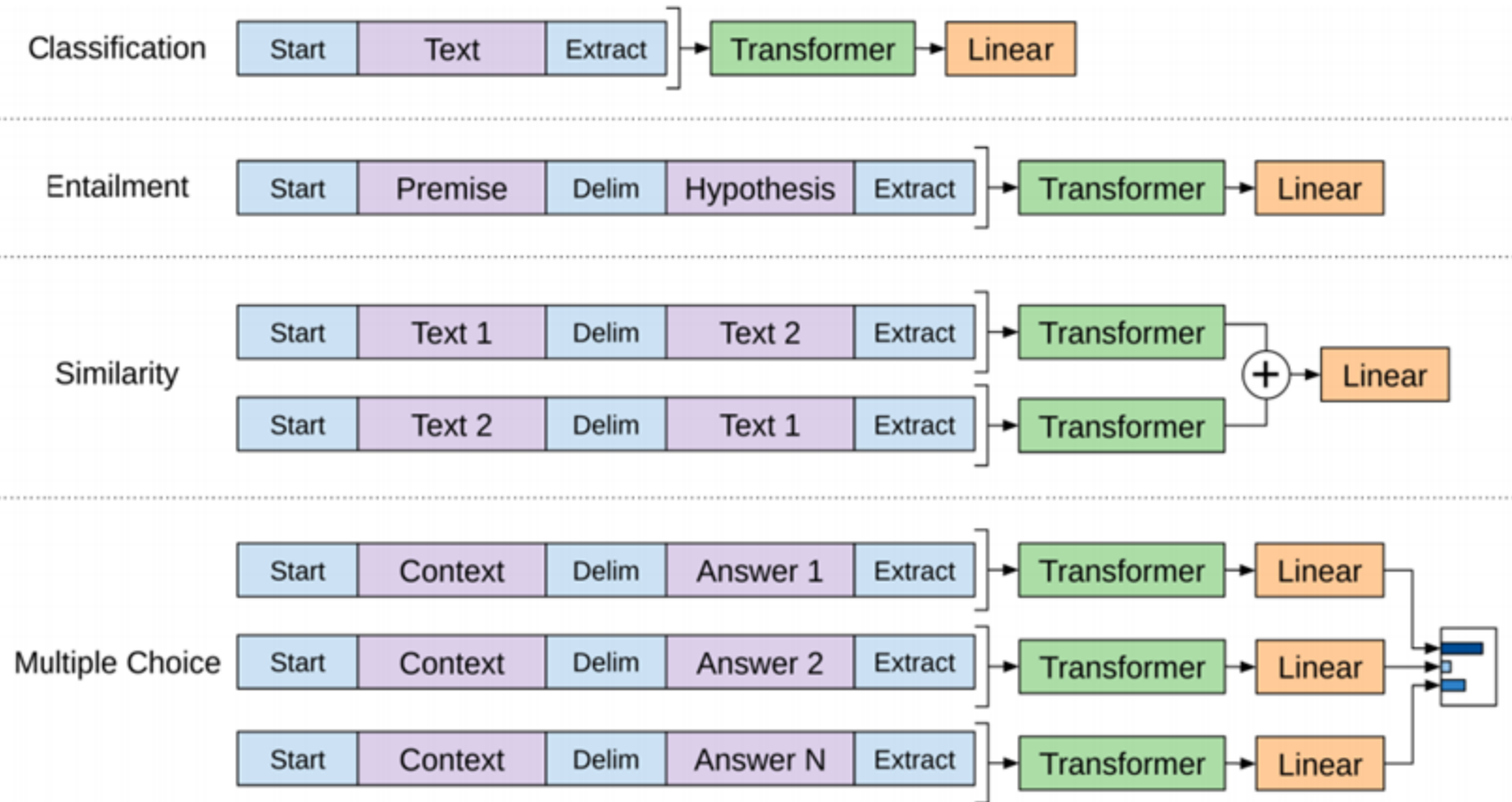
BERT : Pre-training of Deep Bidirectional Transformers for Language Understanding (NAACL 2019, Devlin et al.)

- Denoising Auto Encoder
 - [MASK]: a unique token introduced in the training process to mask some tokens
 - Predict masked tokens based on their context information,
 - Pre-train and fine-tune
- Intuition: representation should be robust to the introduction of noise
- Masked Language Model (MLM)



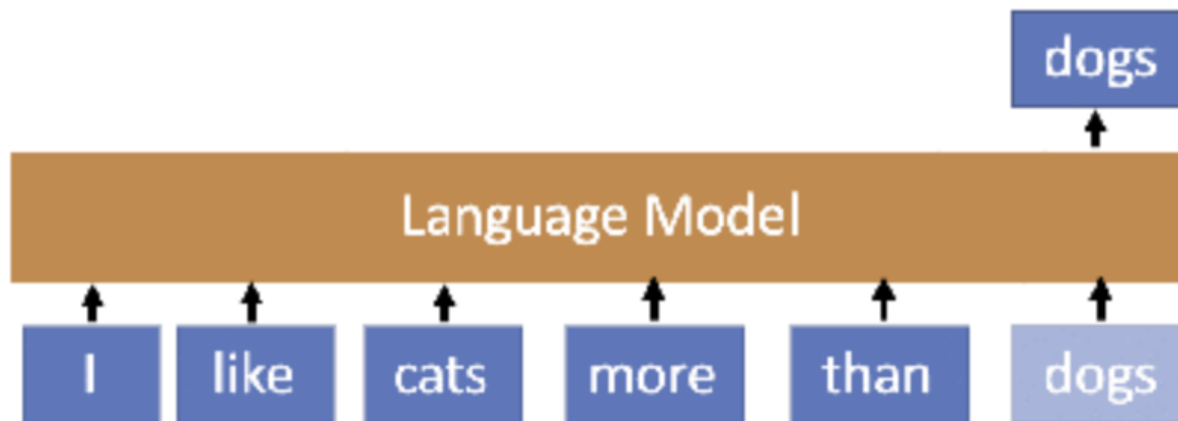
Open AI's GPT-2: 1.5 billion parameters! Trained on 8M pages from reddit

Can use pretrained GPT models for any task. Different tasks use the OpenAI transformer in different ways.



GPT: generative pre-training,

GPT 's architecture is just a transformer's decoder stack.



The prediction scheme for a traditional language model. Shaded words are provided as input to the model while unshaded words are masked out.

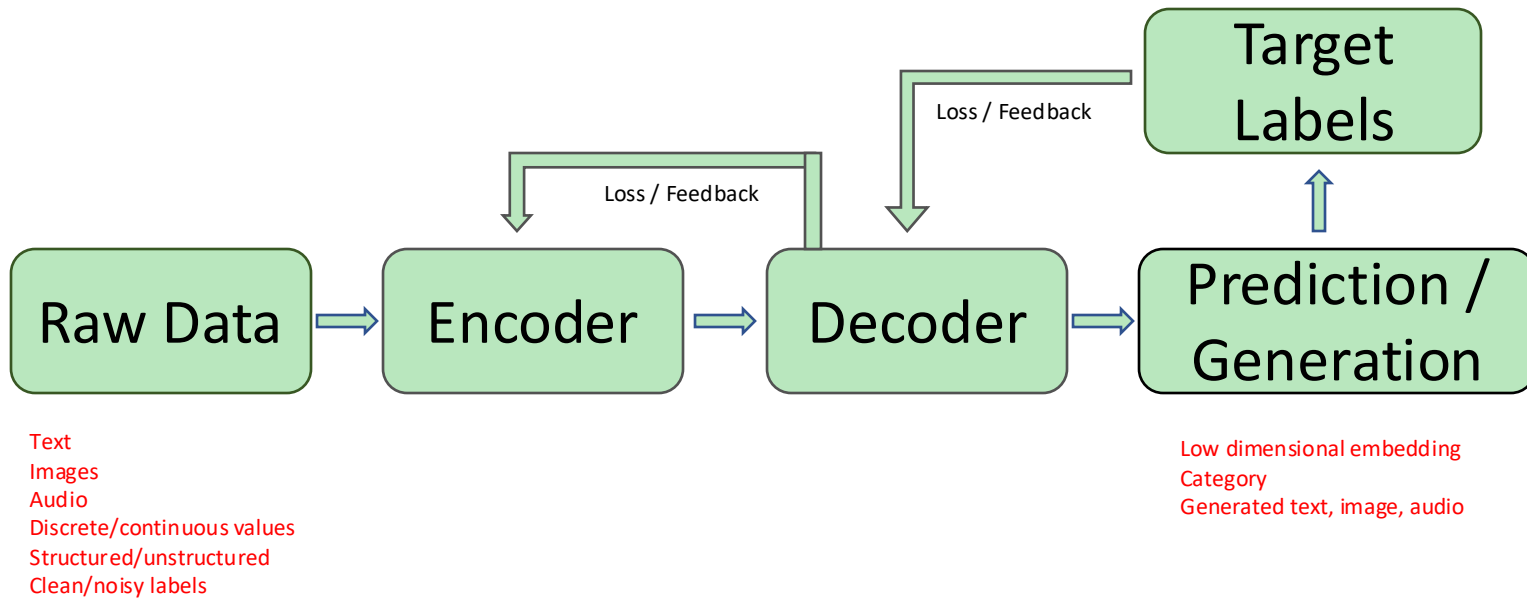
Autoregressive Models

GPT 's pretraining uses
next token prediction loss

$$P(x; \theta) = \prod_{n=1}^N P(x_n | x_{<n}; \theta)$$

- Each factor can be parametrized by θ , which can be shared.
- The variables can be arbitrarily ordered and grouped, as long as the ordering and grouping is consistent.

Summary:

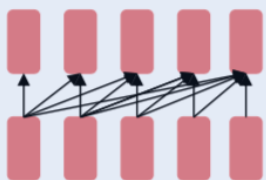


GPT: **Generative Pretraining Models** for Language

CLIP: **Contrastive Language-Image Pretraining** for Vision

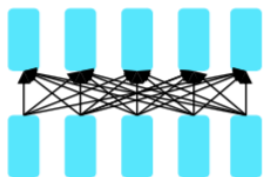
BERT: **Bidirectional Encoder Representations from Transformers**.

Background: Pretraining for three types of architectures



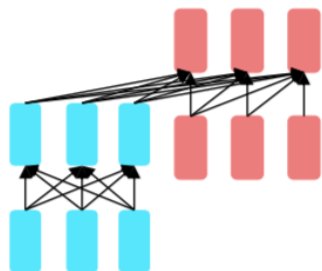
Decoders

- Nice to generate from; can't condition on future words
- **Examples:** GPT-2, GPT-3, LaMDA



Encoders

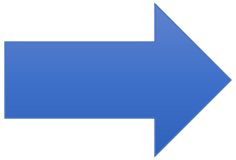
- Gets bidirectional context – can condition on future!
- Wait, how do we pretrain them?
- **Examples:** BERT and its many variants, e.g. RoBERTa



Encoder-Decoders

- Good parts of decoders and encoders?
- What's the best way to pretrain them?
- **Examples:** T5, Meena

S3: Lecture 16: Generative Bayes Classifiers



- ✓ Bayes Classifier (BC)

- Generative Bayes Classifier

- ✓ Naïve Bayes Classifier



- ✓ Gaussian Bayes Classifiers

- Gaussian distribution

- Naïve Gaussian BC

- Not-naïve Gaussian BC ➔ LDA, QDA

(2) Generative BC

$$\arg \max_{c \in C} P(c / \mathbf{X}), \mathcal{C} = \{c_1, \dots, c_L\}$$

$$P(\mathbf{X} | C)$$

$$\mathcal{C} = c_1, \dots, c_L, \mathbf{X} = (X_1, \dots, X_p)$$

$$P(\mathbf{x} | c_1)$$

Generative
Probabilistic Model
for Class 1

$$P(\mathbf{x} | c_2)$$

Generative
Probabilistic Model
for Class 2

...

$$P(\mathbf{x} | c_L)$$

Generative
Probabilistic Model
for Class L

x_1 x_2 ... x_p

x_1 x_2 ... x_p

x_1 x_2 ... x_p

$$\mathbf{X} = (x_1, x_2, \dots, x_p)$$

Review : Bayes' Rule – for Generative Bayes Classifiers

$$\arg \max_{c \in \mathcal{C}} P(c | \mathbf{X}), \mathcal{C} = \{c_1, \dots, c_L\}$$

$$P(C | X) = \frac{P(X | C)P(C)}{P(X)}$$

Prior

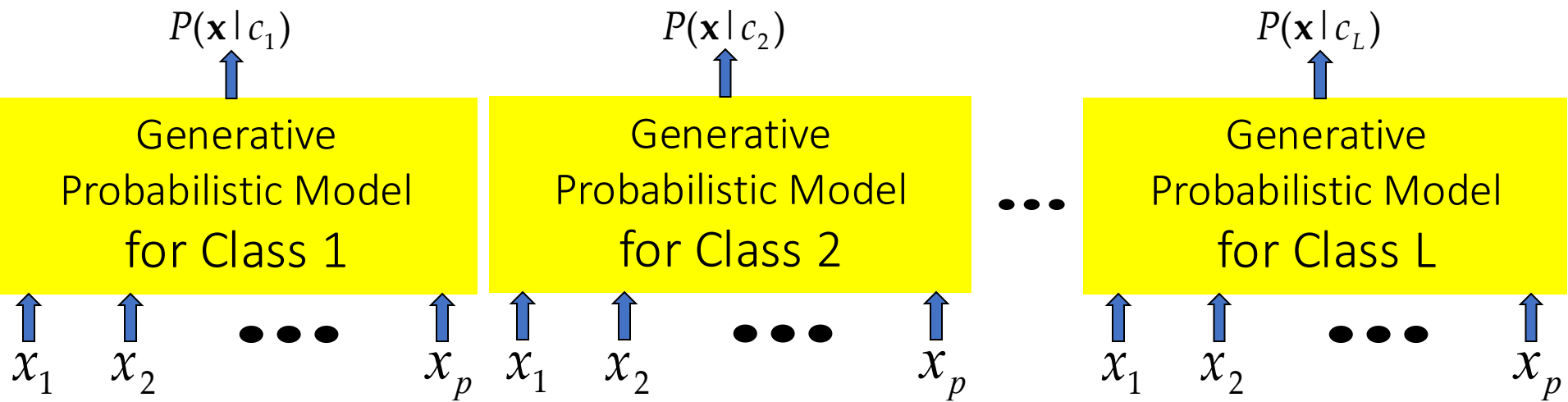
$$P(C_1|x), P(C_2|x), \dots, P(C_L|x)$$

$$P(C_1), P(C_2), \dots, P(C_L)$$

$$P(C_i | \mathbf{X}) = \frac{P(\mathbf{X} | C_i)P(C_i)}{P(\mathbf{X})}$$

Establishing a probabilistic model for classification through generative probabilistic models

$$\underset{C_i}{\operatorname{argmax}} P(C_i | X) = \underset{C_i}{\operatorname{argmax}} P(X, C_i) = \underset{C_i}{\operatorname{argmax}} P(X | C_i) P(C_i)$$



$$\mathbf{X} = (x_1, x_2, \boxed{}, \dots, x_p)$$

An Example

- Example: Play Tennis

PlayTennis: training examples

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny ✓	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain ✓	Mild	High	Weak	Yes
D5	Rain ✓	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

X_1

X_2

X_3

X_4

C

X_1	X_2	X_3	C

Generative Bayes Classifier:

- Learning Phase →

$$\underset{c_j \in \mathcal{C}}{\operatorname{argmax}} P(x_1, x_2, \dots, x_p | c_j) P(c_j)$$

$$P(C_1), P(C_2), \dots, P(C_L)$$

$$P(\text{Play}=\text{Yes}) = 9/14 \quad P(\text{Play}=\text{No}) = 5/14$$

$$P(X_1, X_2, \dots, X_p | C_1), P(X_1, X_2, \dots, X_p | C_2)$$

Outlook (3 values)	Temperature (3 values)	Humidity (2 values)	Wind (2 values)	Play=Yes	Play=No
sunny	hot	high	weak	0/9	1/5
sunny	hot	high	strong	.../9	.../5
sunny	hot	normal	weak	.../9	.../5
sunny	hot	normal	strong	.../9	.../5
....					
....					
....					
....					

$$3 \times 3 \times 2 \times 2 \text{ [conjunctions of attributes]} \times 2 \text{ [two classes]} = 72 \text{ parameters}$$

Generative Bayes Classifier:

- Testing Phase

- Given an unknown instance $\mathbf{X}'_{ts} = (a'_1, \dots, a'_p)$
- Look up tables to assign the label c^* to $\mathbf{X}'_{ts}$ if

Last Page:
the learned
model

$$\hat{P}(a'_1, \dots, a'_p | c^*) \hat{P}(c^*) > \hat{P}(a'_1, \dots, a'_p | c) \hat{P}(c), \forall c \in \mathcal{C}, c \neq c^*, \mathcal{C} = \{c_1, \dots, c_L\}$$

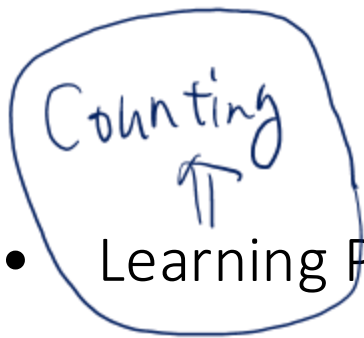
- Given a new instance,
 $\mathbf{x}' = (\text{Outlook}=\text{Sunny}, \text{Temperature}=\text{Cool}, \text{Humidity}=\text{High}, \text{Wind}=\text{Strong})$

Naïve Bayes Classifier

- Naïve Bayes classification
 - Assumption that all input attributes are conditionally independent!

[assumption]

$$P(X_1, X_2, \dots, X_p | C) = P(X_1 | C) P(X_2 | C) \cdots P(X_p | C)$$



- Learning Phase

Estimate $P(X_j = x_{jk} | C = c_i)$ with examples in training;

$P(X_2|C_1), P(X_2|C_2)$

Outlook	Play=Yes	Play=No
Sunny	2/9	3/5
Overcast	4/9	0/5
Rain	3/9	2/5

Temperature	Play=Yes	Play=No
Hot	2/9	2/5
Mild	4/9	2/5
Cool	3/9	1/5

Humidity	Play=Yes	Play=No
High	3/9	4/5
Normal	6/9	1/5

$P(X_4|C_1), P(X_4|C_2)$

Wind	Play=Yes	Play=No
Strong	3/9	3/5
Weak	6/9	2/5

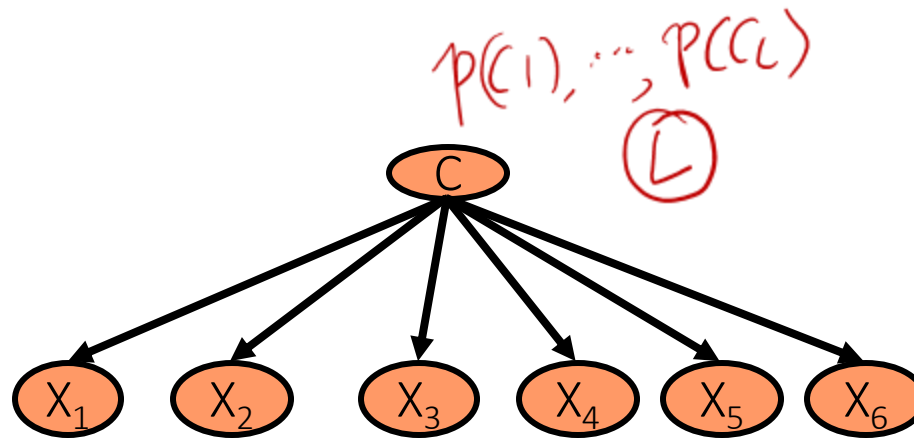
3+3+2+2 [naïve assumption] * 2 [two classes]= 20 parameters

$$P(\text{Play=Yes}) = 9/14$$

$$P(\text{Play=No}) = 5/14$$

$P(C_1), P(C_2), \dots, P(C_L)$

Learning (training) the NBC Model



- maximum likelihood estimates:
 - simply use the frequencies in the data

$$\hat{P}(c_j) = \frac{N(C = c_j)}{N}$$

$$\hat{P}(x_i | c_j) = \frac{N(X_i = x_i, C = c_j)}{N(C = c_j)}$$

S3: Lecture 16: Generative Bayes Classifiers

- Test Phase
$$[\hat{P}(a'_1 | c^*) \cdots \hat{P}(a'_p | c^*)] \hat{P}(c^*) > [\hat{P}(a'_1 | c) \cdots \hat{P}(a'_p | c)] \hat{P}(c)$$

- Given a new instance,

$x' = (\text{Outlook}=\text{Sunny}, \text{Temperature}=\text{Cool}, \text{Humidity}=\text{High}, \text{Wind}=\text{Strong})$

- Look up in conditional-prob tables

$P(\text{Outlook}=\text{Sunny} | \text{Play}=\text{Yes}) = 2/9$

$P(\text{Temperature}=\text{Cool} | \text{Play}=\text{Yes}) = 3/9$

$P(\text{Humidity}=\text{High} | \text{Play}=\text{Yes}) = 3/9$

$P(\text{Wind}=\text{Strong} | \text{Play}=\text{Yes}) = 3/9$

$P(\text{Play}=\text{Yes}) = 9/14$

$P(\text{Outlook}=\text{Sunny} | \text{Play}=\text{No}) = 3/5$

$P(\text{Temperature}=\text{Cool} | \text{Play}=\text{No}) = 1/5$

$P(\text{Humidity}=\text{High} | \text{Play}=\text{No}) = 4/5$

$P(\text{Wind}=\text{Strong} | \text{Play}=\text{No}) = 3/5$

$P(\text{Play}=\text{No}) = 5/14$

- MAP rule

$P(\text{Yes} | x') : [P(\text{Sunny} | \text{Yes})P(\text{Cool} | \text{Yes})P(\text{High} | \text{Yes})P(\text{Strong} | \text{Yes})]P(\text{Play}=\text{Yes}) = 0.0053$

$P(\text{No} | x') : [P(\text{Sunny} | \text{No})P(\text{Cool} | \text{No})P(\text{High} | \text{No})P(\text{Strong} | \text{No})]P(\text{Play}=\text{No}) = 0.0206$



Given the fact $P(\text{Yes} | x') < P(\text{No} | x')$, we label x' to be “No”.

Summary:

Generative Bayes Classifier

Task: Classify a new instance X based on a tuple of attribute values $X = \langle X_1, X_2, \dots, X_p \rangle$ into one of the classes

$$c_{MAP} = \operatorname{argmax}_{c_j \in C} P(c_j \mid x_1, x_2, \dots, x_p)$$

$$= \operatorname{argmax}_{c_j \in C} \frac{P(x_1, x_2, \dots, x_p \mid c_j) P(c_j)}{P(x_1, x_2, \dots, x_p)}$$

$$= \operatorname{argmax}_{\substack{c_j \in C \\ j=1,2,\dots,L}} \underbrace{P(x_1, x_2, \dots, x_p \mid c_j)} \underbrace{P(c_j)}$$

MAP = Maximum A Posteriori

WHY ? Naïve Bayes Assumption

- 
- $P(c_j)$
 - Can be estimated from the frequency of classes in the training examples.
 - $P(x_1, x_2, \dots, x_p / c_j)$
 - $O(|X_1| \cdot |X_2| \cdot |X_3| \dots |X_p| \cdot |C|)$ parameters
 - Could only be estimated if a very, very large number of training examples was available.

- 
- $P(x_k / c_j)$
 - $O(|X_1| + |X_2| + |X_3| \dots + |X_p| \cdot |C|)$ parameters
 - Assume that the probability of observing the conjunction of attributes is equal to the product of the individual probabilities $P(x_i / c_j)$.

Smoothing to Avoid Overfitting

$$\hat{P}(x_i | c_j) = \frac{N(X_i = x_i, C = c_j) + 1}{N(C = c_j) + k_i}$$

□

of values of X_i

- Somewhat more subtle version

overall fraction in data
where $X_i = x_{i,k}$

$$\hat{P}(x_{i,k} | c_j) = \frac{N(X_i = x_{i,k}, C = c_j) + mp_{i,k}}{N(C = c_j) + m}$$

$\rightarrow k \in \{1, 2, \dots, k_i\}$

extent of
“smoothing”



11/06

Today Roadmap

- 1. TA Kefang going over the HW3 solutions
- 2. Prof. Qi to quickly review some course content
 - 2a. the LLM-advanced-survey slide deck ([URL](#))
 - 2b. the NBC for text + SVM basics (move to 11/11, due to project meetings)
- 3. Quiz 10



11/11

Today Roadmap

- 0. A few announcements:
 - HW4 / HW5 / Project due time
 - Survey 3 / Nov.25th class zoom or in-person?
- 1. Prof. Qi to quickly review some course content
 - A. the NBC for text + SVM basics
 - B. the 10 advanced topics on deep learning (TBD)
- 2. Quiz 11

A few announcements

- HW4 / HW5 / Project due time
 - HW4 due moves to: Nov. 15th midnight
 - HW5 due moves to: Dec. 1st midnight
 - Project deliverables:
 - Slide deck and code to be submitted by Dec. 16th midnight
 - Final presentation dates moves to zoom on Dec. 11th / 12th / 15th / 16th (every team has a 25mins session, so a total of 12 hours online sessions --- these sessions are open to everybody in the class!!!)
- Added two more office hours (4 from TA, 1 from me)
- Final Exam: Dec. 9th in class exam
 - We will host course review sessions on Dec. 2nd and 4th
- Survey 3 needs your inputs on:
 - Nov.25th class zoom or in-person?
 - Overall satisfaction of the course experience so far..

S3: Lecture 17: ~~Generative Bayes Classifiers~~ →

Naïve Bayes Classifier for Text Classification

Review: Naïve Bayes Classifier

$$\operatorname{argmax}_C P(C | X) = \operatorname{argmax}_C P(X, C) = \operatorname{argmax}_C P(X | C)P(C)$$

Naïve
Bayes
Classifier

$$P(X_1, X_2, \dots, X_p | C) = P(X_1 | C)P(X_2 | C) \cdots P(X_p | C)$$

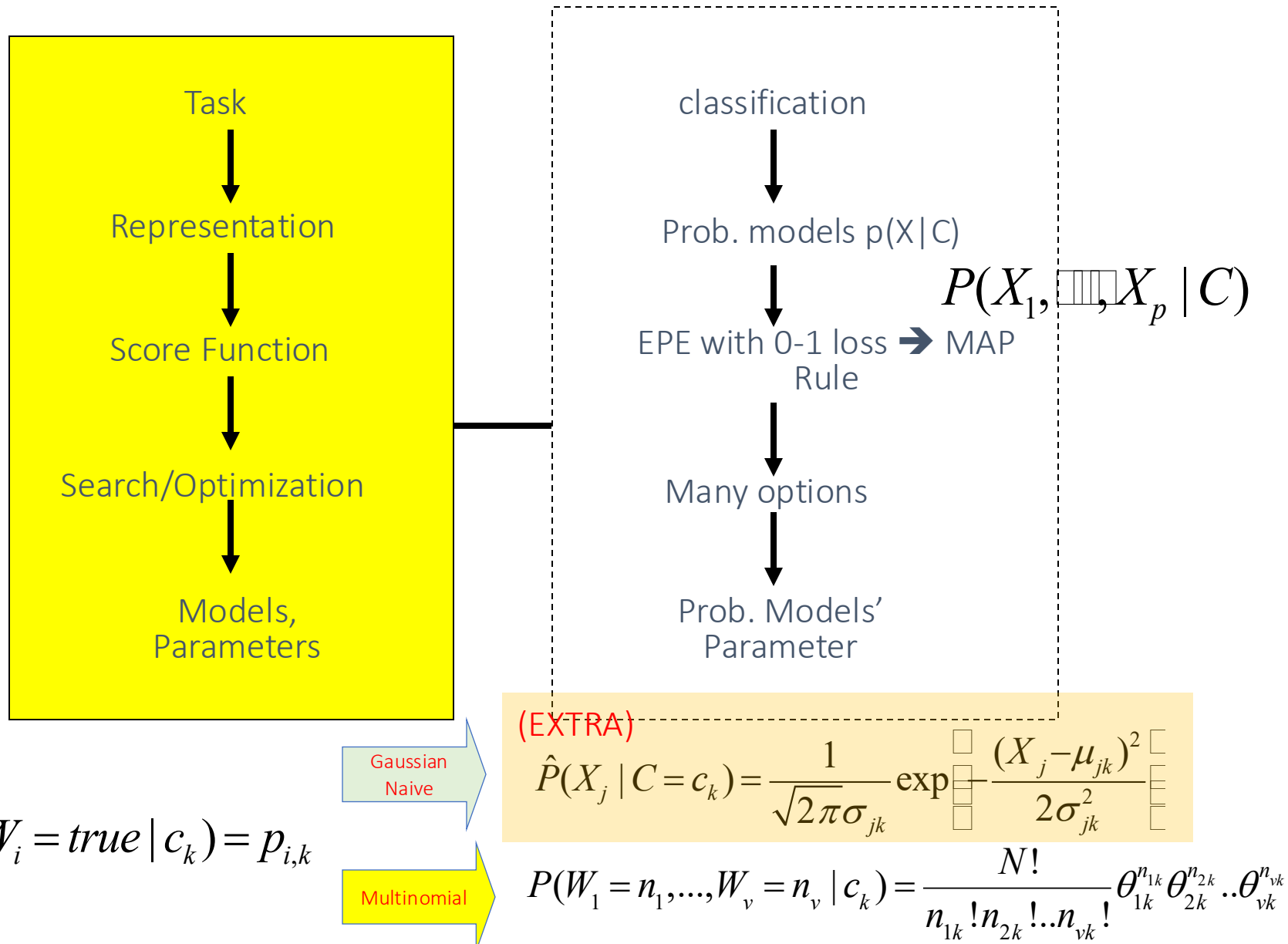
$$c^* = \operatorname{argmax} P(C = c_i | \mathbf{X} = \mathbf{x}) \propto P(\mathbf{X} = \mathbf{x} | C = c_i)P(C = c_i)$$

for $i = 1, 2, \dots, L$

Assuming all input
attributes are
conditionally
independent given a
specific class label!

$$\operatorname{argmax}_k P(C = k | X) = \operatorname{argmax}_k P(X, C) = \operatorname{argmax}_k P(X | C) P(C)$$

Generative Bayes Classifiers



Bernoulli Naïve

$$p(W_i = \text{true} | c_k) = p_{i,k}$$

L17 : Naïve Bayes Classifier for Text

- ✓ Dictionary based Vector space representation of text article
- ✓ Multivariate Bernoulli vs. Multinomial
- ✓ Multivariate Bernoulli naïve Bayes classifier
 - Testing
 - Training With Maximum Likelihood Estimation for estimating parameters
- ✓ Multinomial naïve Bayes classifier
 - Testing
 - Training With Maximum Likelihood Estimation for estimating parameters
 - Multinomial naïve Bayes classifier as Conditional Stochastic Language Models (Extra)

The bag of words representation

$f(\text{I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet.}) = C$

The bag of words representation

$$f(\text{table}) = C$$

great	2
love	2
recommend	1
laugh	1
happy	1
...	...

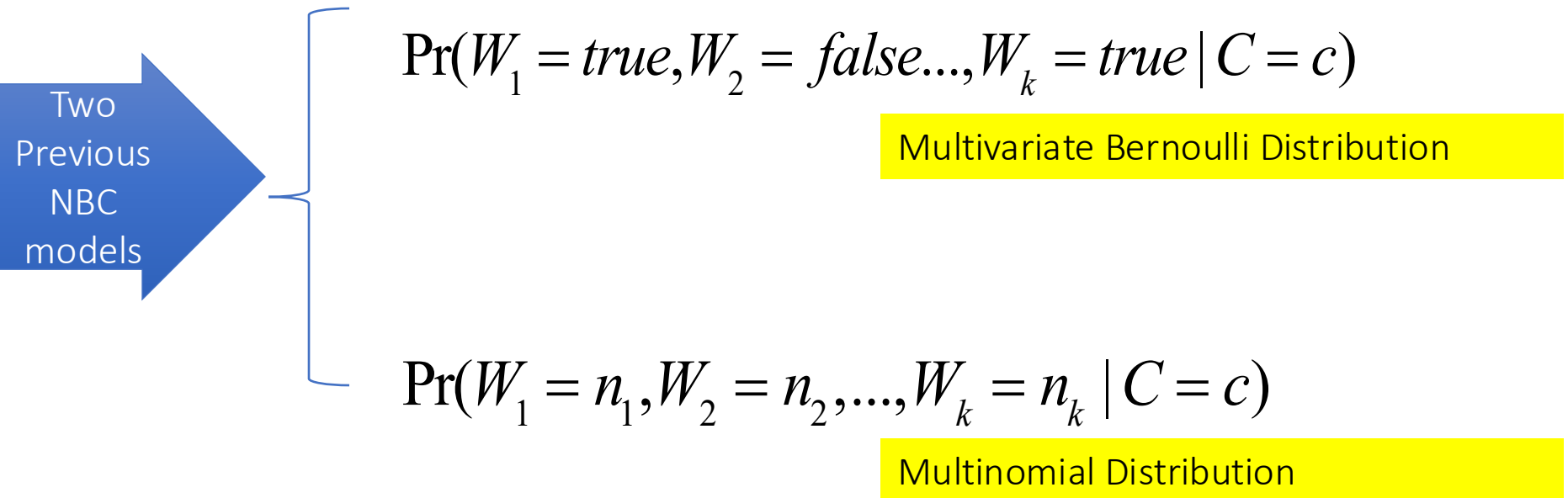
Common refinements: **remove stopwords**, **stemming**, collapsing multiple occurrences of words into one....

Unknown Words

- How to handle words in the **test** corpus that did not occur in the training data, i.e. out of vocabulary (OOV) words?
- Train a model that includes an **explicit** symbol for an unknown word (<**UNK**>).
 - Choose a vocabulary in advance and replace **other** (i.e. not in vocabulary) words in the corpus with <**UNK**>.
 - Very often, <UNK> also used to replace **rare** words

Naïve Probabilistic Models of text documents

$$\Pr(D \mid C = c) =$$



Model 1: Multivariate Bernoulli

- Model 1: Multivariate Bernoulli
 - For each word in a dictionary, feature
 - $X_w = \text{true}$ in document d if w appears in d
- Naive Bayes assumption:
 - Given the document's topic class label, appearance of one word in the document tells us nothing about chances that another word appears

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet.



word	Boolean
great	Yes
love	Yes
recommend	Yes
laugh	Yes
happy	Yes
hate	No

X_w

Model 2: Multinomial Naïve Bayes

- 'Bag of words' representation of text

word	frequency
great	2
love	2
recommend	1
laugh	1
happy	1
...	.

$$\Pr(W_1 = n_1, \dots, W_k = n_k \mid C = c)$$

Can be represented as a multinomial distribution.

Words = like colored balls, there are K possible type of them (i.e. from a dictionary of K words)

A Document = contains N words,
each word occurs n_i times (like a bag of N colored balls)

Training: Parameter estimation

Multinomial model:

$$\hat{P}(X_{\underline{i}} = w_{\underline{i}} | c_j) =$$

(Handwritten red annotations: an arrow points from θ_{w_i, c_j} to w_i in the equation)

fraction of times in which each dictionary word w appears across all documents of class c_j

- Can create a mega-document for class j by concatenating all documents on this class,
- Use frequency of w in mega-document

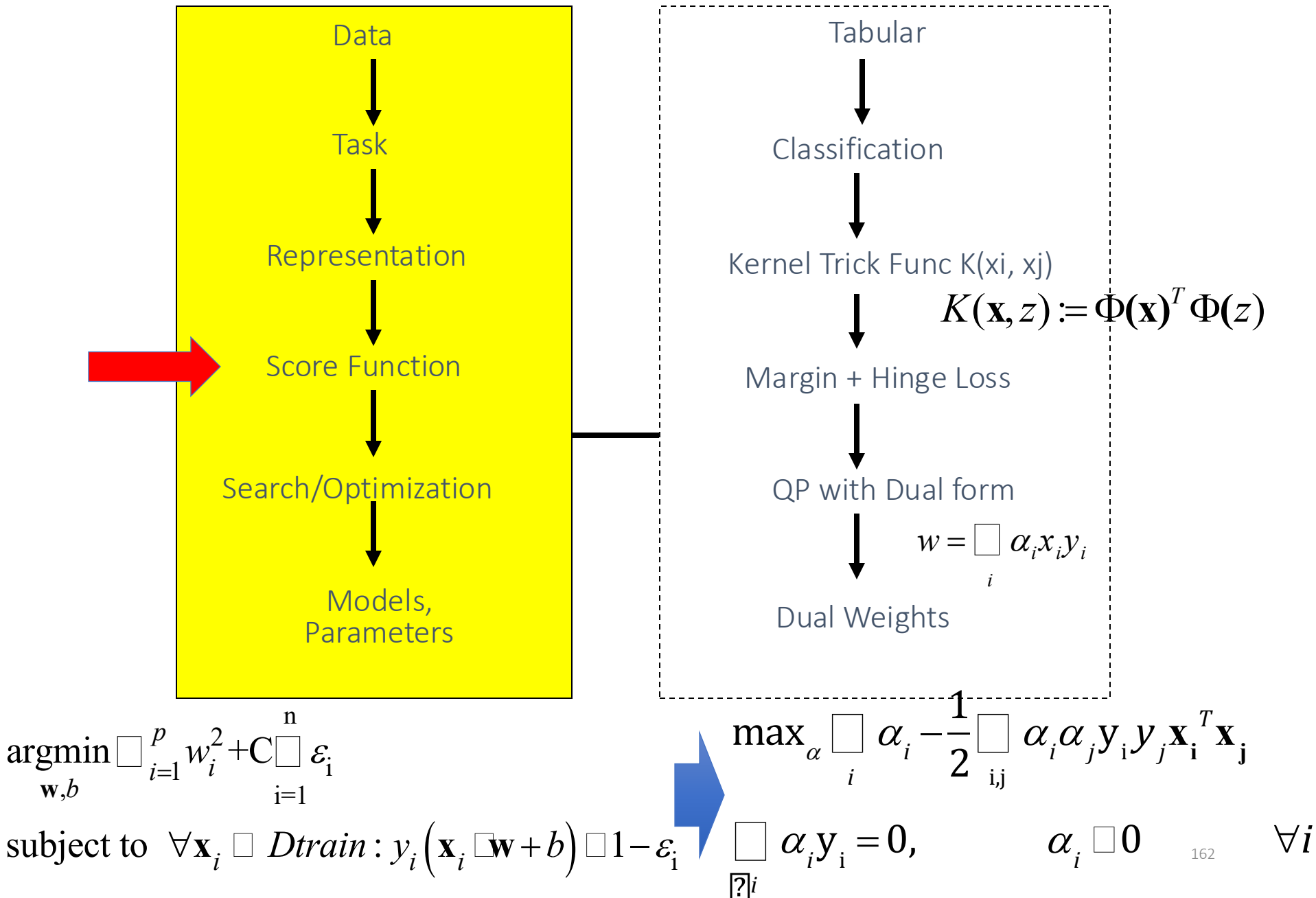
Model 2: Multinomial Naïve Bayes

- 'Bag of words' – TESTING Stage

word	frequency
great	2
love	2
recommend	1
laugh	1
happy	1
...	.

$$\arg \max_c P(W_1 = n_1, \dots, W_k = n_k, c)$$
$$= \arg \max_c \{ p(c) * \theta_{1,c}^{n_1} \theta_{2,c}^{n_2} \dots \theta_{k,c}^{n_k} \}$$

L20: Basic Support Vector Machine

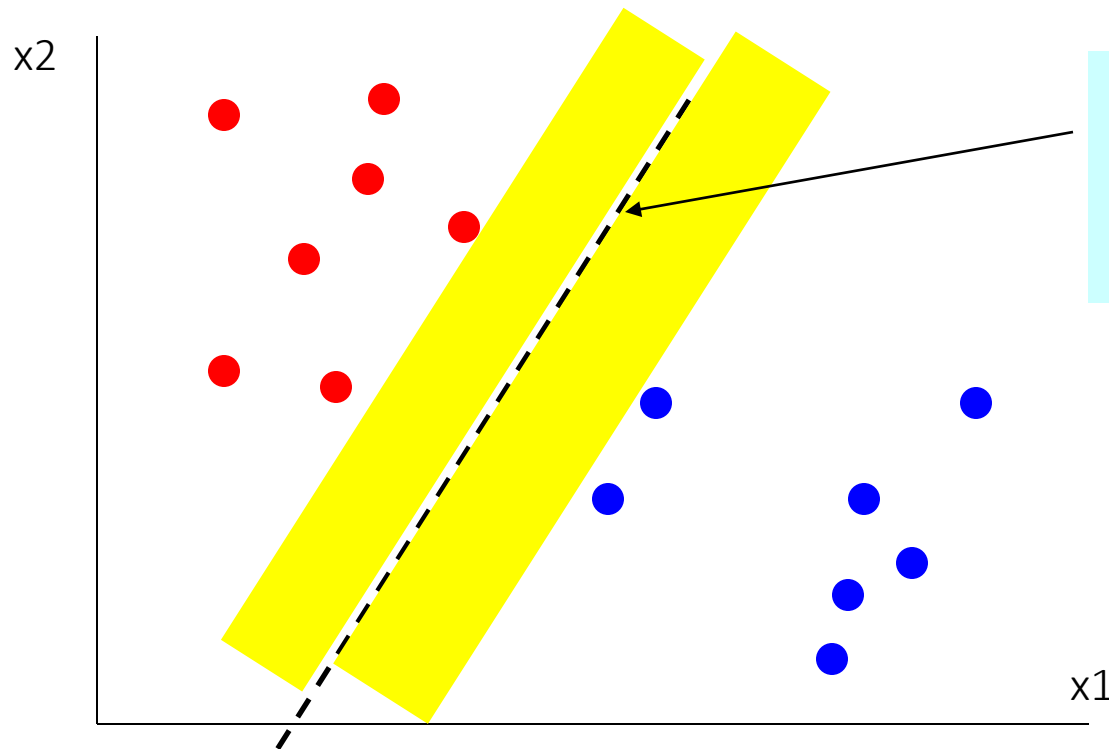


MNIST

Classifier	Test Error Rate (%)	References
Linear classifier (1-layer neural net)	12.0	LeCun et al. (1998)
K-nearest-neighbors, Euclidean (L2)	5.0	LeCun et al. (1998)
2-Layer neural net, 300 hidden units, mean square error	4.7	LeCun et al. (1998)
Support vector machine, Gaussian kernel	1.4	MNIST Website
Convolutional net, LeNet-5 (no distortions)	0.95	LeCun et al. (1998)
Methods using distortions		
Virtual support vector machine, deg-9 polynomial, (2-pixel jittered and deskewing)	0.56	DeCoste and Scholkopf (2002)
Convolutional neural net (elastic distortions)	0.4	Simard, Steinkraus, and Platt (2003)
6-Layer feedforward neural net (on GPU) (elastic distortions)	0.35	Ciresan, Meier, Gambardella, and Schmidhuber (2010)
Large/deep convolutional neural net (elastic distortions)	0.35	Ciresan, Meier, Masci, Maria Gambardella, and Schmidhuber (2011)
Committee of 35 convolutional networks (elastic distortions)	0.23	Ciresan, Meier, and Schmidhuber (2012)

Max margin classifiers

- Instead of fitting all points, focus on boundary points
- Learn a boundary that leads to the largest margin from both sets of points

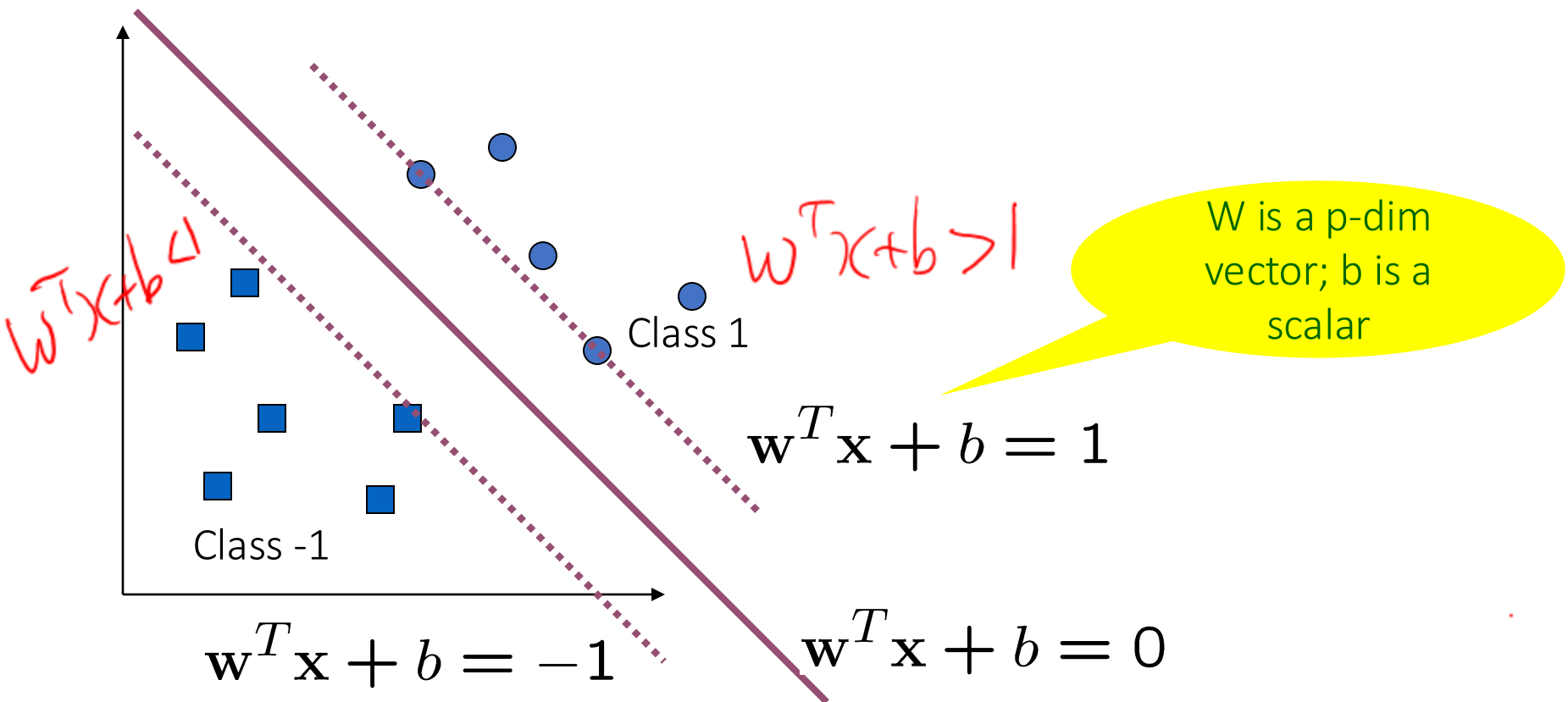


From all the possible boundary lines, this leads to the largest margin on both sides

$$f(x, \mathbf{w}, b) = \text{sign}(\mathbf{w}^T \mathbf{x} + b)$$

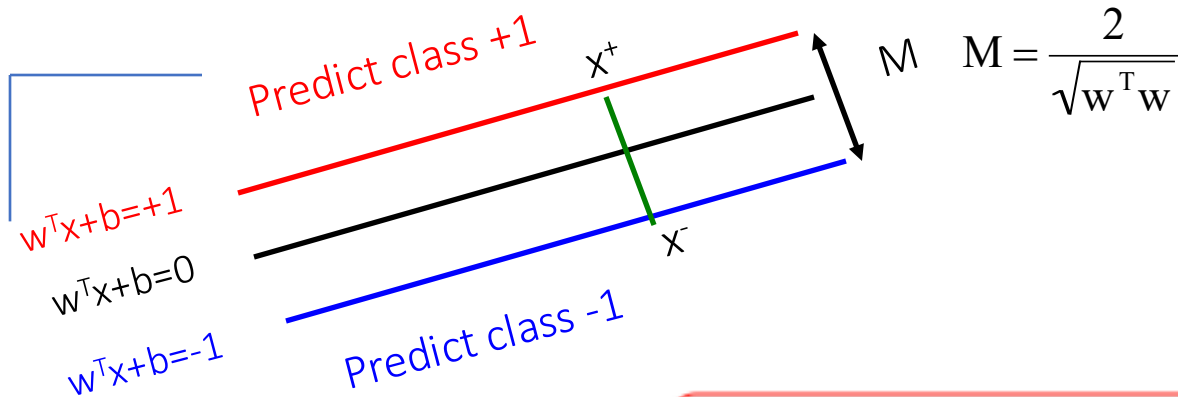
Max-margin & Decision Boundary $\mathbf{w}^T \mathbf{x} + b = 0$

- The decision boundary should be as far away from the data of both classes as possible



Optimization Step

i.e. learning optimal parameter for SVM



1. Correctly classifies all points
2. Maximizes the margin (or equivalently minimizes $w^T w$)

$\left\{ \begin{array}{l} \text{Min } (w^T w)/2 \\ \text{subject to the following constraints:} \end{array} \right.$

$\left\{ \begin{array}{l} \text{For all } x \text{ in class } +1 \\ w^T x + b \geq 1 \\ \text{For all } x \text{ in class } -1 \\ w^T x + b \leq -1 \end{array} \right.$

A total of n constraints if we have n training samples

Optimization Reformulation

$$f(x, w, b) = \text{sign}(w^T x + b)$$

1. Correctly classifies all points
2. Maximizes the margin (or equivalently minimizes $w^T w$)

$$\text{Min } (w^T w)/2$$

subject to the following constraints:

For all x in class + 1

$$w^T x + b \geq 1$$

For all x in class - 1

$$w^T x + b \leq -1$$



A total of n constraints if we have n input samples



$$\underset{w, b}{\text{argmin}} \sum_{i=1}^p w_i^2$$

$$\text{subject to } \forall \mathbf{x}_i \in D_{\text{train}}: y_i (\mathbf{w}^T \mathbf{x}_i + b) \geq 1$$

$$y_i \in \{+1, -1\}$$

Quadratic Objective

$$\underbrace{w^T x}_{|x|} \underbrace{+ b}_{|x|} \underbrace{\geq 1}_{|x|}$$

Quadratic programming i.e.,

- Quadratic objective
- Linear constraints

Two optimization problems: For the **separable** and **non separable** cases

$$\text{Min } (w^T w)/2$$

For all x in class + 1

$$w^T x + b \geq 1$$

For all x in class - 1

$$w^T x + b \leq -1$$

$$\min_w \frac{w^T w}{2} + C \sum_{i=1}^n \varepsilon_i$$

For all x_i in class + 1

$$w^T x_i + b \geq 1 - \varepsilon_i$$

For all x_i in class - 1

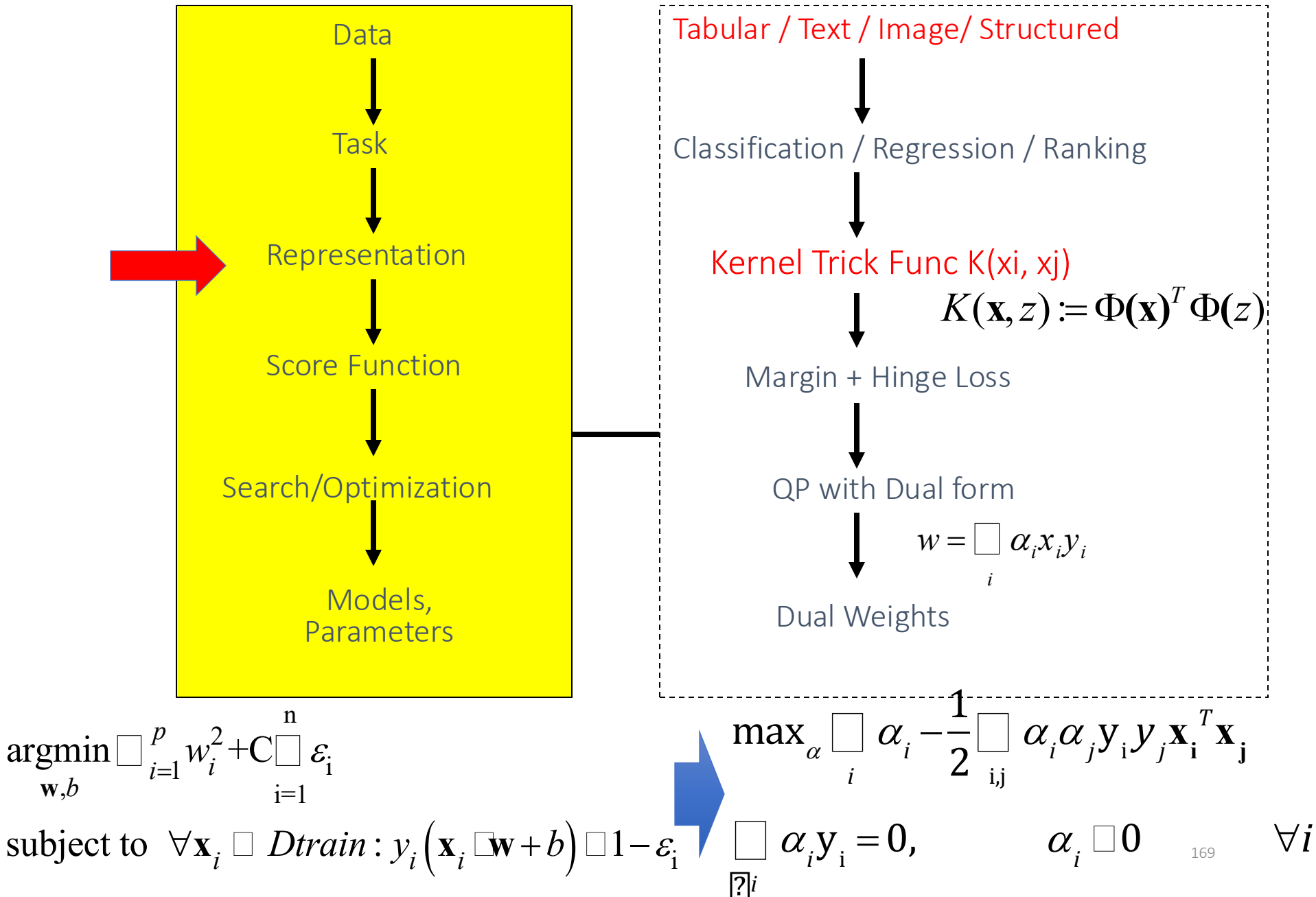
$$w^T x_i + b \leq -1 + \varepsilon_i$$

For all i

$$\varepsilon_i \geq 0$$

- Instead of solving these QPs directly we will solve a **dual formulation of the SVM optimization problem**
- The main reason for switching to this type of representation is that it would allow us to use a **neat trick that will make our lives easier (and the run time faster)**

L21: Kernel Support Vector Machine

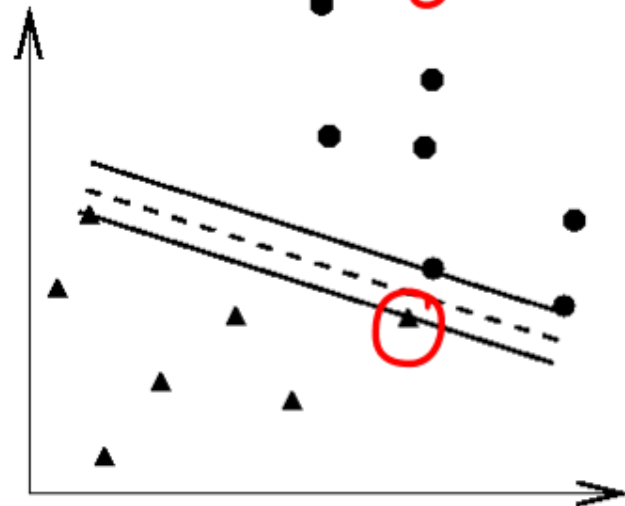


Model Selection, find right C

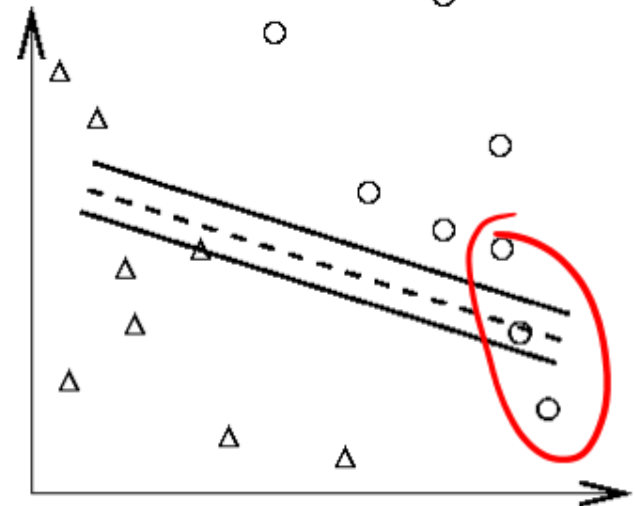
Training

Test

large C



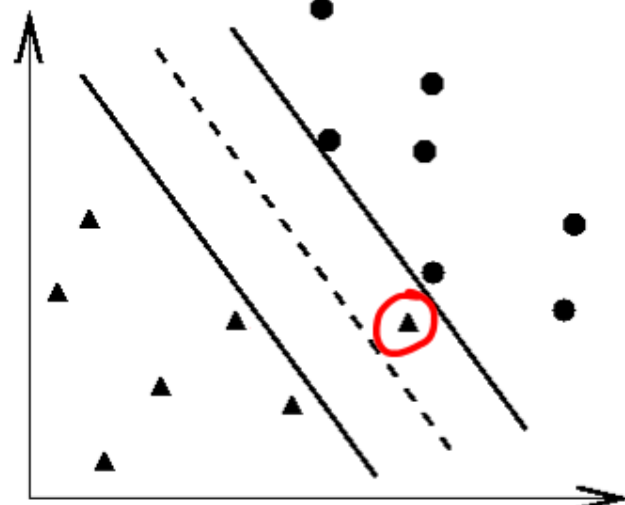
(a) Training data and an overfitting classifier



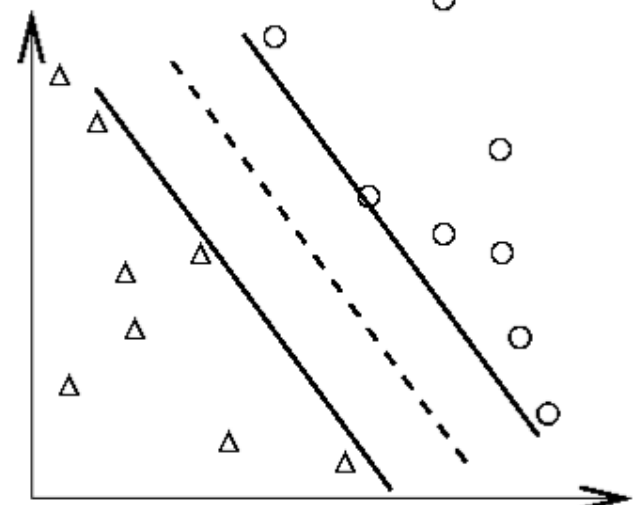
(b) Applying an overfitting classifier on testing data

Select the right penalty parameter C

small C



(c) Training data and a better classifier



(d) Applying a better classifier on testing data

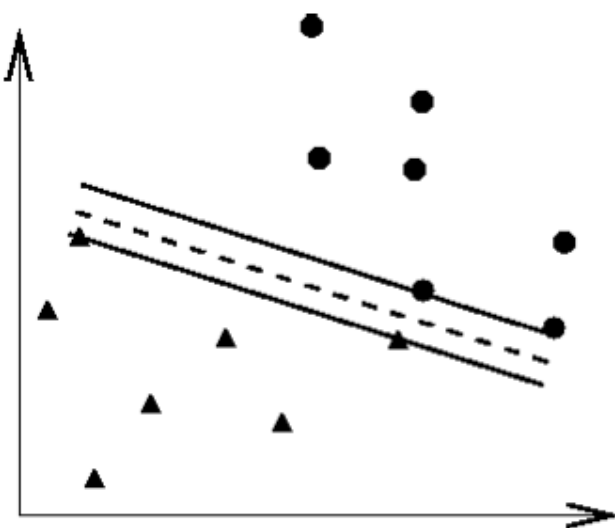
Model Selection, find right C

large C

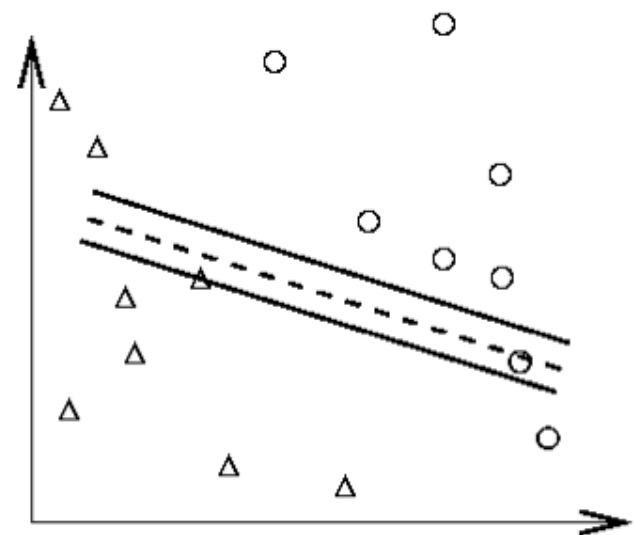
A large value of C means that misclassifications are bad - resulting in smaller margins and less training error (but more expected true error).

may lead

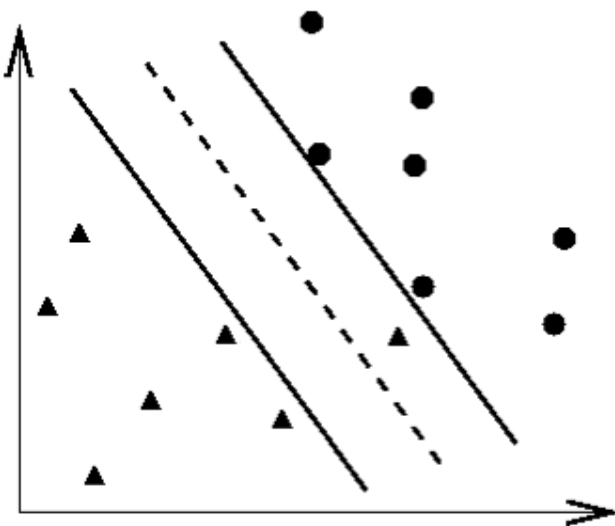
small C



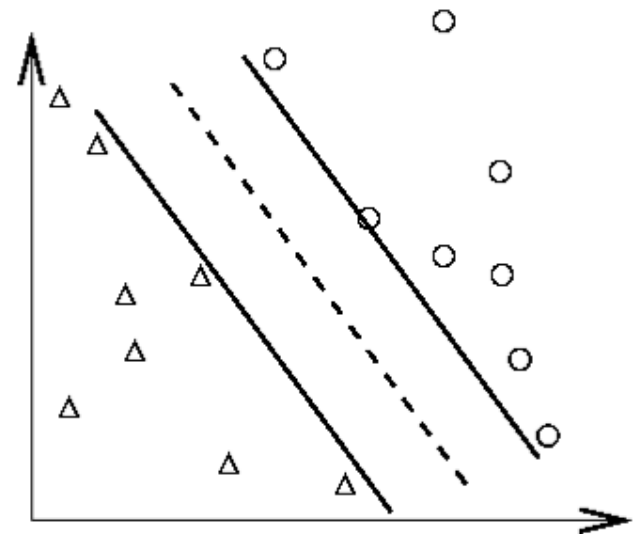
(a) Training data and an overfitting classifier



(b) Applying an overfitting classifier on testing data



(c) Training data and a better classifier



(d) Applying a better classifier on testing data

Hinge Loss for Soft SVM

$$\sum_{i=1}^n \varepsilon_i$$


$$\min_w \frac{\mathbf{w}^T \mathbf{w}}{2} + C \sum_{i=1}^n \varepsilon_i$$

For all $\mathbf{x}_i$ in class + 1

$$\mathbf{w}^T \mathbf{x}_i + b \geq 1 - \varepsilon_i$$

For all $\mathbf{x}_i$ in class - 1

$$\mathbf{w}^T \mathbf{x}_i + b \leq -1 + \varepsilon_i$$

For all i

$$\varepsilon_i \geq 0$$

soft

$$\sum_{i=1}^n \max(0, 1 - y_i f(\mathbf{x}_i))$$

$$\begin{aligned} & \operatorname{argmin}_{\mathbf{w}, b} \frac{\mathbf{w}^T \mathbf{w}}{2} + C \sum_{i=1}^n \max(0, 1 - y_i (\mathbf{w}^T \mathbf{x}_i + b)) \\ & \text{subject to:} \\ & y_i (\mathbf{w}^T \mathbf{x}_i + b) \geq 1 - \varepsilon_i \\ & \varepsilon_i \geq 0 \end{aligned}$$

vs. Hard
SVM

$$\operatorname{argmin}_{\mathbf{w}, b} \sum_{i=1}^p w_i^2 / 2$$

$$\text{subject to } \forall \mathbf{x}_i \in D_{\text{train}} : y_i (\mathbf{w}^T \mathbf{x}_i + b) \geq 1$$

A little bit theory:

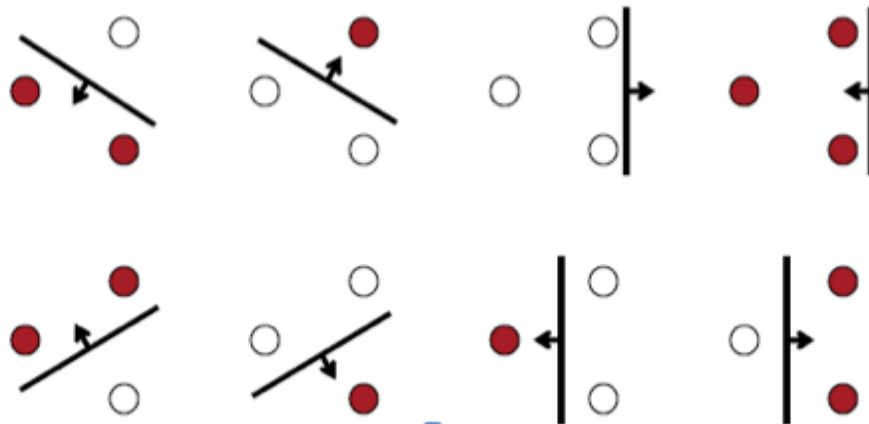
Vapnik-Chervonenkis (VC) dimension

$X \rightarrow \phi(X)$
 $\geq N-1$

If data is mapped into sufficiently high dimension, then samples will in general be linearly separable;

N data points are in general separable in a space of N-1 dimensions or more!!!

- VC dimension of the set of oriented lines in R^2 is 3
 - It can be shown that the VC dimension of the family of oriented separating hyperplanes in R^N is at least N+1

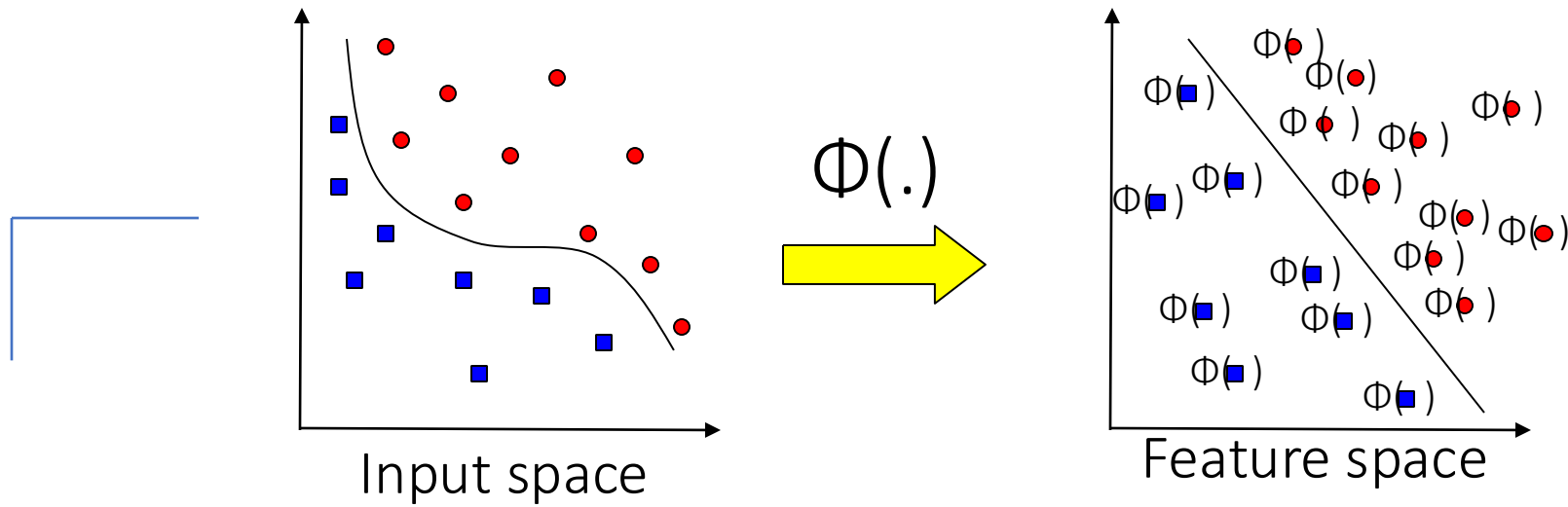


If data is mapped into sufficiently high dimension, then samples will in general be linearly separable;

N data points are in general separable in a space of N-1 dimensions or more!!!

$$X \rightarrow \Phi(X)$$

Linearly separated into
two classes $\{+1, -1\}$



SVM solves these two issues simultaneously

- “Kernel tricks” for efficient computation
- Dual formulation only assigns parameters to samples, not to features

$K(\mathbf{x}_i, \mathbf{x}_j) \equiv \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j)$ is called the kernel function.

- Linear kernel (we've seen it)

$$K(\mathbf{x}, \mathbf{z}) = \mathbf{x}^T \mathbf{z}$$

$$\begin{cases} \mathbf{x} \in \mathbb{R}^p \\ \mathbf{z} \in \mathbb{R}^p \end{cases}$$

- Polynomial kernel (we will see an example)

$$K(\mathbf{x}, \mathbf{z}) = \left(1 + \mathbf{x}^T \mathbf{z}\right)^d = \underbrace{\Phi_p(\mathbf{x})^T}_{O(p)} \underbrace{\Phi_p(\mathbf{z})}_{p \rightarrow O(p^d)}$$

where $d = 2, 3, \dots$. To get the feature vectors we concatenate all d th order polynomial terms of the components of $\mathbf{x}$ (weighted appropriately)

- Radial basis kernel

$$K(\mathbf{x}, \mathbf{z}) = \exp\left(-r \|\mathbf{x} - \mathbf{z}\|^2\right) = \underbrace{\Phi_r(\mathbf{x})^T}_{O(p)} \underbrace{\Phi_r(\mathbf{z})}_{p=\infty}$$

In this case, r is hyperpara. The feature space of the RBF kernel has an infinite number of dimensions

Never represent features explicitly

☐ Compute dot products with a closed form

Very interesting theory – Reproducing Kernel Hilbert Spaces

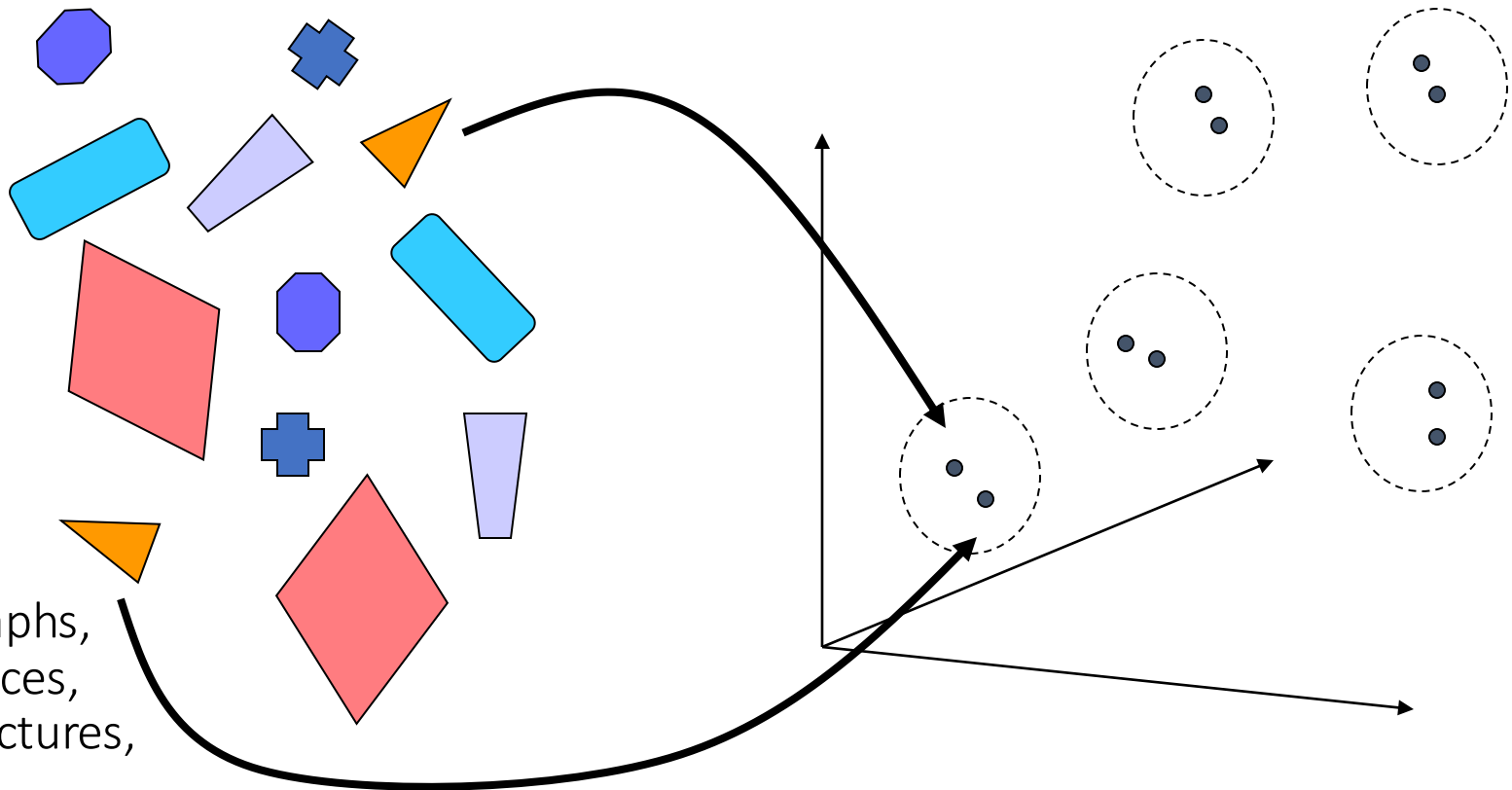
☐ Not covered in detail here

Kernel trick has helped Non-traditional data like strings and trees able to be used as input to SVM, instead of feature vectors

When numerical x and z do not exist, we can calculate

Vector vs. Relational data

$$K(x, z)$$



e.g. Graphs,
Sequences,
3D structures,

Summary:

Modification Due to Kernel Trick

- Change all inner products to kernel functions
- For training,

Original
Linear

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j$$

$$\sum_i \alpha_i y_i = 0$$

$$C > \alpha_i \geq 0, \forall i \in \text{train}$$

With kernel
function -
nonlinear

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j)$$

$$\sum_i \alpha_i y_i = 0$$

$$C > \alpha_i \geq 0, \forall i \in \text{train}$$

VC: $\phi(x)$
large

Support vectors: non-zero α_i

- only a few α_i can be nonzero!!

$$\forall i \Rightarrow \alpha_i (y_i (w^T x_i + b) - 1) = 0, i = 1, \dots, n$$

Handwritten diagram illustrating the margin maximization problem in SVM. It shows three parallel lines representing the decision boundary and margins. The top solid line is labeled $y_i(w^T x_i + b) = 1$ and the bottom solid line is labeled $y_i(w^T x_i + b) = -1$. The region between them is the margin. A dashed line in the middle represents the decision boundary. The distance from the decision boundary to the margins is labeled α_i . The top margin is labeled $\alpha_i = 0$ and the bottom margin is labeled $\alpha_i = 0$. The region between the margins is labeled $\alpha_i > 0$.

Dual SVM– Testing

To evaluate a new sample $\mathbf{x}_{ts}$ we need to compute:

$$\hat{y}_{ts} = \text{sign}(w^T \mathbf{x}_{ts} + b) = \text{sign}\left(\sum_{i=1..n} \alpha_i y_i \mathbf{x}_i^T \mathbf{x}_{ts} + b\right)$$

Dot product with (“all” ??)
training samples


$$\hat{y}_{ts} = \text{sign}\left(\sum_{i \in \text{SupportVectors}} \alpha_i y_i (\mathbf{x}_i^T \mathbf{x}_{ts}) + b\right)$$

↓
 $\alpha_i > 0$

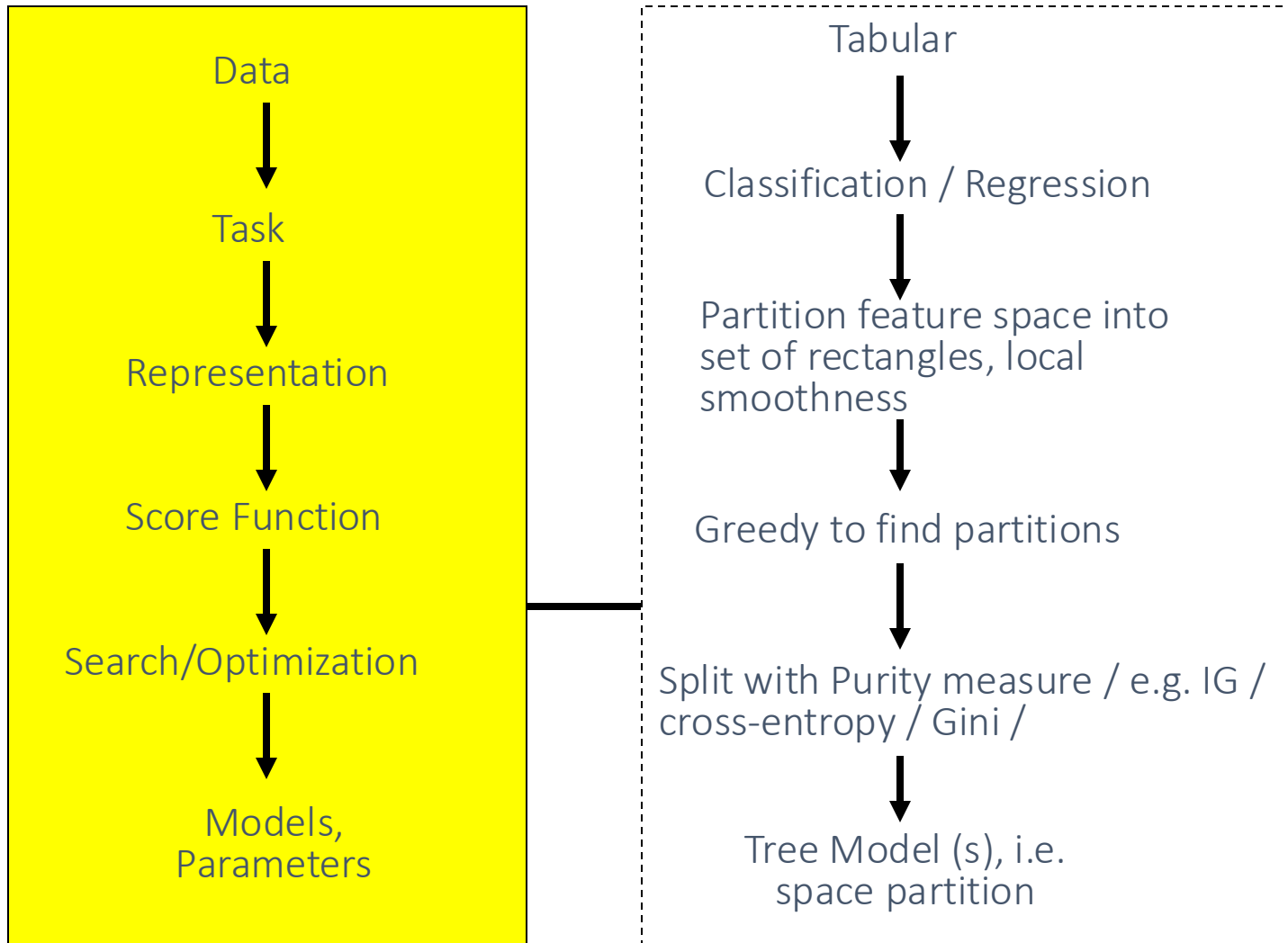
For α_i that are 0,
no influence

Why do SVMs work?

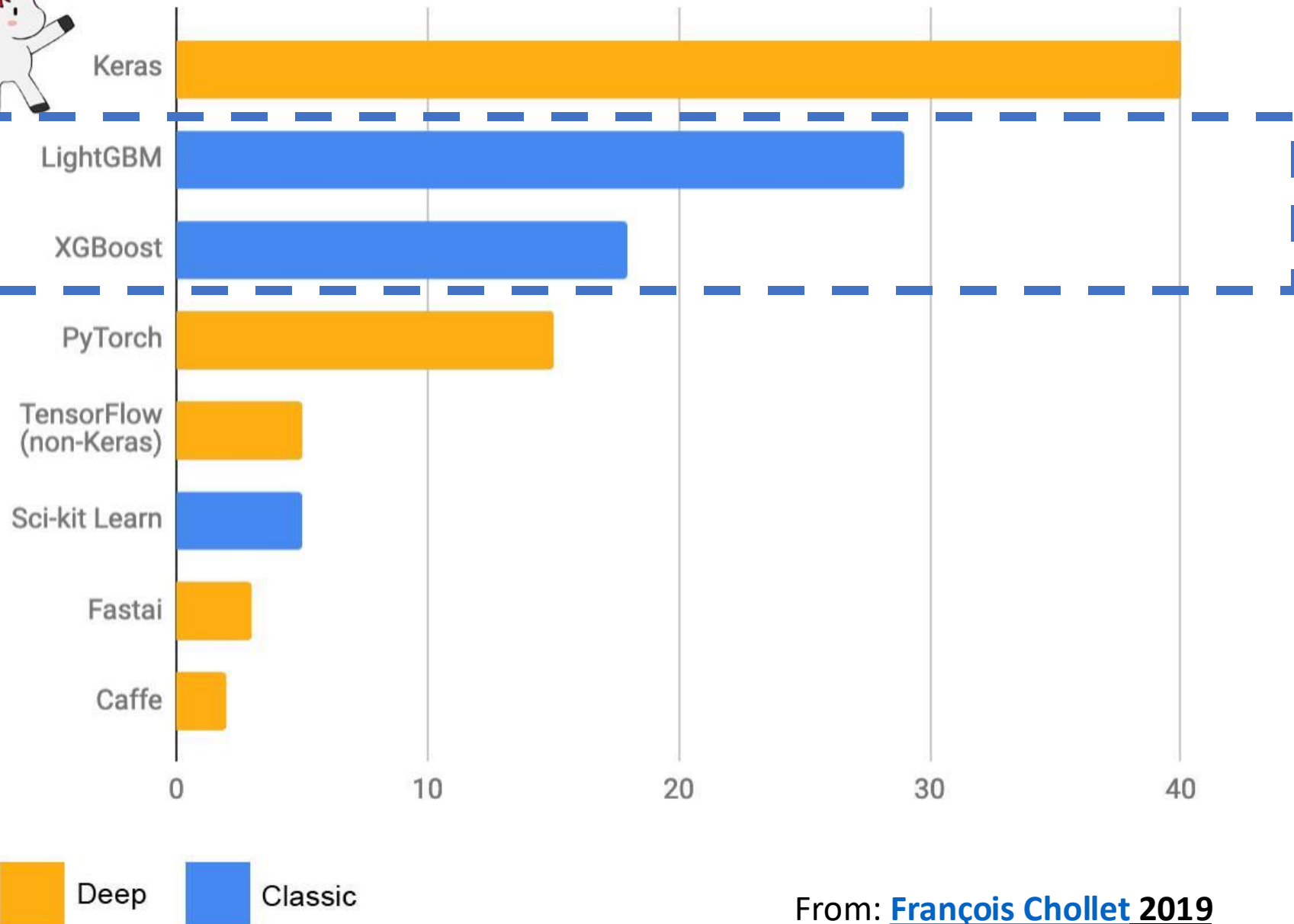
$x \rightarrow \phi(x)$ e.g. RBF

- ❑ If we are using huge features spaces (e.g., with kernels), how come we are not overfitting the data?
 - ✓ Number of parameters remains the same (and most are set to 0)
 $O(1)$, α_i $i=1, \dots, n$
 - ✓ While we have a lot of inputs, at the end we only care about the support vectors and these are usually a small group of samples
 - ✓ The maximizing of the margin acts as a sort of regularization term leading to reduced overfitting

L22: Decision Tree / Bagged DT/ Random Forest



Primary ML software tool used by top-5 teams on Kaggle in each competition (n=120)



From: [François Chollet 2019](#)

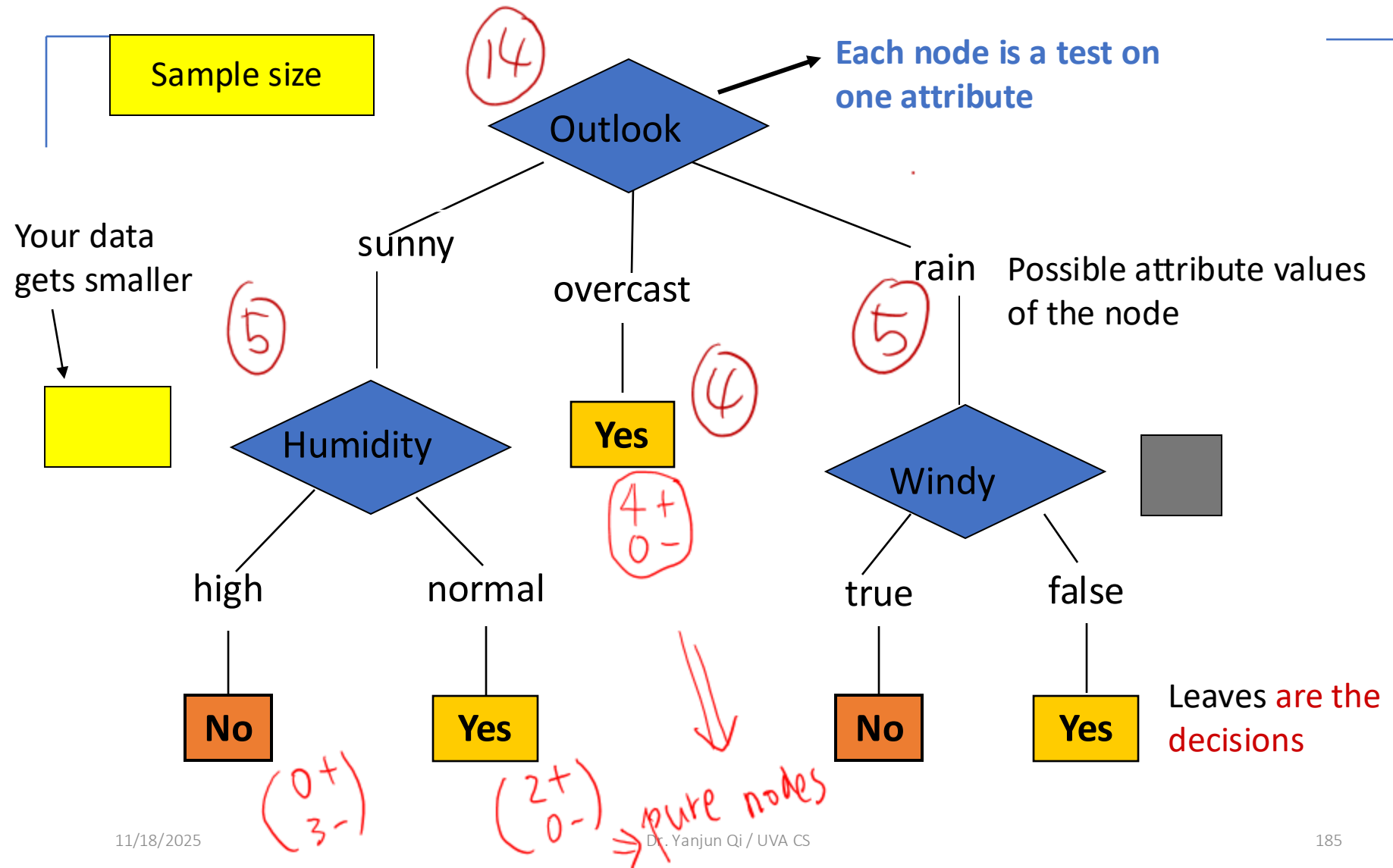
Example

- Example: Play Tennis

PlayTennis: training examples

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes ←
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes ←
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes ←
D13	Overcast	Hot	Normal	Weak	Yes ←
D14	Rain	Mild	High	Strong	No

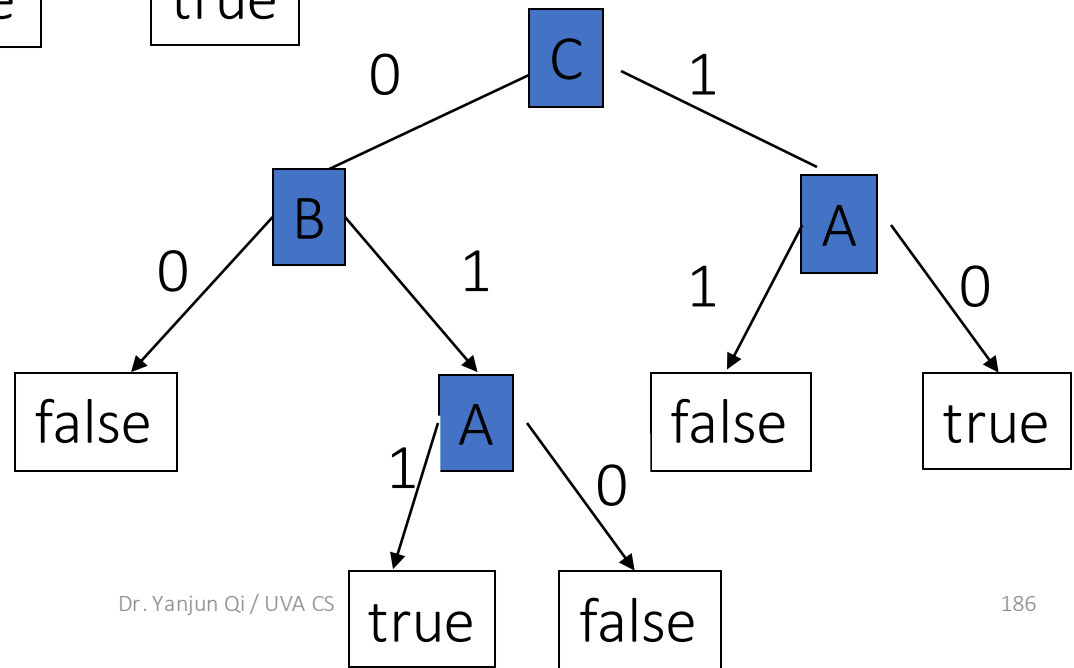
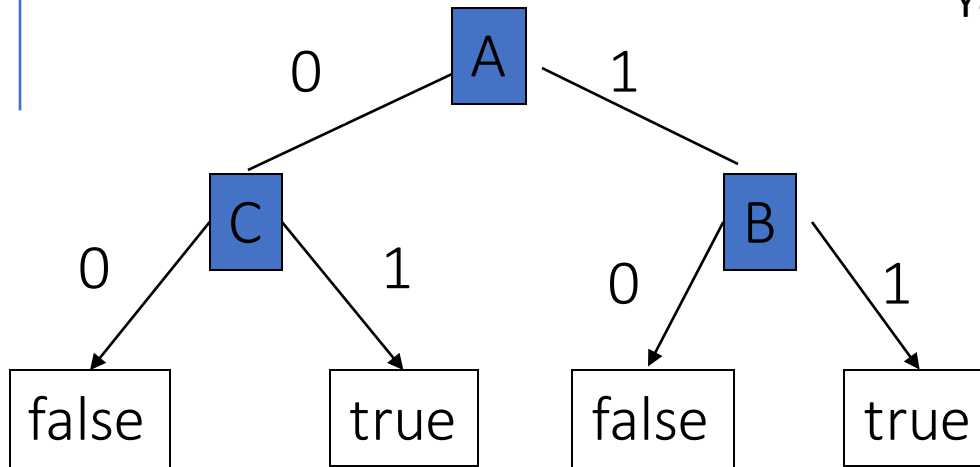
Anatomy of a decision tree



Same concept / different representation

$$Y = ((A \text{ and } B) \text{ or } ((\text{not } A) \text{ and } C))$$

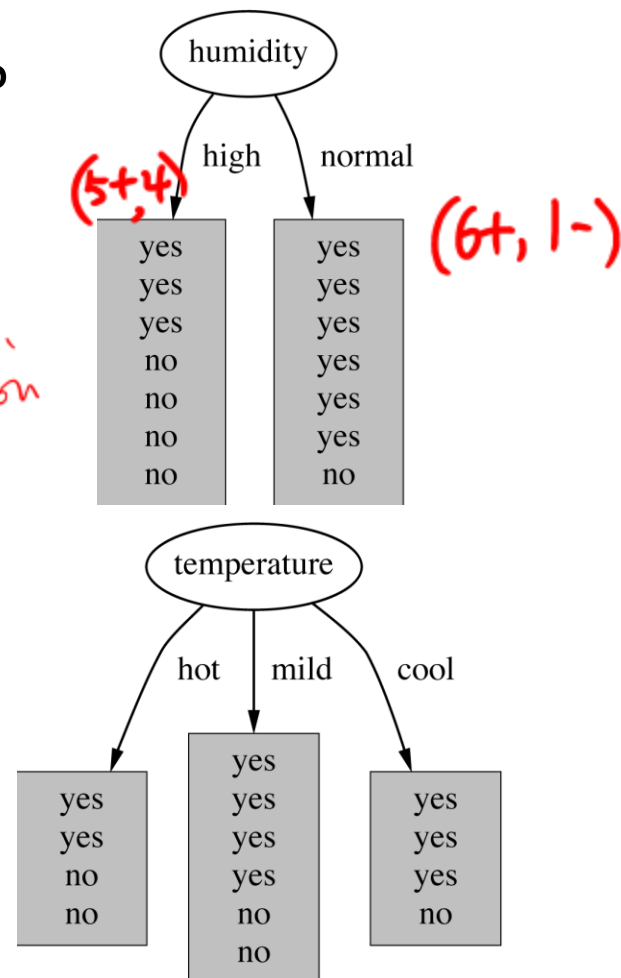
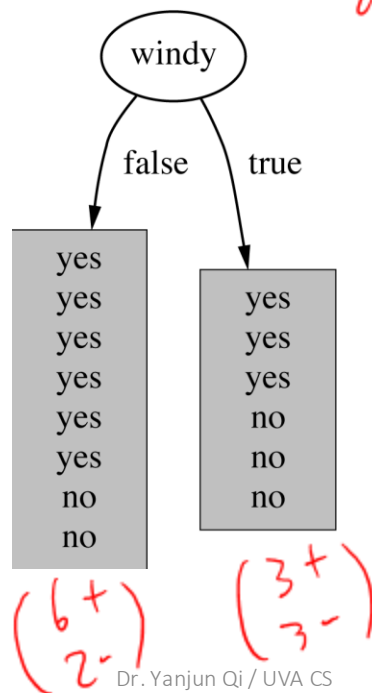
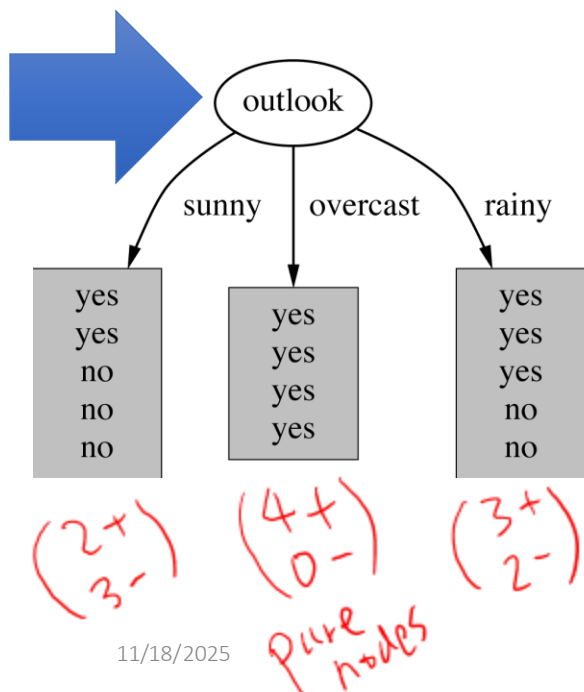
not unique



How do we choose which attribute to split ?

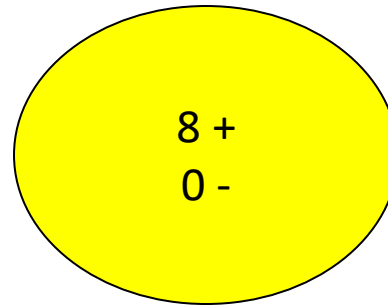
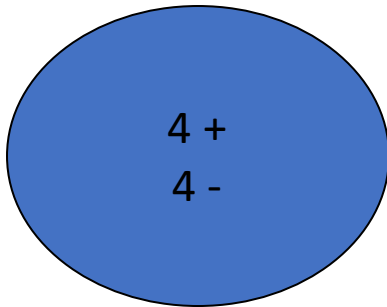
Which attribute should be used first to test?

Intuitively, you would prefer the one that *separates* the training examples as much as possible. *→ Wrt. class distribution*



Entropy Lower $\Rightarrow$ better purity

- Entropy measures the purity



The distribution is less uniform
Entropy is lower
The node is purer

Information gain

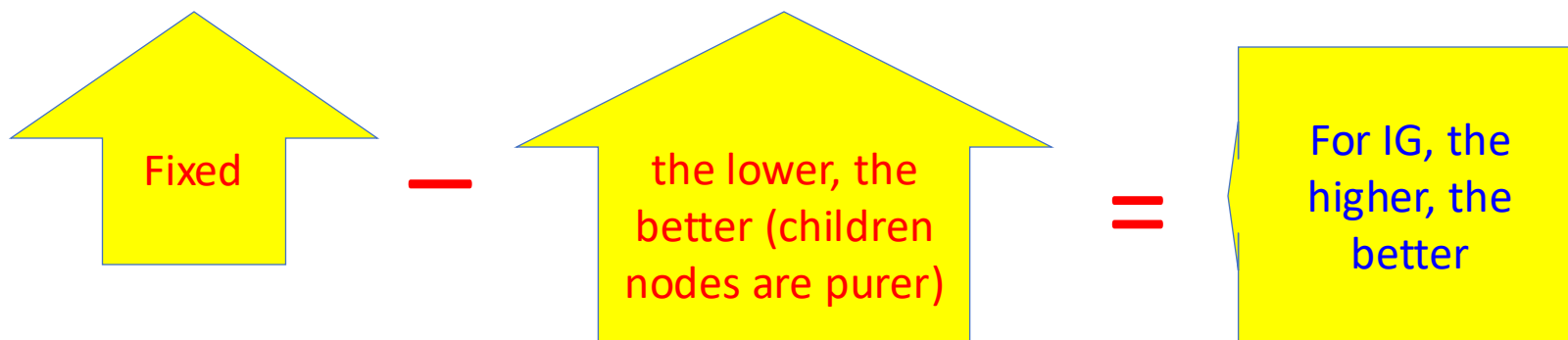
- $IG(X,Y)=H(Y)-H(Y|X)$

Reduction in uncertainty of Y by knowing a feature variable X

Information gain:

= (information before split) – (information after split)

= entropy(parent) – [average entropy(children)]



Information gain

- $IG(X,Y)=H(Y)-H(Y|X)$

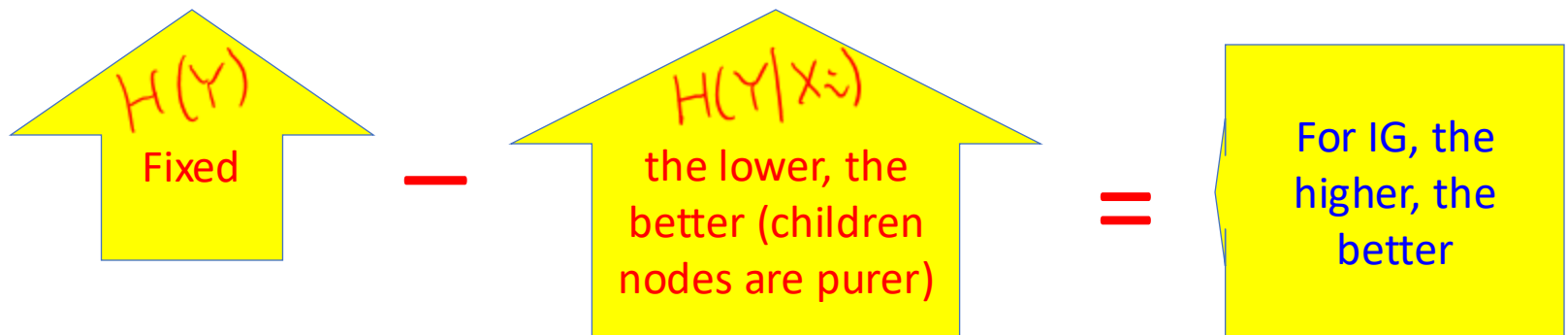
→ pick x_i
BY
 $\arg\max_{x_i}$ $\left\{ \begin{array}{l} IG(x_1, Y) \\ IG(x_2, Y) \\ \vdots \\ IG(x_m, Y) \end{array} \right.$

Reduction in uncertainty of Y by knowing a feature variable X

Information gain:

= (information before split) – (information after split)

= entropy(parent) – [average entropy(children)]



Conditional entropy

$$H(Y) = - \sum_i p(y_i) \log_2 p(y_i)$$

$$H(Y | \underbrace{X = x_j}) = - \sum_i p(y_i | x_j) \log_2 p(y_i | x_j)$$

$$H(Y | X) = \sum_j \underbrace{p(x_j)} H(Y | X = x_j)$$

$$= - \sum_j p(x_j) \sum_i p(y_i | x_j) \log_2 p(y_i | x_j)$$

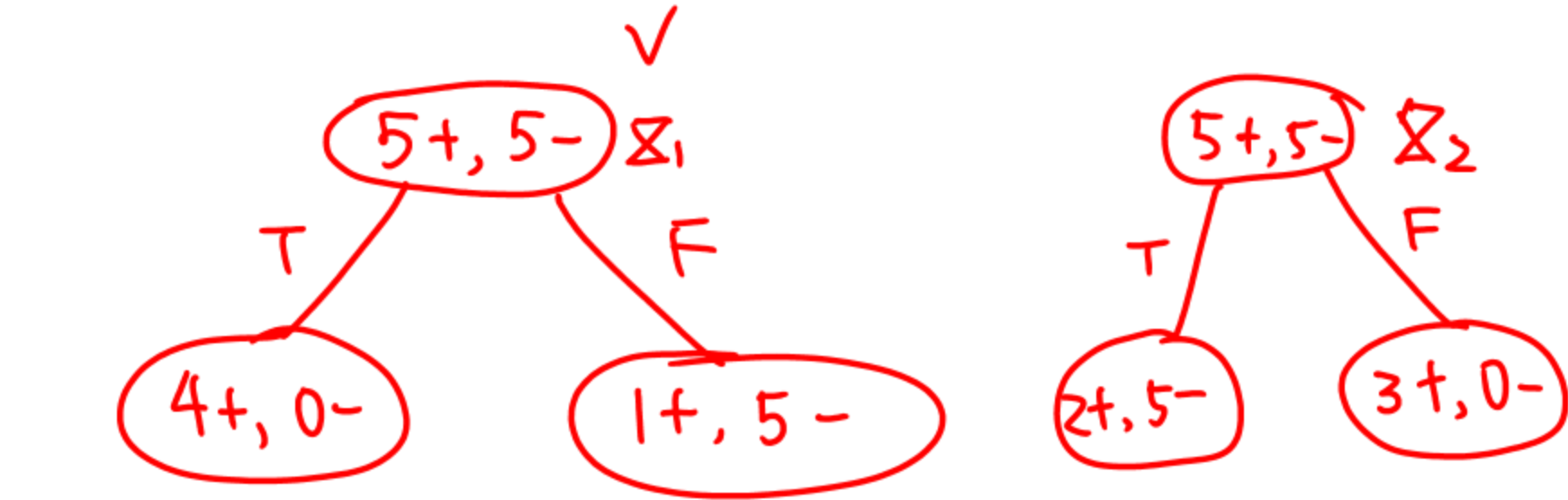
Example

Attributes Labels

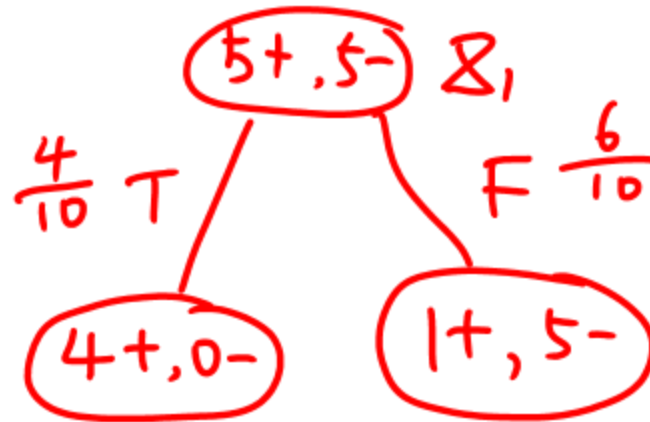
X1	X2	Y	Count
T	T	+	2
T	F	+	2
F	T	-	5
F	F	+	1

Which one do we choose

X1 or X2?



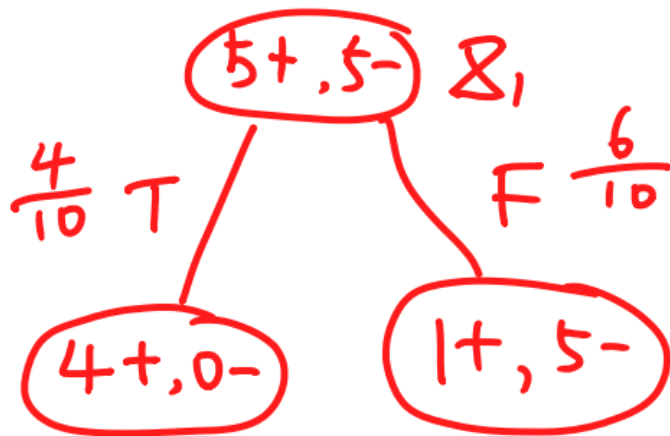
X1	X2	Y	Count
T	T	+	2
T	F	+	2
F	T	-	5
F	F	+	1



$$H(Y | X_1 = T) = - \left\{ p(Y=+ | X_1=T) \log p(Y=+ | X_1=T) \right. \\ \left. \begin{matrix} \text{4+, 0-} \Rightarrow \\ + p(Y=- | X_1=T) \log p(Y=- | X_1=T) \end{matrix} \right\} \\ = 0$$

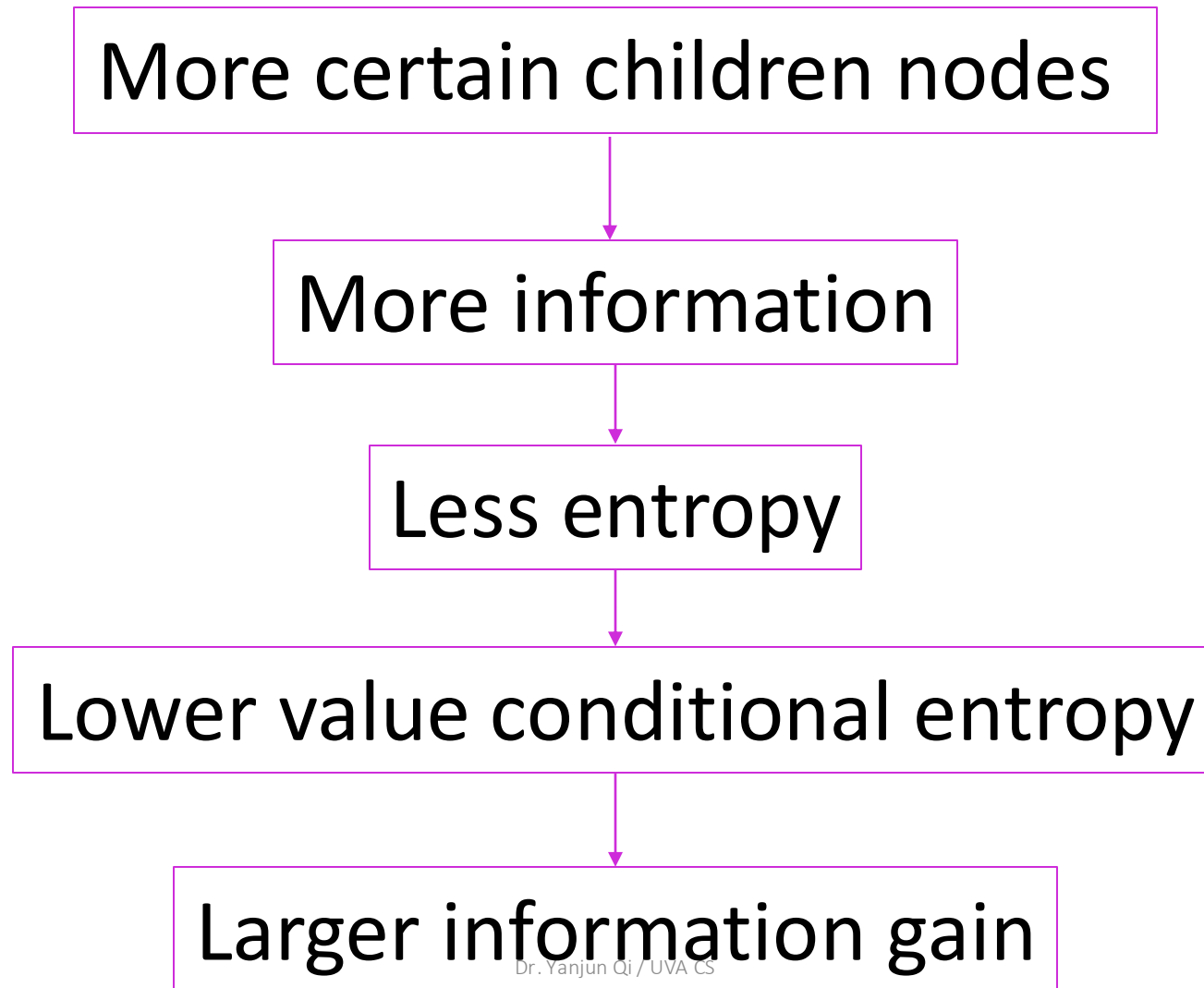
$$H(Y|X_1=T) = \begin{pmatrix} 4+ \\ 0- \end{pmatrix} \Rightarrow -(P(+) \log P(+) + P(-) \log P(-)) \\ = -(1 \lg 1 + 0 \log 0) = 0$$

$$H(Y|X_1=F) = \begin{pmatrix} 1+ \\ 5- \end{pmatrix} \Rightarrow -(P(+) \log P(+) + P(-) \log P(-)) \\ = -\left(\frac{1}{6} \log \frac{1}{6} + \frac{5}{6} \log \frac{5}{6}\right)$$



$$H(Y|X_1) = \frac{4}{10} H(Y|X_1=T) + \frac{6}{10} H(Y|X_1=F)$$

Intuition of Node Splitting



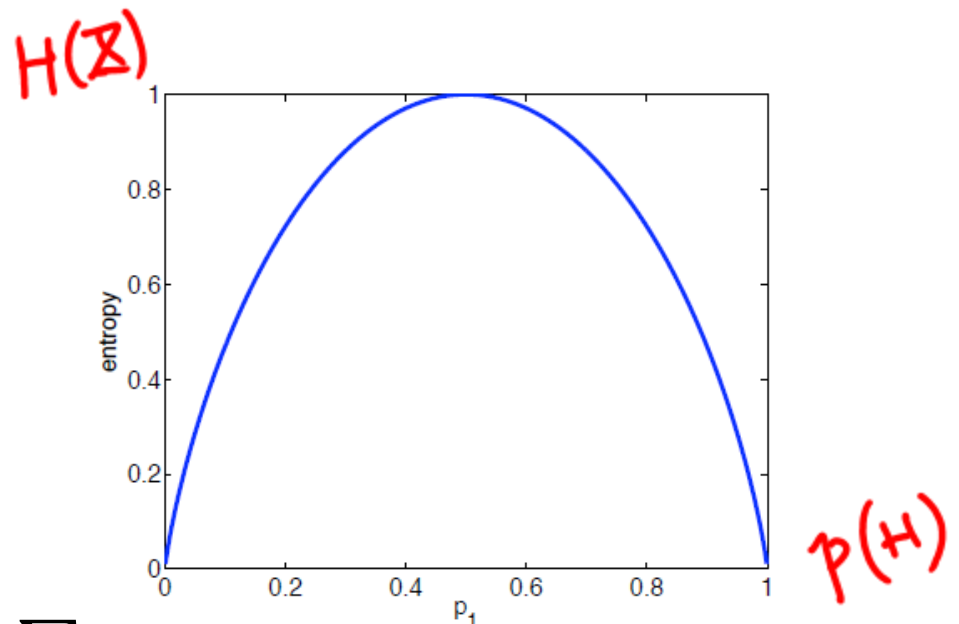
Entropy

- If there are k possible outcomes

$$H(X) \leq \log_2 k$$

- Equality holds when all outcomes are equally likely

- The more the probability distribution that deviates from uniformity, the lower the entropy



$$H(X) = E(I(X)) = \sum_i p(x_i) I(x_i) = -\sum_i p(x_i) \log_2 p(x_i)$$

e.g. for a random binary variable

Many tree building algorithms...(EXTRA)

Feature	C4.5	CART	CHAID	CRUISE	GUIDE	QUEST
Unbiased Splits				✓	✓	✓
Split Type	<i>u</i>	<i>u,l</i>	<i>u</i>	<i>u,l</i>	<i>u,l</i>	<i>u,l</i>
Branches/Split	≥ 2	2	≥ 2	≥ 2	2	2
Interaction Tests				✓	✓	
Pruning	✓	✓		✓	✓	✓
User-specified Costs		✓	✓	✓	✓	✓
User-specified Priors		✓		✓	✓	✓
Variable Ranking		✓			✓	
Node Models	<i>c</i>	<i>c</i>	<i>c</i>	<i>c,d</i>	<i>c,k,n</i>	<i>c</i>
Bagging & Ensembles					✓	
Missing Values	<i>w</i>	<i>s</i>	<i>b</i>	<i>i,s</i>	<i>m</i>	<i>i</i>

b, missing value branch; *c*, constant model; *d*, discriminant model; *i*, missing value imputation; *k*, kernel density model; *l*, linear splits; *m*, missing value category; *n*, nearest neighbor model; *u*, univariate splits; *s*, surrogate splits; *w*, probability weights



11/18

Today Roadmap

- 0. To Remind:
 - Dec. 9th Final Exam in Person (notes only!)
 - Next week: Nov.25th class zoom (100%! Based on Survey 3)
 - Please sign up your project final presentation time slot ASAP!
 - HW5 in Canvas! Project Due entry in Canvas!
- 1. Review some course content
 - A. Hier. Clustering (an example to run through)
 - B. Boosting
 - C. Decision Tree (how to calculate IG, an example to run through!)
- 2. Quiz 12
 - FYI: Q11 is the hardest over all past quizzes ... So no worries if not full scored
 - Makeup quizzes Q14 + Q15 in the week of Dec. 2nd
 - We will use your top10 quiz grade ... only top 10 !!

S4: Lecture 23 : : Ensemble

- Framework of Ensemble:

- 1. Get a set of classifiers $f_1(x), f_2(x), f_3(x), \dots$

They should be diverse.

$f_B(x)$

How to have different training data sets

- Re-sampling your training data to form a new set
- Re-weighting your training data to form a new set

blend

- 2. Aggregate the classifiers (*properly*)



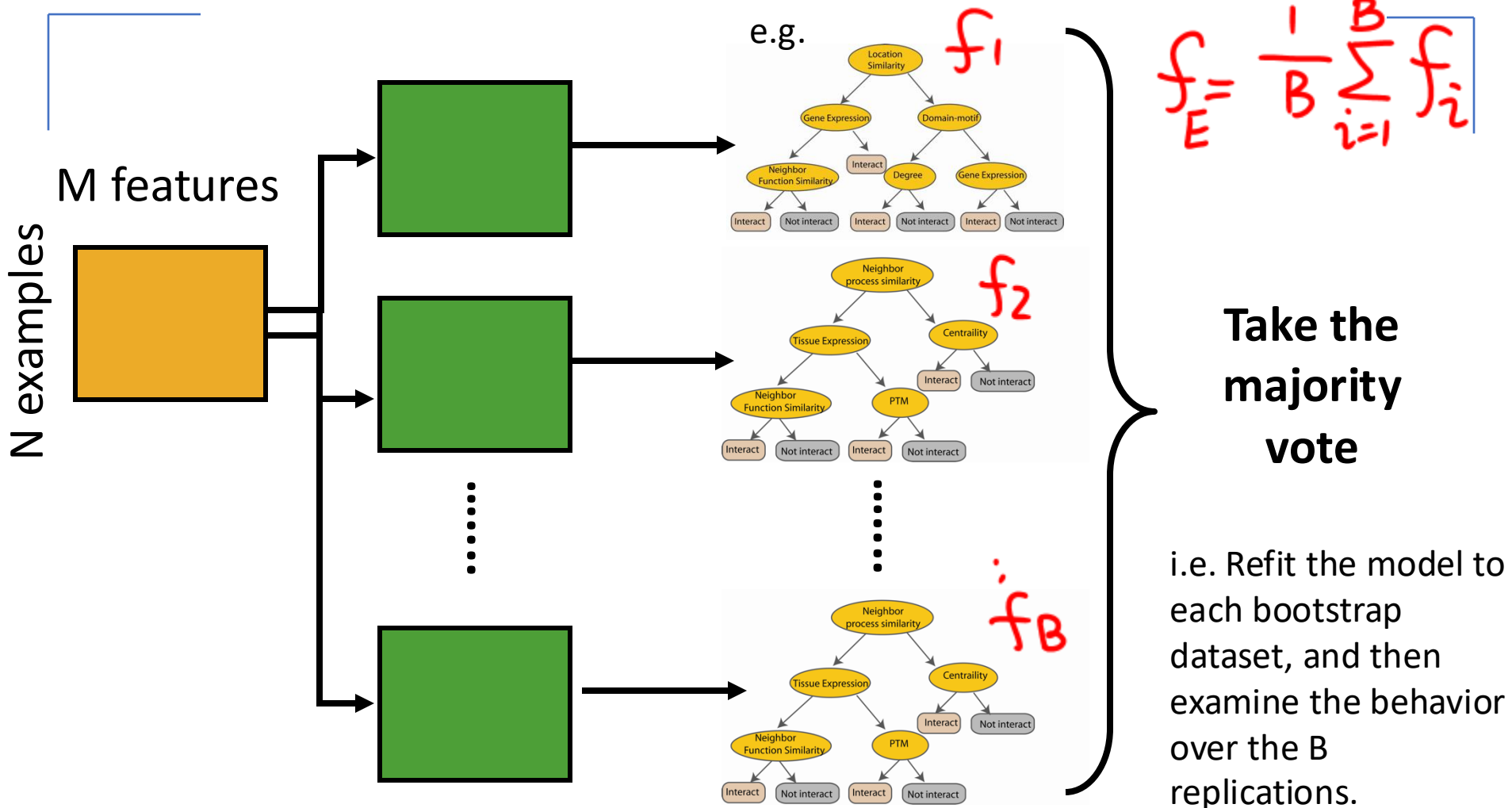
S4: Lecture 23 :

Random Forest / Boosting : Ensemble

- Bagging / reduce Variance
 - Bagged Decision Tree
 - Random forests:
- Boosting
 - Adaboost / reduce bias , reduce Variance
 - Xgboost
- Stacking / Gradient Boosting



Bagging of DT Classifiers



With vs Without Replacement



- **Bootstrap with replacement** can keep the **sampling size the same as the original size** for every repeated sampling. The sampled data groups are independent on each other.
- **Bootstrap without replacement** cannot keep the sampling size the same as the original size for every repeated sampling. The sampled data groups are dependent on each other.

$$\text{Var}(f_i)$$

vs.

$$\text{Var}\left(\frac{1}{B} \sum_{i=1}^B f_i\right)$$

Decision Boundary Comparison

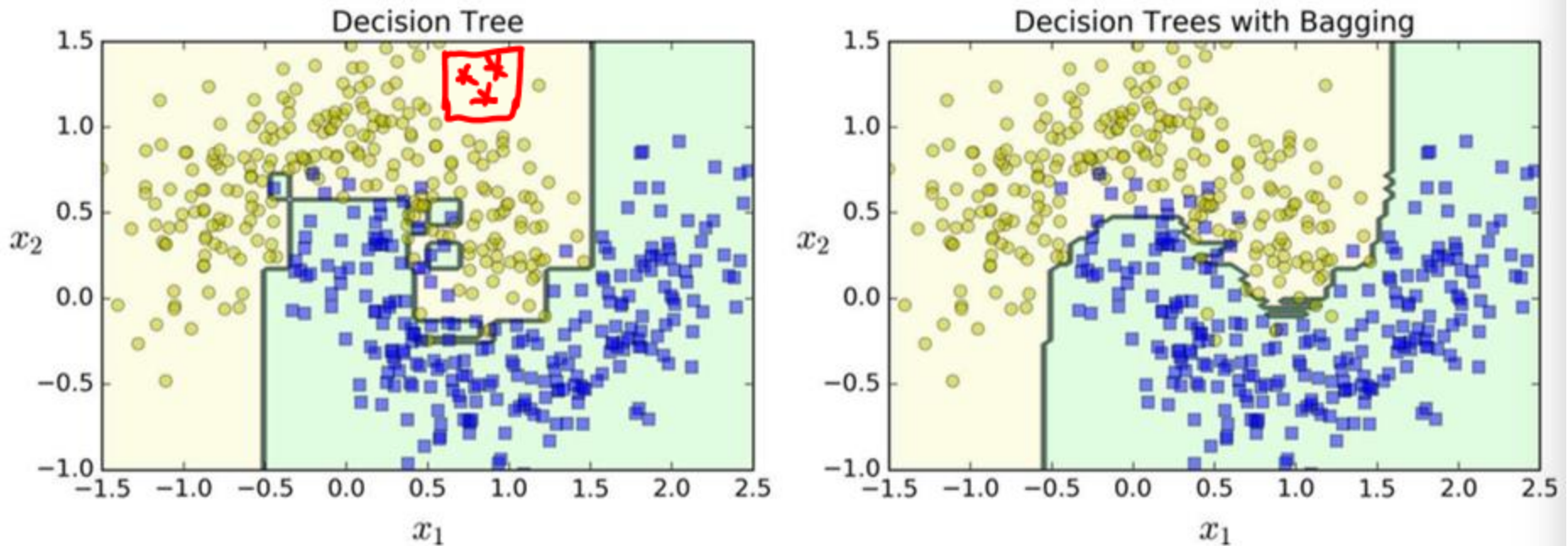
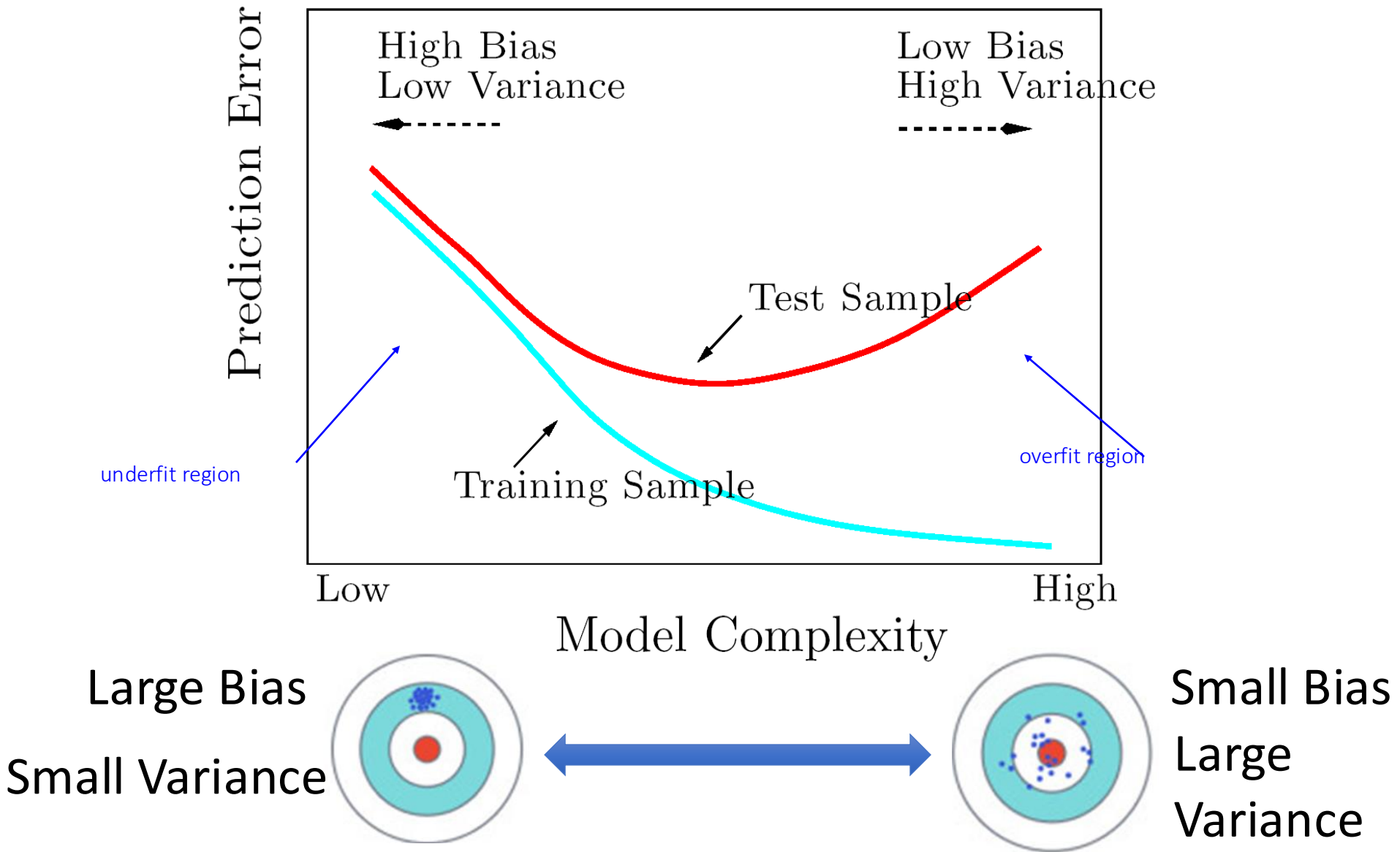


Figure 7-5. A single Decision Tree versus a bagging ensemble of 500 trees

Recap: Bias-Variance Tradeoff / Model Selection



classifiers $f_1(x)$,
 S_1

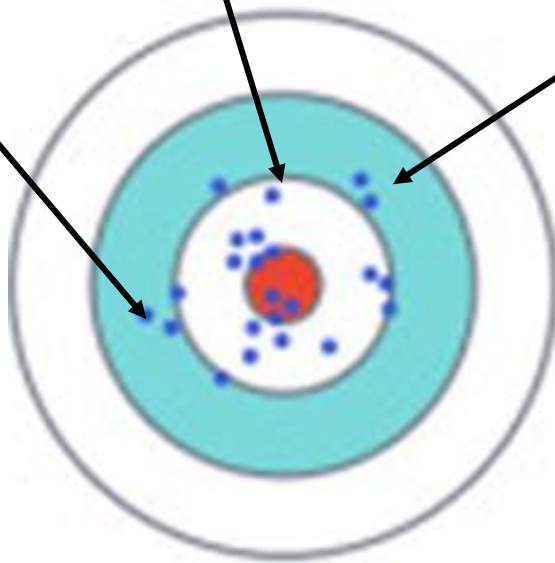
$f_2(x)$,
 S_2

$f_3(x)$,

$f_B(x)$
 S_B

A complex model will
have large variance.

We can average
complex models to
reduce variance.



If we average all the f_i ,
is it close to f^*

$$E[\hat{f}] = f^*$$

Bagged Trees → Random Forests

1. Construct subset $(x_1^*, y_1^*), \dots, (x_n^*, y_n^*)$ by sampling original training set with replacement.
2. Build tree-structured learners $h(x, \Theta_k)$, where at each node, **m predictors at random** are selected before finding the best split.
 - Gini Criterion.
 - No pruning.
3. Combine the predictions (average or majority vote) to get the final result.

Why correlated trees (in Bagging) are not ideal ?

Assuming each tree has variance σ^2

If simply **identically distributed**, then average variance is

$$\rho_{ij} = \text{Corr}(f_i, f_j) = \rho$$
$$\rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$$

As $B \rightarrow \infty$, second term $\rightarrow 0$

Thus, the pairwise correlation always affects the variance

Why correlated trees (in Bagging) are not ideal ?

How to deal?

If we reduce m (the number of dimensions we actually consider in each splitting),

then we reduce the pairwise tree correlation

Thus, variance will be reduced.

S4: Lecture 23 :

Random Forest / Boosting : Ensemble

➤ Bagging / reduce Variance

➤ Bagged Decision Tree

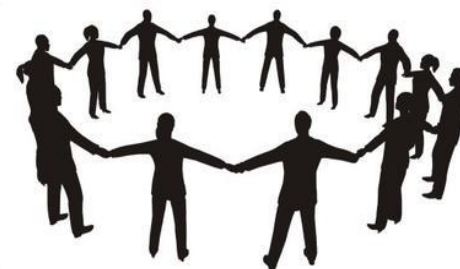
➤ Random forests:

➤ Boosting

➤ Adaboost / reduce bias , reduce Variance

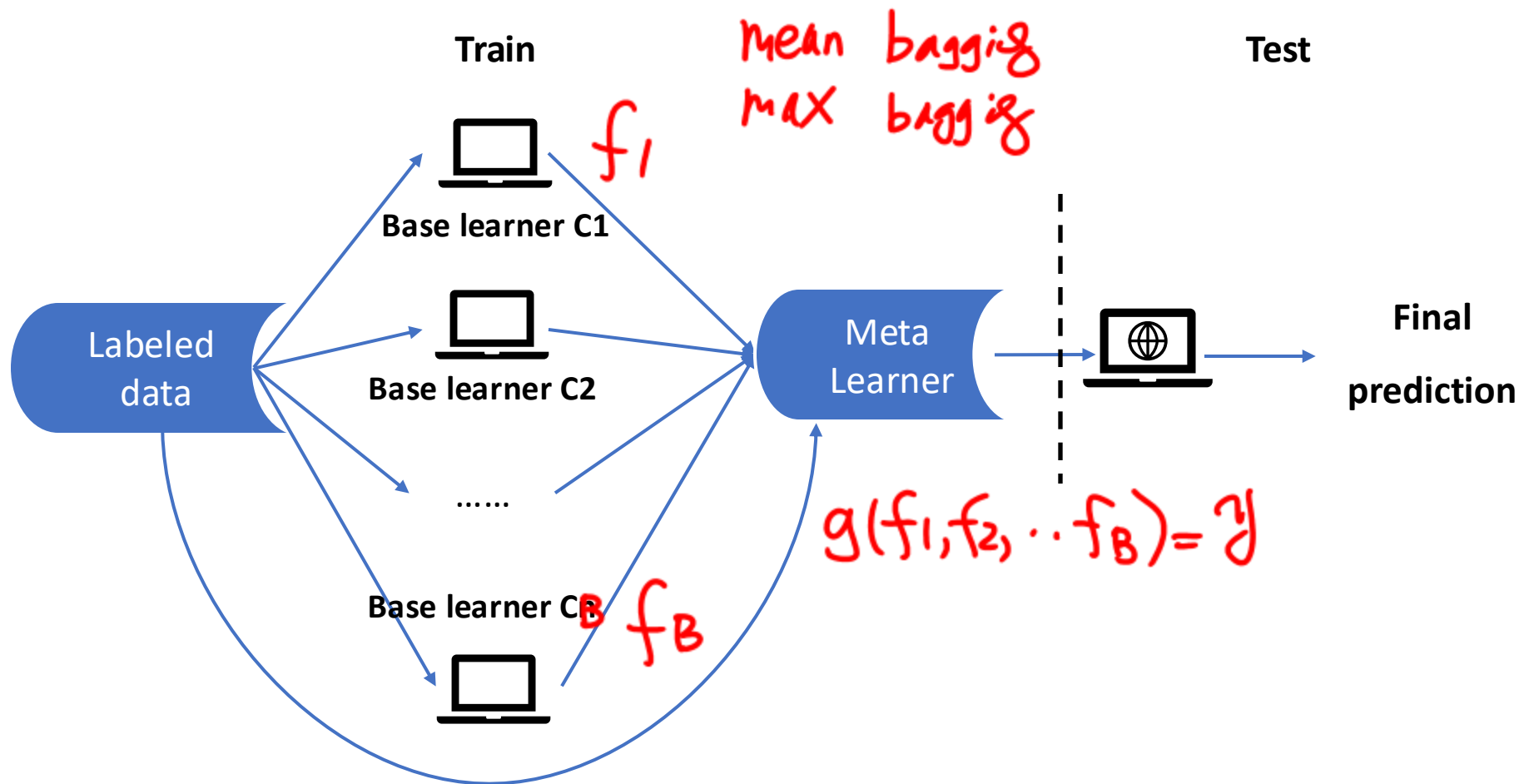
➤ Xgboost

➔ ➤ Stacking / Gradient Boosting



Stacking

- Main Idea: Learn and combine multiple classifiers



Generating Base and Meta Learners

- Base model—efficiency, accuracy and diversity
 - Sampling training examples
 - Sampling features
 - Using different learning models
 - Meta learner
 - Majority voting
 - Weighted averaging
 -
 - Higher level classifier — Supervised (e.g. Xgboost as blender)
- Unsupervised
- 

S4: Lecture 23 :

Random Forest / Boosting : Ensemble

➤ Bagging / reduce Variance

➤ Bagged Decision Tree

➤ Random forests:

➔ ➤ Boosting

➤ Adaboost / reduce bias , reduce Variance

➤ Xgboost

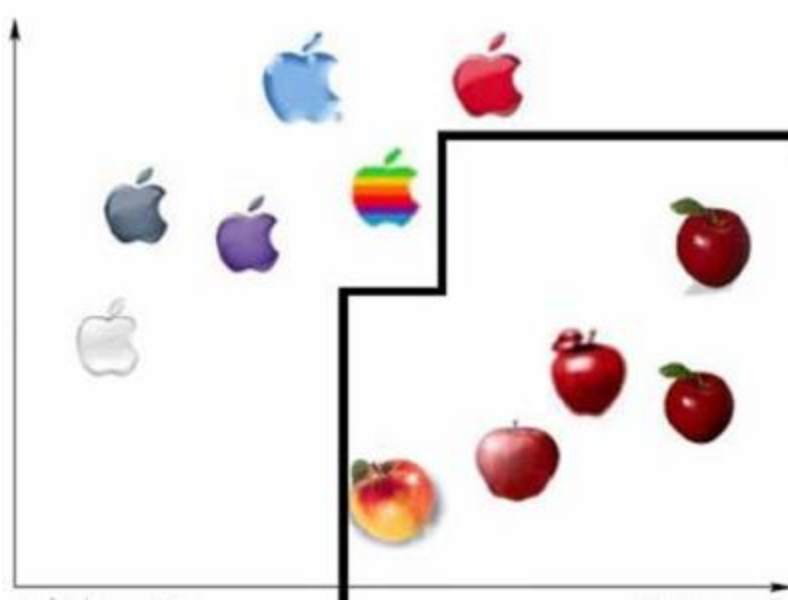
➤ Stacking / Gradient Boosting



Boosting Strategies

1. Have many rules (base classifiers) to **vote** on the decision
 1. Base learners are **shallow decision trees!!!**
2. **Sequentially** train base classifiers that **corrects** mistakes of previous → focus on **hard** examples
3. Give higher **weight** to better rules

Final Classifier is the additive combination of base rules:



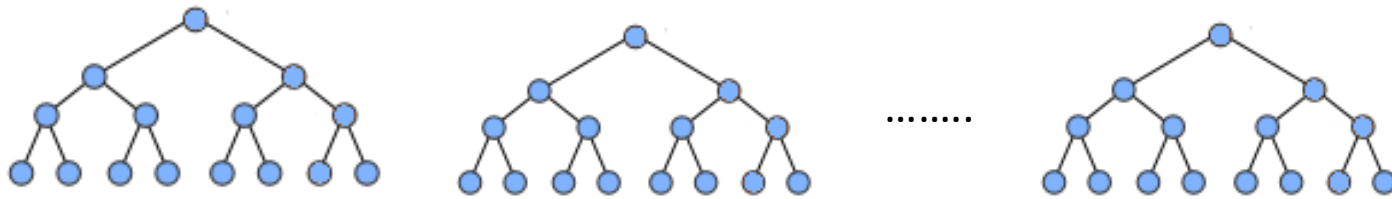
Boosting vs. Bagging

- Similar to bagging, boosting combines a weighted sum of many classifiers, thus **both reduce variance**.
- One key difference: unlike bagging, boosting fit the tree to the entire training set, and adaptively weight the examples.
- **Boosting** tries to do better at each iteration, (by making model a bit more complex), thus **it reduces bias**.

XGBoost

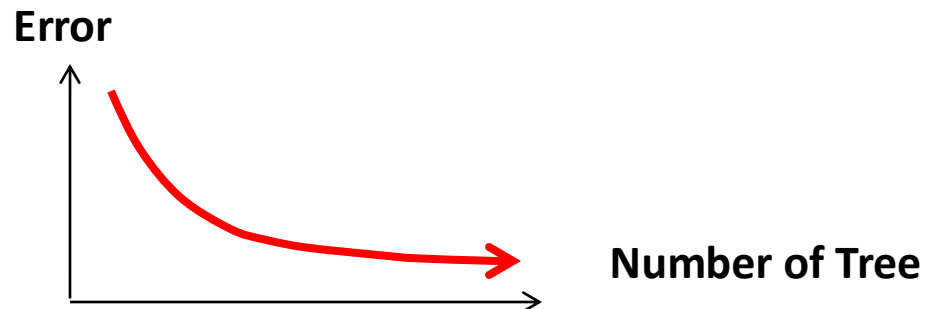
- Additive tree model: add new trees that complement the already-built ones
- Response is the optimal linear combination of all decision trees
- Popular in Kaggle Competitions for efficiency and accuracy

Additive tree model



More in 18c-
extraBoosting
Slides

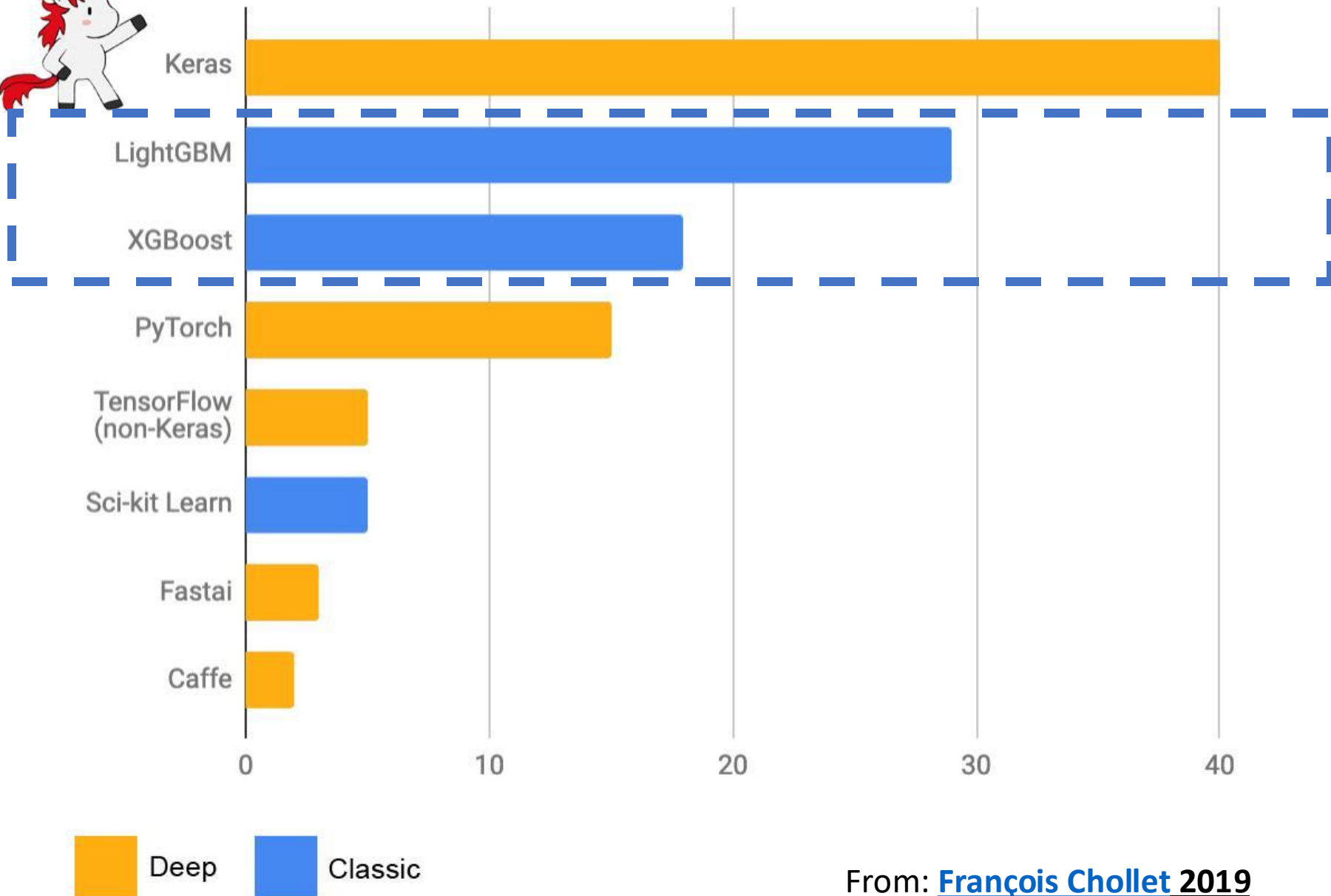
Greedy Algorithm



XGBoost Implementation

- XGBoost is a **very efficient Gradient Boosting Decision Tree implementation** with some interesting features:
- **Regularization:** Can use L1 or L2 regularization.
- **Handling sparse data:** Incorporates a sparsity-aware split finding algorithm to handle different types of sparsity patterns in the data.
- **Weighted quantile sketch:** Uses distributed weighted quantile sketch algorithm to effectively handle weighted data.
- **Block structure for parallel learning:** **Makes use of multiple cores on the CPU**, possible because of a block structure in its system design. Block structure enables the data layout to be reused.
- **Cache awareness:** Allocates internal buffers in each thread, where the gradient statistics can be stored.
- **Out-of-core computing:** **Optimizes the available disk space and maximizes its usage when handling huge datasets that do not fit into memory.**

Primary ML software tool used by top-5 teams on Kaggle in each competition (n=120)

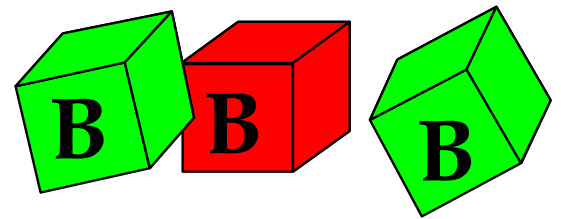
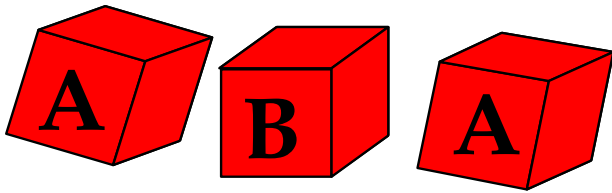


From: [François Chollet 2019](#)

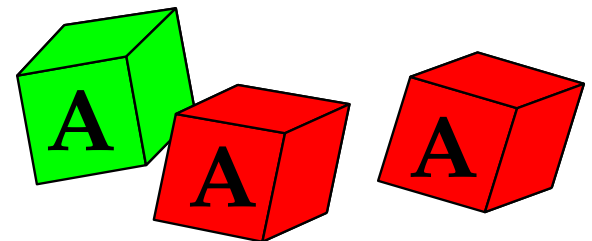
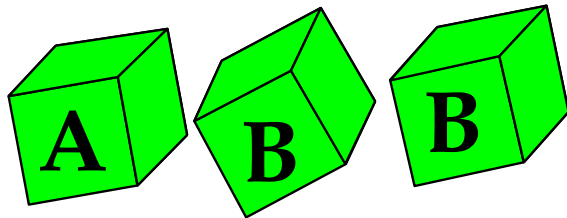
S5: Lecture 24:

Unsupervised Clustering (I): Hierarchical

example: clustering is subjective



Two possible Solutions...



Course Content Regarding Tasks

 ~~Regression (supervised)~~

Y is a continuous

 ~~Learning theory~~

About $f()$

 ~~Classification (supervised)~~

Y is a discrete

 Unsupervised models

NO Y

 ~~Graphical models~~

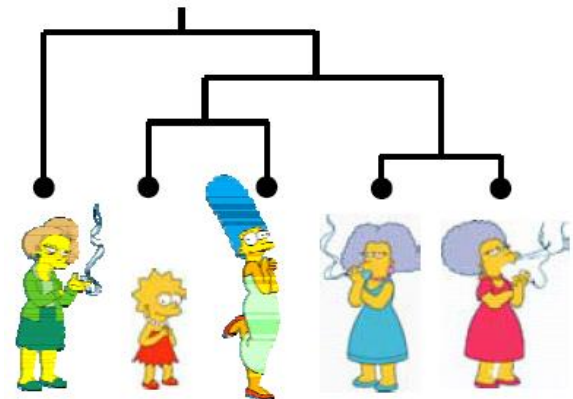
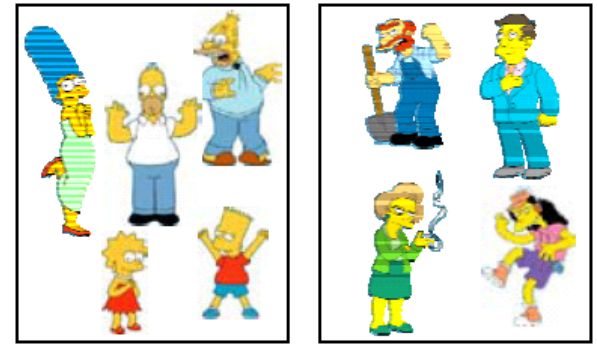
About interactions among $Y, X_1, \dots, X_p$

 Reinforcement Learning

Learn to Interact with environment

Clustering Algorithms

- Partitional algorithms
 - Usually start with a random (partial) partitioning
 - Refine it iteratively
 - K means clustering
 - Mixture-Model based clustering
- Hierarchical algorithms
 - Bottom-up, agglomerative
 - Top-down, divisive



(How-to) Hierarchical Clustering

- Given: a set of objects and the pairwise distance matrix
- Find: a tree that optimally hierarchical clustering objects?
 - Globally optimal: exhaustively enumerate all tree
 - Effective heuristic methods:

(How-to) Hierarchical Clustering

The number of dendrograms with n leafs
 $= (2n - 3)! / [(2^{n-2}) (n - 2)!]$

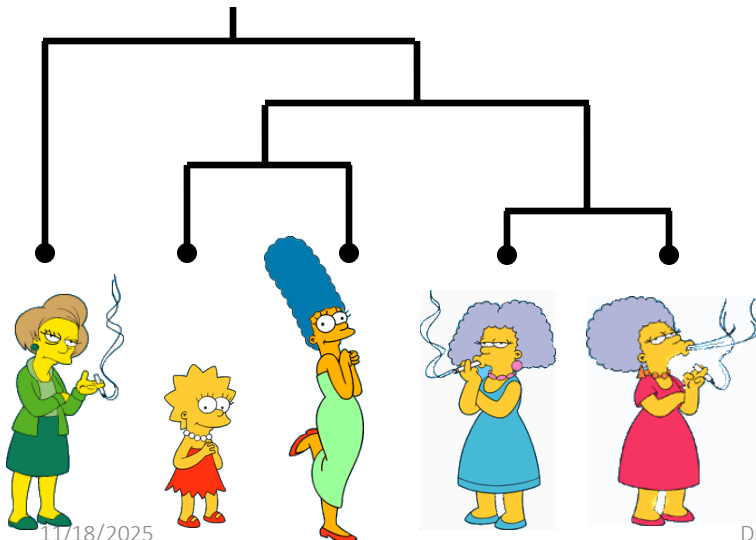
Number of Leafs	Number of Possible Dendrograms
2	1
3	3
4	15
5	105
...	...
10	34,459,425

Bottom-Up (agglomerative): Starting with each item in its own cluster, find the best pair to merge into a new cluster. Repeat until all clusters are fused together.



Clustering: the process of grouping a set of objects into classes of similar objects →

high intra-class similarity
low inter-class similarity



How to decide the distances between clusters ?

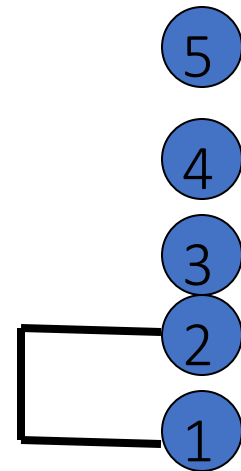
- Single-Link
 - Nearest Neighbor: their closest members.
- Complete-Link
 - Furthest Neighbor: their furthest members.
- Average:
 - average of all cross-cluster pairs.

Summary of Hierarchical Clustering Methods

- No need to specify the number of clusters in advance.
- Hierarchical structure maps nicely onto human intuition for some domains
- They do not scale well: time complexity of at least $O(n^2)$, where n is the number of total objects.
- Like any heuristic search algorithms, local optima are a problem.
- Interpretation of results is (very) subjective.

Example: single link

	1	2	3	4	5
1	0				
2	2	0			
3	6	3	0		
4	10	9	7	0	
5	9	8	5	4	0



Extra: Cost analysis

	1	2	3	4	5
1	0				
2	2	0			
3	6	3	0		
4	10	9	7	0	
5	9	8	5	4	0

$N \times N$

$(1,2) \downarrow$

	1	2	3	4	5
1	0				
2	2	0			
3	6	3	0		
4	10	9	7	0	
5	9	8	5	4	0

$\Rightarrow O(N)$

$$\vec{x}_i \in \mathbb{R}^p, (\vec{x}_1, \vec{x}_2, \dots, \vec{x}_N)$$

$$\text{time: } O(\text{dist}(\vec{x}_i, \vec{x}_j)) \sim O(p)$$

$$O(\text{pairwise Matrix}) \sim O(pN^2)$$

$$\text{BestCluster} \sim O(N^2)$$

